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# Matryoshka market: Nested convictions in equities


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## Summary


- **Geopolitical resolution is the master assumption.** Equity markets are pricing a framework agreement by September, consistent with the prediction market consensus and the pronounced kink in VIX term structure at Q3. That bet has logic: the global (political) cost of ongoing conflict and mid-term elections make a deal valuable to the US administration by autumn. But a genuine agreement requires Iran to provide credible security guarantees for the Strait of Hormuz. Tail and downside risks are visible in options but not in equity valuations. In our downside scenario, equities could correct by 25% and 20%, respectively, in Europe and the US. Indeed, if resolution fails and oil remains structurally elevated, every subsequent layer inherits a more adverse starting point: margins compress, central banks are constrained and the stagflation risk premium returns not as a tail scenario but as a working assumption.

- **The earnings story is concentrated exposure to some sector tailwinds, not broad-based resilience.** The S&P 500 has delivered double-digit EPS growth for six consecutive quarters, and European corporates surprised with Q1 growth of +11.5% y/y. Record margins should be read as limited remaining buffer. Beyond the headline figure, approximately 17pps of S&P 500 EPS growth in Q2 2026 originate from the IT sector alone – growing at an estimated +66% y/y and representing roughly 31% of total index earnings. Excluding technology, the rest of the S&P 500 expands at around +12%, in line with a 6% nominal GDP world. Europe tells a parallel story: exclude energy, the direct beneficiary of elevated crude, and STOXX 600 EPS growth falls from +11.5% to approximately +7%. Furthermore, a 20% equity correction would additionally impair US consumption by an estimated 1-1.5pp of GDP through the wealth-effect channel – a feedback loop markets are pricing as exogenous.

- **Non-cyclical index composition concentrates rate sensitivity rather than reducing it and the pending mega-IPO wave deepens the crowding.** The shift in major equity indices away from cyclicals toward technology, healthcare, utilities and high-dividend defensives is read as structural insulation. Non-cyclical sectors divide into bond-proxy profiles (utilities, healthcare) and long-duration growth profiles (large-cap technology): both are materially sensitive to real interest rates, through yield competition and discount rate mechanics respectively. The index has not removed rate risk, it has hidden it in less visible form. The pending mega-IPO wave deepens this structural vulnerability. SpaceX (USD1.75trn), OpenAI (USD850bn) and Anthropic (USD850bn) – combined USD3.45trn – would represent approximately 4-5% of S&P 500 market capitalization at index inclusion, making their combined entry the largest supply event in US equity market history. All three would land in or adjacent to technology, further crowding the most rate-sensitive index segment.

- **The real equity risk premium is a breakeven, not a buffer.** The negative nominal equity risk premium (ERP) is rationalized two ways: equities offer inflation protection that nominal bonds do not, and dominant technology constituents carry a structural growth premium that justifies a negative nominal spread, by analogy with long-duration bonds whose value sits in distant cash flows. On the real ERP measure (i.e. earnings yield minus real risk-free rate) the picture is thinner still, sitting at approximately 2.5%. That is not a buffer, it is a

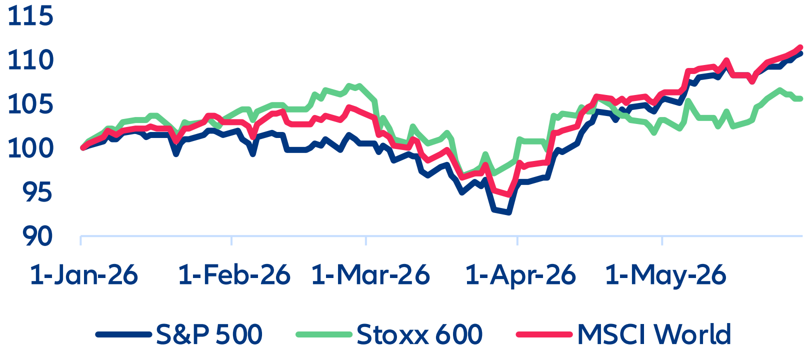
breakeven. A rise in real rates of 50–75bps eliminates it entirely. At that level, the reallocation from equities into investment-grade credit becomes a mechanical trade for liability-matched institutional investors. The multiple stays compressed until either real rates fall or earnings growth widens the premium. In a scenario where geopolitics and growth have simultaneously deteriorated, neither condition arrives quickly.

- **Technical positioning amplifies fundamental moves non-linearly while the mega-IPO pipeline adds further risk.** The 2025–2026 rally has been amplified by structural technical tailwinds: CTA trend-following models near maximum long, option dealers suppressing intraday volatility and FINRA margin debt near cycle highs. Each is mechanically reversible, and the reversal is self-reinforcing rather than gradual. A 5-7% fundamental-driven decline can cascade into a 15–20% technical drawdown before positional overhang clears. The mega-IPO pipeline introduces another amplifier poorly captured by conventional risk models. Passive vehicles must rebalance to new index weights on a defined schedule, selling existing holdings irrespective of valuation or momentum. These rebalancing flow would represent some of the largest single-session institutional selling pressure seen outside of crisis conditions, enough to breach the technical thresholds that convert a managed correction into a self-sustaining cascade without any prior fundamental deterioration. A lockup expiry cliff at approximately 180 days post-listing adds a second, later-dated supply wave.
- **In the adverse scenario, all three policy levers are simultaneously constrained.** Since 2010, equity pricing has reflected not only fundamental value but policy optionality – the assumption that a geopolitical deal, fiscal stimulus or central bank rate cuts are available in any deterioration. The underpriced risk is not the absence of any single lever but the correlation of constraints across all three. The geopolitical channel has been partially activated - ceasefire negotiations are themselves a policy response. The fiscal channel has structural limits: Europe's substantial post-2022 commitments to energy security and industrial policy did not fully offset structural competitiveness damage from sustained energy repricing. The central bank channel requires either sovereign market dysfunction or a banking stress event to activate – high-bar triggers in an inflationary environment.

## A geopolitical resolution is the master assumption of equity markets

**As of May 2026, the S&P 500 and Euro Stoxx 600 sit within a few percent of their all-time highs.** Implied volatility has spiked on event days but mean-reverted quickly. It seems that equity markets are looking through the war in Iran – one of the most geopolitically significant events in the Middle East in a generation (see Figure 1). However, the market is not ignoring the war. It is pricing a specific outcome: resolution over the summer. That outcome is also holding a set of convictions on other key issues from interest rates to earnings to economic policies. In that sense equity markets can be seen as a Matryoshka doll: it is not a deceptive object, but the outer vivid layer hides how many layers remain and what sits at the center.

Figure 1: Equity performance (Jan-26 = 100)




*Sources: LSEG Workspace, Allianz Research*

**The assumption of a resolution makes sense but it comes with risks.** The options market's implied volatility term structure shows a pronounced kink around Q3 2026: elevated near-term volatility fading sharply into year-end, consistent with pricing a binary resolution event. Confirming this conviction, the prediction market for an Iran ceasefire or framework agreement by September sits above 60%. Implicitly, what market participants and analysts are expecting is that the political cost of the ongoing crisis will lead to a resolution – one way or the other. Surging inflation, fuel shortages, the potential social and economic risks coupled with the perspective of US mid-term elections make a deal by autumn "inevitable". This is not an irrational bet. But it is a specific, time-bound, politically-conditioned bet. If it was to break down, it would have more consequences down the line.

**The critical issue is the concession structure required for a genuine agreement.** A durable deal requires Iran to move simultaneously on two fronts: a verifiable freeze or rollback of its nuclear program, and some form of security guarantees for the strait of Hormuz. A partial framework agreement is plausible and may be sufficient to sustain equity markets temporarily. But a nominal settlement that unravels, a spoiler event – proxy escalation, third-party strike – or a US administration that claims resolution without genuine Iranian compliance, would leave the problem intact. The probability-weighted payoff from being long equities in a world where a September resolution fails is underrepresented in current levels. If the conflict persists and oil remains structurally elevated, the stagflation risk premium returns. In our downside scenario, equities could correct by 25% and 20%, respectively, in Europe and the US. Brent reprices higher, energy input cost pass-through hits margins and inflation persistence that follows constrains central bank flexibility. Every subsequent layer of analysis would inherit a more adverse starting point.

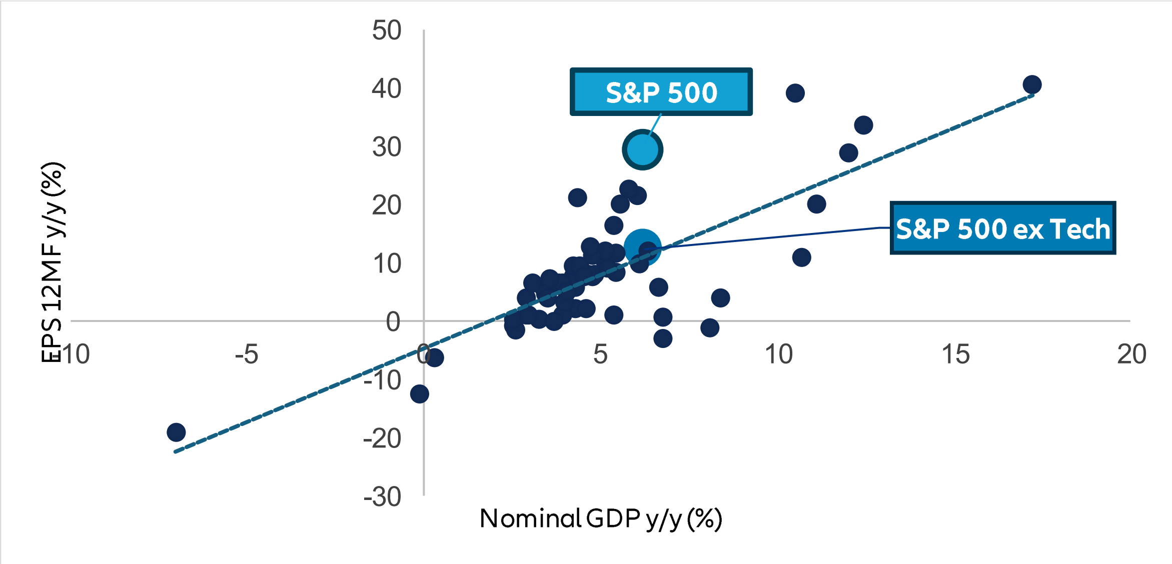
## Record margins and earnings signal asymmetric downside

**The earnings picture entering 2026 has been much better than feared.** The S&P 500 index has been recording double-digit earnings growth for over six consecutive quarters now, and the trend is expected to continue throughout 2026. On the other side of the Atlantic, the STOXX Europe 600 surprised with Q1 earnings jumping by +11.5% y/y (Figure 3). Europe, often dismissed as structurally lagging, is increasingly demonstrating that earnings resilience is becoming a cyclical force of its own. Revisions have been net positive for two consecutive quarters. Growth forecasts have held – US GDP is tracking above 2%, European growth has recovered modestly from the 2024-25 soft patch and inflation has remained below the 3.5-4% threshold that historically triggers meaningful demand destruction. Corporate margins are at or near record highs. The market reads this as durable fundamental support: a solid earnings floor. Inflation is treated as a manageable pass-through phenomenon that is fueling turnover and not hurting volume significantly. The implicit assumption is that margin resilience reflects a relative pricing power.

**Record margins are a sign of limited remaining buffer.** When margins are at historical peaks, the distribution of outcomes is skewed: the scope for positive earnings surprise is narrow, while the scope for compression if costs rise or volumes soften is substantial. Markets are treating elevated margins as a baseline from which to project forward earnings. They should instead be read as a vulnerability indicator – the starting point from which any adverse shock produces a larger-than-average revision cycle.

**Behind the headline: one sector is doing the heavy lifting**. Aggregate earnings growth can provide valuable insights, though it is important to consider the nuances to make a fully informed assessment. Figure 2 breaks down the S&P 500's current 12-month forward EPS trajectory, comparing it to the historical relationship between corporate earnings and nominal GDP growth. This relationship has shown remarkable stability across cycles from 2012 to 2025. The conclusion is clear: if we exclude the technology sector, the remaining 69% of S&P earnings align with the regression line, with an estimated growth rate of around 12% year-on-year, in line with the 6% nominal GDP growth rate. This indicates a stable and predictable performance, consistent with market cycles. The macro is performing its intended function. The addition of technology has led to a 29% increase in EPS growth, a level that would typically require an 8-10% nominal GDP to maintain. The index is not being priced according to the economic conditions of the market it is intended to represent. It is being priced for one sector's ability to defy the economy it inhabits.

Figure 2: EPS 12-month forward vs nominal GDP (Q1 2012 – Q2 2026)




*Sources: LSEG, Allianz Research*

**The arithmetic behind this discrepancy matters.** It is estimated that approximately 17pps of headline S&P EPS growth in Q2 2026 originate from the IT sector, accounting for roughly 31% of total index earnings. This sector is projected to expand at an estimated rate of +66% year-on-year. This pace is not indicative of the broader earnings cycle, but rather a capex monetization thesis, concentrated in a select group of names, that is meaningfully ahead of any macro anchor. The remaining components of the index are expanding at a rate that aligns with the broader economic environment. However, this implies limited potential for further growth and limited capacity to offset deficiencies in other components of the index.

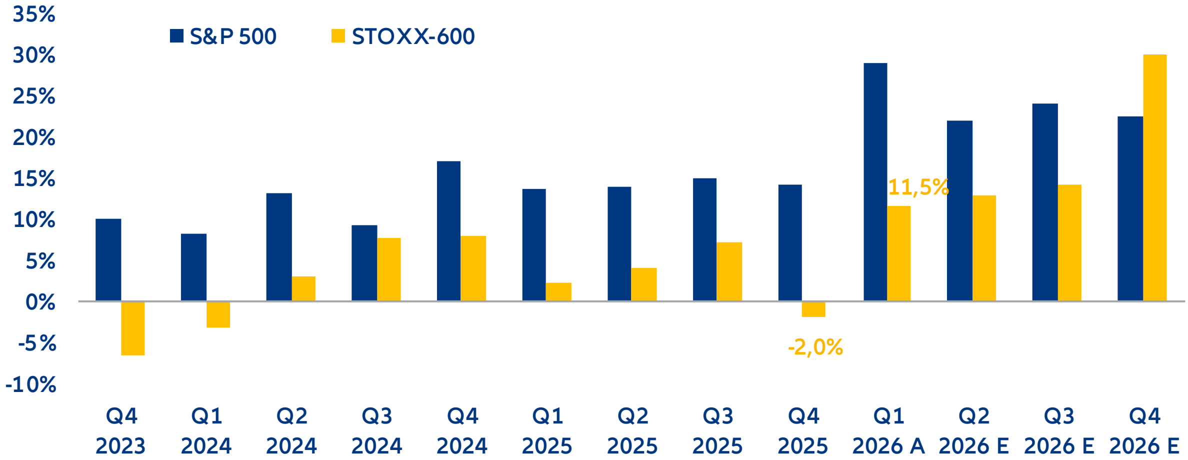
**This reframing of the asymmetry argument provides greater precision.** The concern is not that margins will compress uniformly across the index; it is that a single sector will decelerate. A moderation from +66% to a still-exceptional +30% IT earnings growth – the kind of deceleration that would accompany any credible concern about AI capital spending returns, hyperscaler spending fatigue or monetization delays – would have a materially negative effect on the headline S&P EPS number, with limited offset from the rest of the market. The index's earnings recovery, which is currently supported by a broad range of stakeholders, is, in essence, a highly concentrated exposure to a single factor: the continued compounding of AI capex monetization at its current rate. The "record margin" story is, arithmetically, a single-sector story – strip out IT and the index is operating in line with a 6% nominal GDP world.

**One underappreciated risk is the circular reference in the current macro configuration.** US equity markets and US consumption are not independent variables. Household wealth effects from equity performance materially influence consumer spending, which in turn supports corporate revenues and earnings. A 20% equity market correction – the order of magnitude in our downside scenario – would impair US consumption by an estimated 1-1.5pp of GDP. That growth impairment feeds back into earnings revisions, which would extend the equity correction. Markets are pricing equities as though consumption is an exogenous growth driver but it is partly endogenous to equity performance itself.

## Sector composition of index hides heterogenous trends and concentrate risks

**Q1 2026 corporate results and earnings revisions paints a two-speed market.** While both regions show improving forward EPS trajectories into 2026, sector dynamics are playing a huge role in the diverging performance across the Atlantic. While the continued earnings expansion of the S&P 500 remains largely concentrated within the technology sector (EPS: +55% growth in Q1) – primarily driven by AI-related capex acceleration and hyperscaler monetization dynamics – the STOXX Europe 600 (historically weaker and more volatile) has recently shown a meaningful improvement in market sentiment and earnings expectations, despite Europe's historically more fragile and cyclical macro backdrop. After delivering the worst earnings growth in seven quarters in Q4 2025 (-2.0% y/y), European corporates are heading to the best quarter in three years, recording a double digit growth of +11.5% y/y not seen since Q1 2023 (Figure 3). Yet, this noteworthy growth is fundamentally explained by one single sector: energy, registering an EPS growth of around +50% in Q1. Excluding this sector, the STOXX-600's average EPS growth rate is around +7%. The escalation of geopolitical tensions in the Middle East has sustained elevated crude prices and widened upstream profitability, allowing major oil & gas companies to deliver a substantial expansion in operating margins and cash-flow generation, ultimately translating into significantly stronger P&L performance across the sector. Furthermore, consensus earnings expectations for the upcoming quarters of 2026 have materially improved as well, supported by the strong momentum in the European energy complex. By the end of 2025, EPS estimates for Q2 and Q3 growth pointed to +3.1% and +6.6%, respectively, while current estimates point to double-digit figures of +12.8% and +14.2%. This upward revision reflects not only a sustained period of elevated crude prices following more than three months of geopolitical escalation in the Middle East, but also the increasing likelihood that heightened energy market volatility will persist well into 2026, supporting structurally stronger upstream earnings and reinforcing the positive revisions momentum across the sector.

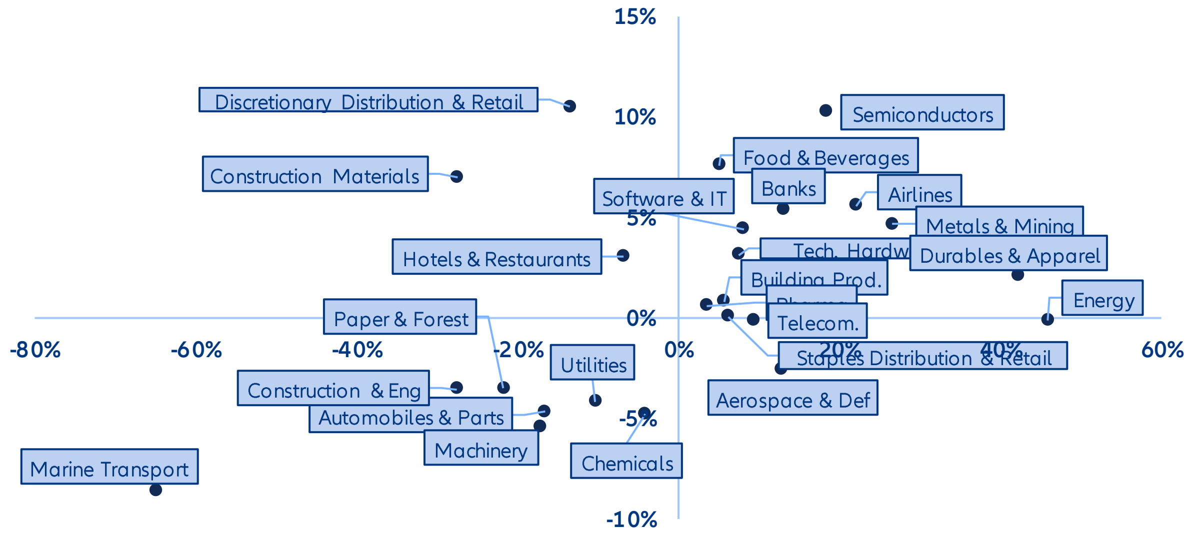
Figure 3: EPS y/y growth rates for companies in the S&P 500 vs STOXX 600 indices




*Sources: LSEG I/B/ES, Allianz Research*

**European Q1 2026 earnings dynamics point to a market that is resilient on the surface but highly differentiated at the sector level.** The entire universe of European corporates have reported aggregated EPS growth of +10.3% y/y, achieved despite an exceptionally subdued top-line environment, with revenues expanding only +0.3%, underscoring the extent to which European corporates have leaned on pricing discipline, cost-out programs and operating leverage to preserve margins in a still-fragile macro and sentiment backdrop. Sector dispersion has been a key feature: Energy stands out as the primary earnings engine, benefitting from the sustained geopolitical tensions and the elevated crude dynamics, while marine transport delivered outsized profitability amid high bunker oil prices and limited pricing power. At the same time, several traditionally cyclical, industrial-heavy segments – including chemicals, machinery, paper & forest products and automobiles (Figure 4) – emerged as notable underperformers, reflecting a combination of structurally elevated fixed-cost bases (particularly energy-linked input costs in chemicals and paper) and higher demand elasticity, which together amplified margin pressure over the period.

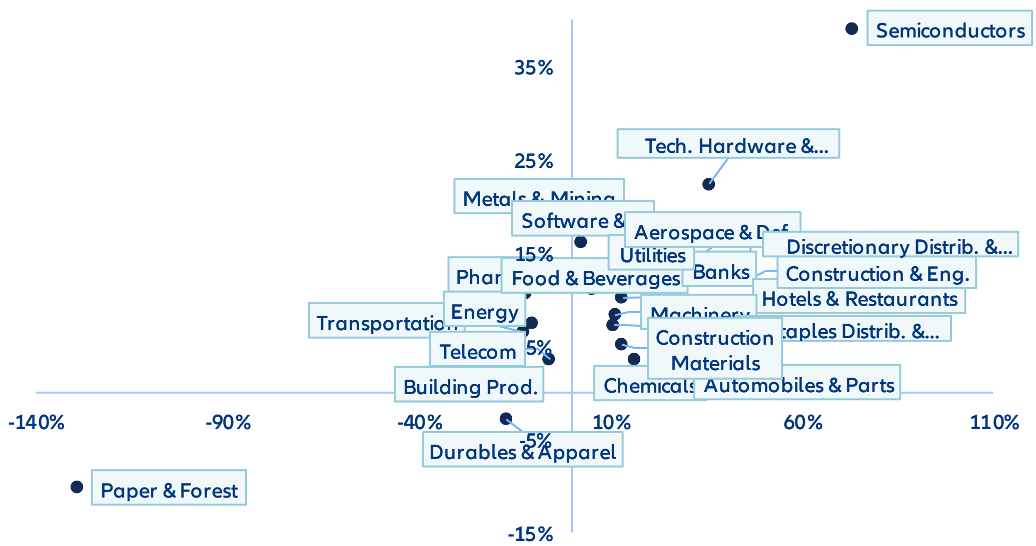
Figure 4: Revenue (Y axis) and EPS (X axis) y/y growth rates for European corporates in Q1 2026, by sector




*Sources: Refinitiv Datastream (as of 27 May 2026), Allianz Research*

**Unlike the US market where earnings expansion remains heavily concentrated in a narrow set of tech-driven growth engines, Europe's earnings resilience appears materially broader across the real economy.** This reduces concentration risk at the index level, but simultaneously leaves the region more exposed to cyclical and commodity-sensitive dynamics rather than to a single dominant secular growth theme. The divergence is particularly evident in the composition of earnings leadership. In Europe, upgrade momentum has been driven predominantly by energy and other macro-sensitive sectors, reflecting the region's higher operating leverage to commodity prices and geopolitical developments. By comparison, the US earnings profile remains structurally more durable and less dependent on commodity-linked upside, with growth continuing to be anchored by technology-intensive industries. As shown in Figure 5, semiconductors (+39.1% y/y) and technology hardware (+22.4%) recorded the strongest revenue growth rates of any US sector in Q1, reinforcing the extent to which AI-related capital expenditure and digital infrastructure investment continue to dominate the US corporate earnings cycle. Paper & forest products stood out as the clearest earnings casualty of the quarter in the US, caught between deteriorating pricing dynamics, structurally elevated energy and freight costs, and weakening end-market demand, a combination that effectively erased operating leverage across the industry.

Figure 5: Revenue (Y axis) and EPS (X axis) y/y growth rates for US corporates in Q1 2026, by sector




*Sources: Refinitiv Datastream (as of 29 May 2026), Allianz Research*

**Non-cyclical composition concentrates rate sensitivity.** The S&P 500 and Euro Stoxx 600 carry a significantly higher weight in non-cyclical sectors than in prior cycles. Non-cyclical sectors can be divided into two groups: bond-proxy profiles (utilities, healthcare, high-dividend energy stocks) and long-duration growth profiles (large-cap Technology). Both are materially sensitive to real interest rates – the former through yield competition and leverage cost, the latter through discount rate effects on terminal value. The index shift away from cyclicals has not reduced rate sensitivity but it has concentrated it in a less visible form. In a rising real rate environment, the "defensive" parts of the index are among the most exposed and the earnings quality argument for holding them dissolves precisely when it is most needed.

**The anticipated listings of OpenAI and Anthropic, alongside the already-discussed SpaceX entry, represent a structurally distinct event from the ordinary IPO cycle.** Together, these three companies carry private-market valuations of USD1.75trn (SpaceX) and USD850bn ( for each OpenAI and Anthropic) – a combined USD3.45trn, roughly equivalent to the entire current weight of the financial sector in the S&P 500. This trend signals continued risk appetite and is expected to deepen index liquidity, attract fresh capital inflows and sustain the momentum dynamic that has supported equity returns in early 2026. Their combined entry into capitalization-weighted indices would be the largest supply event in US equity market history, and their sectoral concentration compounds the risk. All three would land in, or adjacent to, technology – the index segment already identified as carrying the most concentrated rate sensitivity. The simultaneous addition of three megacap AI-and-infrastructure names does not diversify the index; it deepens a crowding that already exists.

**The mechanics are straightforward but their scale is not.** Passive and semi-passive vehicles – which now account for over 55% of US equity AUM – are rules-bound buyers at index inclusion. They have no discretion on timing or price: they must acquire the weight the index assigns, funded by proportional sales of every other existing constituent. For additions of this size, the required reallocation is not a rounding error. It is a structural flow event, one that persists for the adjustment period and leaves the index systematically lighter in every other sector. At current index weights, a combined entry at this scale would represent 4-5% of S&P 500 market capitalization on day one of full inclusion – forcing capital redistribution that would suppress every incumbent constituent's relative performance during the rebalancing window – regardless of fundamentals. The AI premium that has inflated Technology valuations through 2025 and into 2026 would, paradoxically, be the mechanism through which the rest of the index is diluted. Anticipation of these listings partially prices in the demand side, the supply-side redistribution impact on incumbents remains largely absent from current consensus.

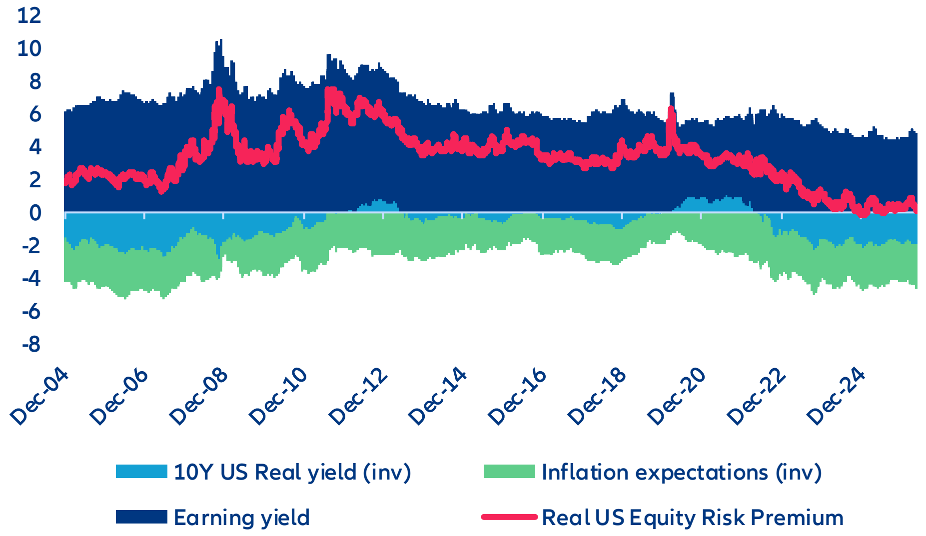
## Valuation and rates

**The real equity risk premium is a breakeven, not a buffer.** The negative nominal equity risk premium is rationalized two ways. Equities are real assets whose cash flows index to inflation, while nominal bonds do not – in an inflationary regime, the inflation-protection argument compresses the ERP markets are willing to accept. The structural growth premium of dominant technology constituents is treated as deserving of a negative nominal premium, by analogy with long-duration bonds where value sits in distant cash flows. On the real ERP measure – earnings yield minus real (inflation-adjusted) risk-free rate – the picture looks more supportive, sitting at approximately 2.5%. This is thin by historical standards but positive, and the market treats it as sufficient justification for current institutional equity weights.

**The inflation-hedge argument is valid, but only while real rates remain contained.** It breaks down precisely when it is most needed: in a regime where inflation is persistent and central banks respond by keeping real rates elevated. In that environment, the inflation-hedge and the rate-headwind operate on the same asset simultaneously, and the rate headwind dominates for non-cyclical, long-duration equity profiles – which is most of the index. The competitive pressure from credit is more immediate than equity multiples suggest. Investment-grade credit at BBB spreads of approximately 120bps, or high-yield at approximately 180bps, delivers real returns comparable to equities, with materially lower volatility, senior claim in the capital structure, and defined maturity.

**The 2.5% real ERP is not a buffer; it is a breakeven.** A rise in real rates of 50-75bps eliminates it. At that point, the reallocation from equities into investment-grade credit is a mechanical, structurally-driven trade – not a sentiment event, and not reversible until real rates fall or earnings growth widens the premium again. The de-rating that follows is structural rather than cyclical, driven by the rate level and not by risk appetite. The multiple stays compressed until the underlying arithmetic turns; in a scenario where geopolitics and growth have also deteriorated, neither condition arrives quickly.

Figure 6: S&P500 earnings yields and risk premium (%)

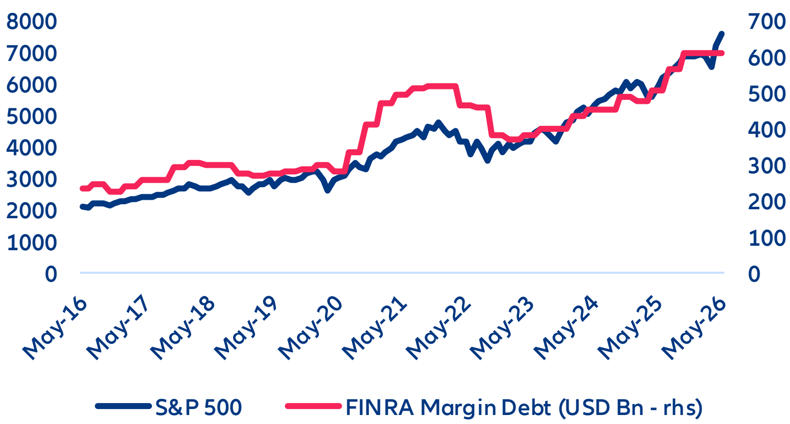



*Sources: LSEG Workspace, Allianz Research*

## The same technical mechanics that amplified the rally can worsen a correction.

**Technical positioning amplifies fundamental moves non-linearly.** The 2025-2026 rally has been supported by structural technical tailwinds as well as by fundamentals. Commodity Trading Advisors (CTA) trend-following models are near maximum long exposure, reflecting the sustained upward trend in major indices. Options dealer positioning – driven by concentrated retail options activity at round-number strikes – reinforces trend directionality. FINRA margin debt (i.e. funds that investors have borrowed from brokers) has recovered from its 2022 lows to levels near cycle highs (see Figure 7). Markets read these conditions as healthy indicators of broad participation and sustainable momentum.

Figure 7: S&P 500 and FINRA margin debt




*Sources: LSEG Workspace, Allianz Research*

**Each of these tailwinds is mechanically reversible, and the reversal dynamic is self-reinforcing rather than gradual.** CTA momentum models do not taper when trend signals reverse, they flip. The transition from maximum long to maximum short can occur within days, concentrating supply at precisely the moment other sellers are also emerging; a relatively modest move can initiate selling that amplifies into a substantially larger one. Margin debt creates a third channel: leveraged holders facing margin calls cannot wait for fundamental improvement, and the supply they generate is price-indifferent. The combined effect is a non-linear response to adverse fundamentals. A 5-7% fundamental-driven decline can cascade into a 15-20% technical drawdown before the positional overhang clears. Recovery requires not just fundamental stabilization but a full clearing of technical positions – CTAs rebuilding long exposure, margin debt being reduced etc. – a process that typically extends over weeks to months, well beyond what fundamental analysis alone would project.

**The mega-IPO pipeline introduces a technical amplifier that operates through a distinct channel that is poorly captured by conventional risk models.** Large index inclusions generate predictable, time-compressed, price-insensitive flows: passive vehicles must rebalance to the new index weights on a defined schedule, selling existing holdings in proportion to their current weight irrespective of valuation or momentum. This is mechanical selling driven by the arithmetic of capitalization weighting. For listings at the scale now under discussion, the rebalancing flow would represent some of the largest single-session institutional selling pressure seen in US markets outside of crisis conditions.

**The interaction with existing technical positioning creates an adverse compounding effect.** CTA momentum models hold existing large-cap technology names near maximum long. In such rebalancing-driven supply event in precisely those names (forced, scheduled, and price-indifferent), the ordinary self-reinforcing dynamics take hold: dealers sell into weakness, CTAs follow trend signals, margin calls cascade. The IPO supply event would not need to be the proximate cause of a broader market dislocation, it only needs to be large enough to move prices through the technical thresholds that convert a controlled rebalancing into a self-sustaining cascade. The lockup expiry cliff – typically 180 days post-listing, at which point insiders and pre-IPO institutional holders become sellers – adds a second, later-dated supply wave that extends the technical overhang well beyond the initial listing date. Markets are not pricing either wave.

## In a downside scenario, all policy levers are simultaneously constrained.

**The policy safety net is thinner than markets assume.** A defining feature of equity behavior since 2010 has been the implicit assumption that policy will intervene when markets deteriorate sufficiently. This has been validated repeatedly: QE programs, fiscal stimulus, emergency rate cuts and geopolitically-motivated pivots have all been deployed in ways that supported equities. The result is that pricing reflects not just fundamental value but policy optionality. In 2026, markets appear to be pricing at least one of three possible interventions as available in any adverse path: a geopolitical deal (the TACO put), fiscal stimulus or central bank rate cuts. The implicit assumption is that whichever lever activates, the response will be sufficient to prevent a full cascade.

**The underpriced risk is not the absence of any single put, it is the correlation of constraints across all three.** The geopolitical channel has been partially activated: the ceasefire negotiations themselves are the response. But a full activation requires concessions exceeding prior pivots (e.g. verifiable nuclear rollback from Iran plus a reparations framework for Iran) and an administration that claims a partial settlement as a complete victory would exhaust this option without resolving the underlying risk. The fiscal channel has structural limits that Europe has already demonstrated. Substantial fiscal commitments to energy security, defense and industrial policy since 2022 provided some support, but they could not offset the structural competitiveness damage from sustained energy-price repricing, not for lack of fiscal support, but because energy-intensive sector damage is not curable by spending alone.

**The central bank channel is the most powerful but the most constrained in this specific adverse scenario.** Monetary easing at scale requires either dysfunction in sovereign bond markets or a banking-sector stress event – high-bar triggers. In an environment of persistent inflation, the direct consequence of sustained oil-driven energy costs, the central bank cannot cut pre-emptively without material credibility cost. The put is available, but its strike is deep out of the money; the cost of waiting for conditions to deteriorate enough to activate it is borne by equity holders in the interim. By the time intervention arrives, the technical amplifier dynamic described in the previous section has already moved prices well past the level early action would have stabilized.

## The four potential regimes going forward

**Our six layers of analysis converge on two primary variables that govern the outcome space:** whether the geopolitical situation resolves in the near term, and whether the real rate environment remains contained. The four combinations of these variables define materially distinct regimes for equity markets (see Figure 8).

Figure 8. Scenario matrix

| | Rates falling / stable | Rates sticky / rising |
|---|---|---|
| **Geopolitical resolution** | **Soft landing confirmed.**<br><br>Supply normalizes, inflation fades, central bank retains easing optionality. Policy support intact. Technical amplifiers operate constructively. Fundamental and valuation support both remain. | **Stagflation lite.**<br><br>Earnings resilient but real rates compress multiples. Non-cyclical rate sensitivity exposed. Index rangebound with elevated sector dispersion. Policy put available. |
| **Conflict persists** | **Geopolitical drag, monetary relief.**<br><br>Growth slows, consumption weakens, but central bank cut cycle is available as inflation pressures ease. Technical overhang clears as monetary pivot anchors. Near-term negative, recoverable over a medium horizon. | **Adverse scenario.**<br><br>All layers deteriorate simultaneously. Earnings compress, multiples de-rate, technical amplifiers cascade, all three policy channels constrained. Structural drawdown without an available circuit breaker. |

*Source: Allianz Research*

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,
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Allianz Trade

# Weed out the laggards: Sustainability as a credit filter

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Allianz Research

# Content

# Executive Summary


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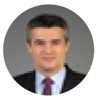

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- **Financial fundamentals remain the primary drivers of credit risk.** Across more than 7,400 companies and 280,000 firm-year observations, profitability (ROA), leverage and size collectively explain the vast majority of variation in credit risk. Sustainability is statistically significant but smaller in magnitude: financial fundamentals carry coefficients 3-7 times larger than the sustainability variable in our global sample. A firm's balance sheet, earnings power and scale determine whether it can service its debt - sustainability adds an incremental but real signal on top of that financial bedrock.
- **Sustainability acts as a credit-risk filter, but its signal concentrates at the bottom of the distribution.** Weaker sustainability is consistently associated with higher default risk. A one-decile improvement in sustainability (roughly +7.5 points on a 0-100 scale) corresponds to an approximately 0.25pp reduction in default probability - a 12-25% relative reduction for firms in the worst default-risk decile. The relationship is non-linear: moving from poor to average sustainability materially improves credit outcomes; moving from average to best-in-class delivers little additional benefit.
- **Environmental performance is the clearest predictor of default risk.** A 10-point improvement in the environmental score is associated with a 0.9-point gain in the Altman Z-score - enough to move borderline firms out of the distress zone. Governance, by contrast, is the key driver of broader credit quality, consistent with financial discipline and transparency underpinning financial strength.
- **European companies lead on sustainability scores and show the greatest sensitivity to ESG factors.** Regional baselines differ sharply: Europe leads (average combined score in the 50s), the US trails (40s, with environmental scores near 30) and Japan scores well on environment (near the high-40s) but weaker on social metrics. Environmental leadership carries the strongest credit premium in emerging markets, where best-in-class issuers benefit from a scarcity premium as standards tighten. In the US, holistic ESG profiles matter more than any single pillar; in Europe, the combined score links to both credit risk and credit quality.
- **The clearest ESG–credit links cluster in sectors exposed to energy transition, operational and reputational risk - notably communication services, consumer staples and energy.** Our sector analysis finds meaningful sustainability–credit relationships in seven of eleven sectors, with the largest effects in these three. Communication/software tends to concentrate financially strong firms that can fund sustainability investment and are rewarded for risk management that encompasses ESG. Consumer staples and energy include carbon-intensive sub-sectors - textiles to oil & gas - where pollution, labor issues, carbon pricing and stranded-asset dynamics can directly hit cash flows, asset values and funding access. The relationship is weaker in materials and IT, reinforcing that sustainability integration in credit requires a sector materiality lens, not a blanket score overlay.

- **These findings are robust across data providers and confirmed by our internal credit assessments.** Replicating the analysis with alternative sustainability scores yields virtually identical results. Using our proprietary internal credit grades, we find that sustainability is already implicitly embedded in analysts' assessments - the sustainability score and environmental and social pillars are all statistically significant predictors of internal grades. Analysts are capturing sustainability-related risks in practice, a more systematic approach could sharpen that signal further.
- **The practical implication is clear: screen out the weakest.** Banks should embed sustainability as a negative screen in credit pricing and due diligence. Credit investors should use it primarily as a downside protection tool. Corporate issuers face a real credit penalty for sitting below the sustainability threshold - and no proportionate reward for exceeding it. The imperative for all market participants is the same: clear the floor. Chasing the ceiling is a marginal bonus.

# Unsustainable practices can generate financial stress

**Sustainability has moved from a niche concern to a mainstream factor in extra-financial evaluation of corporates over the past decade.** The underlying rationale is simple: sustainability issues directly or indirectly influence the fundamentals that eventually determine a company's ability to generate revenues, make a profit or just repay debt. If a company faces environmental fines or a factory is shut down by floods or wildfires, its revenues suffer and costs rise – impeding debt service. If poor labor practices lead to strikes or a consumer boycott, cash flows can dry up. And if weak governance leads to fraud or compliance breaches, a firm can quickly find itself facing legal penalties and loss of investor confidence. All these sustainability-related events hit the company's financial position, translating into traditional credit risk and increasing the probability of default.

**Climate change and environmental management are now recognized as material credit factors.** Physical risks (e.g. extreme weather, droughts, rising sea levels) can destroy assets and disrupt supply chains[1]. For example, in 2019, wildfires in California saddled a major utility with staggering liabilities, contributing to its bankruptcy. Transition risks are also mounting: Governments imposing carbon taxes or banning high-emission activities can strand assets and force costly business model changes[2]. Companies heavily reliant on fossil fuels or with poor pollution controls may face higher costs, asset write-downs and weaker financial ratios, all of which hurt credit quality. On the other hand, firms proactive in cutting emissions or investing in cleaner technology can reduce regulatory risks and operating costs over time, bolstering their credit outlook.

1 See our previous report on adaptation

2 See our previous report on stranded assets

**How a company manages its workforce, customers and communities also bears on creditworthiness.** Neglecting safety and labor rights, for instance, can lead to workplace accidents or strikes that interrupt production. In the mining industry, fatal accidents not only incur direct costs but can prompt license suspensions. Similarly, poor product safety or customer data breaches can trigger lawsuits and erode a firm's reputation and revenues. By contrast, companies that invest in human capital (e.g. training employees, maintaining good labor relations) and uphold product quality tend to have more resilient operations and loyal customers. Although spending on social initiatives may add costs in the short term, that stability should translate into steadier cash flows and lower credit risk in the long run. Lastly, corporate governance has long been linked to credit outcomes. From Enron's collapse to more recent corporate scandals, history shows that failures were rooted in poor governance. Lax oversight, misaligned incentives or opaque disclosure can mask mounting problems until it is too late. When governance breaks down, firms are prone to accounting fraud, corruption or reckless risk-taking that can rapidly undermine their financial position. For creditors, strong governance is a first line of defense: it means competent management, prudent financial controls and transparency. In practice, governance issues can crystalize into credit events through regulatory penalties, adverse audit findings or sudden leadership crises. Any of these can spook investors and restrict a firm's access to funding.

# Sustainability matters for credit quality and credit risk

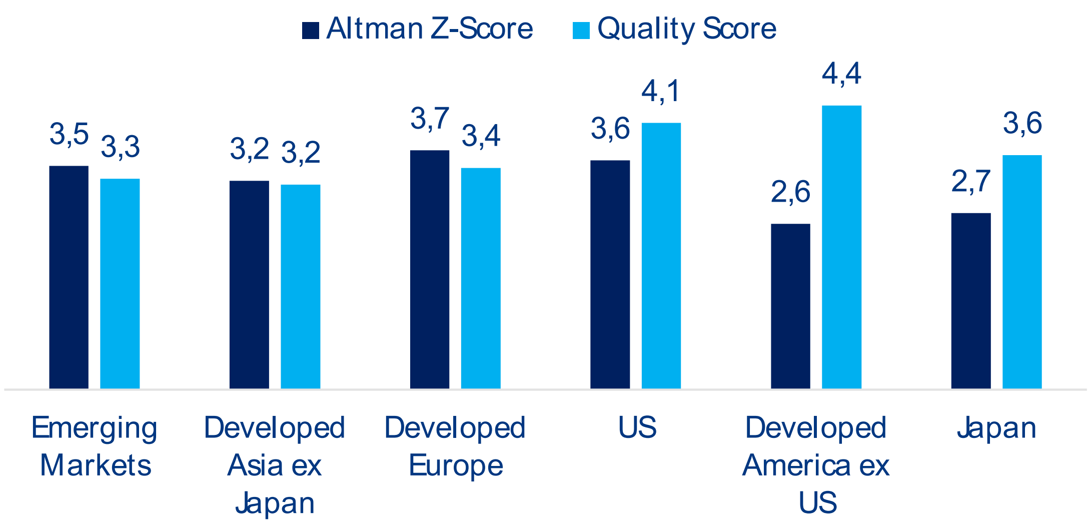
**Over the last ten years, a growing body of research has examined whether strong sustainability actually translates into better credit outcomes.** The evidence, especially from the most recent studies, generally finds that companies with higher ESG ratings tend to have lower credit risk. On the whole, firms that excel on ESG enjoy tangible credit benefits. They often face lower borrowing costs and fewer nasty surprises. For example, multiple studies of corporate bond markets have found that companies with high ESG scores pay narrower credit spreads than peers, even after controlling for their credit ratings. In other words, bond investors demand a smaller risk premium, viewing high-ESG issuers as safer bets. These firms also exhibit lower volatility and reduced tail risk of extreme losses. The upshot is that a strong ESG profile appears to enhance financial resilience, which lenders reward with cheaper debt capital (see appendix for more references on the sustainability-credit risk nexus).

**To further investigate the sustainability-credit risk nexus, we conduct an extensive statistical analysis on a global dataset of companies.** Our study covers firms worldwide over the period 1991-2025, using sustainability scores from Refinitiv (see appendix for more details). This dataset spans both developed and emerging markets, with a broad sector representation. We measured credit risk in two ways: (i) the Altman Z-score, a classic metric of default risk based on financial ratios, and (ii) a credit-quality index capturing broader aspects of financial health (primarily profitability, leverage and cash flow metrics). The Altman Z-score was available for a subset of about 2,100 firms (generally larger companies with complete financial data), while the credit quality index covered a larger sample of 7,400 firms, giving us a comprehensive view of sustainability impacts on creditworthiness. We include firms from North America, Europe, Asia-Pacific and emerging economies. Sustainability scoring coverage has expanded significantly in the past decade; still, not all companies have sustainability ratings or all the inputs for the Altman Z-score. For example, among emerging market companies in our sample, only about 44% have sustainability scores available (as coverage tends to be thinner for smaller EM firms). By contrast, in the US sample almost 67% of firms have sustainability scores, reflecting better disclosure and data coverage in the US. Developed Europe falls in between (around 65% sustainability coverage). These proportions highlight that while sustainability data is now widespread, it is not yet universal. Nevertheless, our total sample comprises over 280,000 firm-year observations, providing a rich basis to detect patterns.

**European companies boast the highest sustainability scores (the average combined score in Europe is about 50 out of 100), compared to the low 40s in emerging markets.** US companies trail Europe: the average US sustainability score is around 40, with particularly low marks on environmental factors (the average US environmental pillar score is barely 30, reflecting that many American firms score poorly on carbon emissions and resource use relative to global peers). Japan, interestingly, scores well on environmental issues (average E score close to 48) but somewhat lower on social metrics. These regional differences suggest that European businesses have led on sustainability, especially on social and environmental dimensions, while North American and some Asia-Pacific firms lag in environmental performance. Does strong sustainability in a region correspond with better credit profiles? The Altman Z-score in our sample averages around 3-4 for most regions (see Figure 1), with higher values indicating safer balance sheets[3]. Developed Europe had a slightly higher mean Z-score (3.7) than the US (3.6) or emerging markets (3.5). These averages put the median company safely out of distress, though the range is wide. We also construct a credit quality index for each firm-year to capture overall financial health. This composite index synthesizes profitability (return on assets) and cash-flow generation (operating cash flow to assets), interest coverage and leverage which are key inputs to credit quality. By design, higher values of this index indicate stronger credit fundamentals (high profits, robust cash flows, less debt). Not surprisingly, developed market firms tend to score better on this quality index on average, given their generally more mature, stable financials.

**We find that a strong sustainability score is associated with lower credit risk.** We employ panel regression analysis to quantify the relationship between sustainability performance and our credit risk measures[4]. In simple terms, when a company's sustainability score is higher, do we subsequently observe better credit outcomes? To answer this, we ran regressions of Altman Z-scores and the credit quality index on sustainability scores, controlling for other factors. Specifically, we include controls for firm size (larger firms often have more diversified risks) and profitability (a higher return on assets naturally boosts credit metrics). We also account for fixed effects (essentially controlling for broad differences across sectors, years and regions) to isolate the impact of sustainability. Importantly, sustainability scores were lagged by one year in the models, meaning we related last year's sustainability performance to this year's credit risk indicators. This helps address causality (i.e. ensuring we are capturing sustainability influencing credit, not the other way around, and also to be consistent with the timing of availability of information, as today we know the sustainability score based on the past year's data). First of all, we need to underline that coefficients for fundamentals are significant and larger across all our regressions compared to sustainability measures. This means that financial metrics are the leading drivers of credit risk. However, we find that sustainability also matters to a significant extent: The coefficient on the Environmental Pillar Score implies that if a firm's environmental score rises by 10 points (say from 50 to 60), its Altman Z-score is expected to be about 0.9 higher on average (see Table 1). Such an effect is meaningful: given many firms' Z-scores hover in the 1.5-3.0 range, a nearly 1 point boost could move a firm from the borderline "grey zone" closer to safety. The overall ESG combined score (which integrates all pillars and accounts for controversies) had a positive coefficient of about 0.05. This suggests that, taken as a whole, strong sustainability performance is associated with better credit health (higher Altman Z-score), although among the individual components, the environmental factor is doing most of the work in explaining that link.

**Figure 1:** Distribution of credit risk and quality scores by regions




Sources: Refinitiv, Allianz Research

3 For context, a Z-score above 2.6 is considered "safe", while below 1.1 is considered "distressed"

4 See appendix for methodological details

**Table 1:** Distribution of credit risk and quality scores by regions

| Sustainability measure vs credit risk (Altman Z-score) | Beta coefficient of the sustainability measure | Beta coefficient of fundamentals | | |
|---|---|---|---|---|
| | | Profitability | Size | Cashflow Strength |
| ESG Combined Score | 0.05* | 0.34*** | 0.28*** | 0.14*** |
| E Pillar Score | 0.09*** | 0.27*** | 0.36*** | 0.18*** |
| S Pillar Score | -0.01 | 0.52*** | 0.42*** | 0.24*** |
| G Pillar Score | 0.02 | 0.49*** | 0.38*** | 0.22*** |

*, ** and *** denote respectively 10%,5% and 1% significance level
Sources: Refinitiv, Allianz Research

**Sustainability is also linked to broad credit quality.** When looking at credit quality, like for credit risk, fundamentals are more important. Nevertheless, we also find here that the combined ESG score has a positive and statistically significant coefficient, indicating that firms with higher overall sustainability performance tend to exhibit better credit quality. In practical terms, a 10-point increase in the ESG score corresponds to about 0.16 higher on our credit quality index – a modest but non-trivial improvement in underlying financial strength (see Table 2). Interestingly, when we break it down into pillars, we see that the governance pillar shows a positive effect. This suggests that good governance practices (board structure, minority shareholder protection, transparency, etc.) are contributing to better credit fundamentals, which is intuitively consistent with the idea that well-governed firms have more disciplined financial management. At a global level, our findings imply that sustainability matters for credit risk and credit quality, but the relative importance of E, S, and G differs between the two. For default risk (Altman Z), environmental performance is the standout (i.e. companies with stronger environmental stewardship tend to be safer from a credit risk perspective). This resonates with the notion that environmental risks (climate, pollution, regulation) have become financially material and can drive default outcomes if unmanaged. In contrast, social and governance factors did not show up in the default risk regression, possibly because their effects on default are indirect or longer-term (or already reflected in other control variables like profitability). For credit quality, we see that having a robust overall sustainability profile helps, and there is a hint that governance (and to an extent social) is playing a role.

**E matters for EMs while overall sustainability is more important for US and Europe.** Looking at different regions[5], we find that environmental pillar impacts are strongest in emerging markets when it comes to credit quality. In this group, a higher environmental score was associated with a significantly better credit quality index. Emerging market firms that are environmental leaders tend to also have stronger credit fundamentals. One interpretation is that in developing economies, where environmental standards and enforcement are rapidly tightening, firms that proactively manage environmental risks gain a credit edge – perhaps by avoiding fines or attracting sustainability-linked capital. Additionally, emerging market companies with strong sustainability ratings might enjoy a reputational premium that improves investor confidence in their financial stability. Moreover, in emerging markets, the governance pillar had a positive relationship with credit quality, suggesting governance reforms are rewarded in those countries with better perceived creditworthiness. In the US, our models showed that overall sustainability performance had a mild positive relationship with credit quality, but none of the individual E, S, G pillars were statistically significant on their own. This might indicate that US credit investors take a holistic view

5 North America (US and other developed Americas), Europe, Asia-Pacific (developed Asia ex-Japan, and Japan) and Emerging Markets.

– i.e. a company has to be well-rounded on sustainability to get credit recognition, and no single pillar being enough. Europe showed a positive link with the overall sustainability performance for both credit risk and credit quality. Europe's long-standing focus on sustainability could mean investors and lenders have integrated these extra-financial criteria in their analysis.

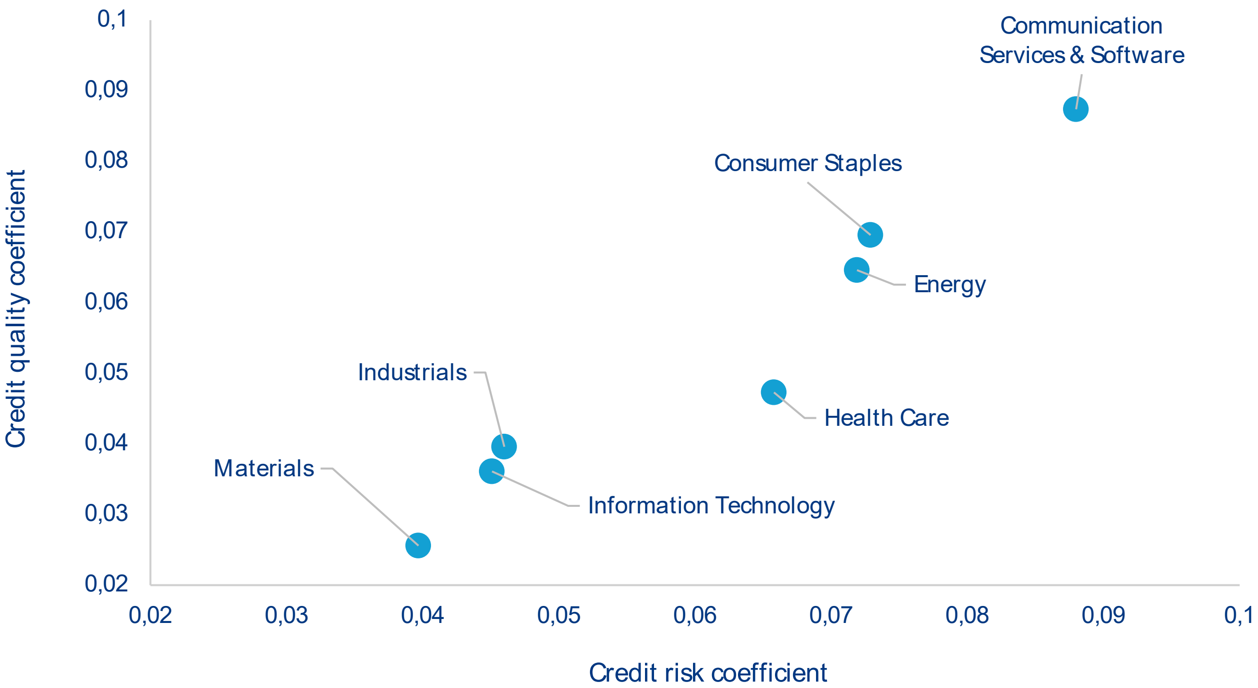
**Sustainability matters most for communication services, consumer staples and energy sector credit.** Overall we find significant links between sustainability and credit risk/quality for seven sectors out of 11, underlining the broad relevance of extra-financial performance for credit worthiness. However, the influence of sustainability is the largest in communication services & software, consumer staples and energy (see Figure 2). For communication services and software, our results most likely stem from the fact that the sector includes financially strong companies that spend and invest in sustainability. Meanwhile, both consumer staples and energy sectors include carbon-intensive segments – from textiles to oil & gas – for which sustainability-related risks represent substantial financial risk, too. For instance, pollution risks and labor related-issues could disrupt firms in textiles while carbon prices and stranded assets are clear hurdles for oil & gas companies. At the other end of the spectrum, although relevant, sustainability scores seem less important for credit metrics for firms in the materials and IT sectors.

**Table 2:** ESG and Credit Quality Index (global panel regression)

| Sustainability measure vs credit quality | Beta coefficient of the sustainability measure | Beta coefficient of fundamentals[i] |
|---|---|---|
| | | Size |
| ESG Combined Score | 0.02*** | 0.14*** |
| E Pillar Score | 0.01 | 0.14*** |
| S Pillar Score | 0.01 | 0.26*** |
| G Pillar Score | 0.01* | 0.19*** |

*, ** and *** denote respectively 10%,5% and 1% significance level
i We do not include profitability and cashflow here as they are used to compute credit quality
Sources: Refinitiv, Allianz Research

**Figure 2:** Sustainability score coefficients for credit risk and credit quality




Sources: Refinitiv & Allianz Research – Note: we only show significant coefficients.

**Our findings hold when using MSCI sustainability data instead of Refinitiv, confirming that the sustainability-credit risk nexus is not an artefact of a single data provider.** A recurring concern in the sustainability-credit literature is that results may be sensitive to the choice of rating provider, given well-documented divergences across sustainability scoring methodologies (Berg, Kölbel & Rigobon, 2022). To address this, we replicate our core Altman Z-score regression using MSCI sustainability scores in place of Refinitiv. The MSCI-based sustainability coefficient is positive and highly significant ($\beta = 0.04$, $p < 0.01$ in Table 3), closely mirroring the Refinitiv result ($\beta = 0.05$ in Table 1). All control variables – profitability, size, leverage, cash-flow strength and interest coverage – retain the expected signs and significance. This consistency across two independent data providers strengthens our confidence that the underlying relationship between sustainability performance and credit risk is genuine and not driven by idiosyncratic scoring choices.

**Using our proprietary internal credit grades, we confirm that sustainability carries credit-relevant information beyond what traditional financial analysis captures.** Beyond third-party credit-risk proxies such as the Altman Z-score, we test whether the sustainability-credit link holds against our own internal credit grading system, which runs from 1 (strongest) to 10 (weakest) and incorporates qualitative analyst judgment alongside financial metrics. Table 4 reports the results using Refinitiv sustainability scores as explanatory variables. The sustainability score, environmental pillar and social pillar all exhibit negative and highly significant coefficients ($p < 0.01$), meaning that companies with stronger sustainability performance receive better internal credit grades. The governance pillar, however, is not statistically significant – likely because governance considerations are already explicitly embedded in our internal credit assessment process. While the economic magnitudes are modest – a move from the worst to the best sustainability score corresponds to roughly a quarter-notch improvement on the 1-to-10 scale – the statistical significance validates that sustainability captures credit-relevant information incremental to what traditional financial analysis already reflects. This supports using sustainability as a complementary signal – particularly as a negative screen, as we demonstrate below – rather than as a standalone driver of credit differentiation.

**Table 3:** Sustainability and Altman Z-score – MSCI data (global panel regression)

| Sustainability measure vs credit risk (Altman Z-score) | Beta coefficient of the sustainability measure | Beta coefficient of fundamentals | | | |
|---|---|---|---|---|---|
| | | Profitability | Size | Leverage | Cashflow Strength |
| MSCI Sustainability Score | 0.04*** | 1.29*** | 0.10*** | −1.06*** | 0.26*** |

*, ** and *** denote respectively 10%,5% and 1% significance level
Sources: Refinitiv, Allianz Research

**Table 4:** Refinitiv sustainability scores and internal credit grades

| Sustainability measure vs internal credit grade | Beta coefficient | Significance |
|---|---|---|
| Sustainability Score | −0.04 | *** |
| Combined Sustainability Score | −0.03 | *** |
| Environmental Pillar Score | −0.03 | *** |
| Social Pillar Score | −0.04 | *** |
| Governance Pillar Score | −0.01 | - |

Negative coefficients indicate better (lower) credit grades. All regressions control for firm size.
*, ** and *** denote respectively 10%, 5% and 1% significance level. Sources: Refinitiv, Allianz Research

**When we split firms into low, medium and high sustainability groups, the credit-risk relationship is concentrated in the bottom tercile – suggesting sustainability is most actionable as a filter to exclude the worst-scoring companies.** To test whether the sustainability-credit link is linear or concentrated in specific segments, we estimate a spline regression that allows the slope to differ across sustainability terciles (using MSCI data). The results are striking (Table 5): the coefficient for the low sustainability group is 0.10 and highly significant at the 1% level, implying that among firms with the weakest sustainability profiles, each additional point of improvement is associated with a meaningful increase in the Altman Z-score. By contrast, the coefficients for the medium (0.01) and high (0.02) groups are small and statistically insignificant. In other words, the marginal credit benefit of better sustainability is strongest at the bottom of the distribution and tapers off once firms reach a moderate standard. This non-linearity has a clear practical implication: rather than rewarding top sustainability performers, the data supports using sustainability primarily as a negative screen – flagging or excluding firms in the lowest sustainability tercile, where poor practices are most strongly associated with elevated credit risk.

**Translating our results into probability-of-default terms, a one-decile improvement in sustainability corresponds to a reduction in default probability of approximately 0.25pp.** To make our findings directly actionable for credit practitioners, we link the sustainability-to-Altman Z relationship to probability of default (PD) through a two-step estimation. First, from our MSCI regression, a one-decile improvement in sustainability score (approximately 7.5 points) is associated with a 0.29-point increase in the Altman Z-score. Second, using a logit regression of observed PD on winsorised Altman Z-scores ($\beta = -0.0096$, $p < 0.01$), we compute the implied change in default probability. The resulting chain implies that a one-decile sustainability improvement – approximately 7.5 points on the 0–100 sustainability scale, for example moving from a score of 12 to 20 for firms in the bottom decile – translates to approximately a 0.25pp reduction in PD. To put this in perspective, observed default probabilities in our sample range from near zero to around 5%. For a company in the worst decile of default risk, where PDs typically reach 1–2% or higher, a 0.25pp reduction represents a 12–25% relative improvement in creditworthiness – a meaningful shift that can alter risk classification. The effect compounds across large credit portfolios and is substantially larger for the environmental pillar using Refinitiv data: the stronger E-pillar coefficient ($\beta = 0.09$ per point, from Table 1) implies that a 10-point improvement in environmental performance corresponds to an even greater PD reduction – underscoring that environmental risk management is the single most credit-relevant dimension of sustainability.

**Table 5:** Sustainability tercile effects on Altman Z-score – Spline regression (MSCI data)

| Sustainability group | Beta coefficient | Significance |
|---|---|---|
| Low (bottom tercile) | 0.10 | *** |
| Medium (middle tercile) | 0.01 | - |
| High (top tercile) | 0.02 | - |

Controls: Profitability, Size, Leverage, Cash-flow Strength, Interest Coverage – all significant at 1%.
Sources: MSCI, Allianz Research

# Implications for financial institutions, investors and businesses

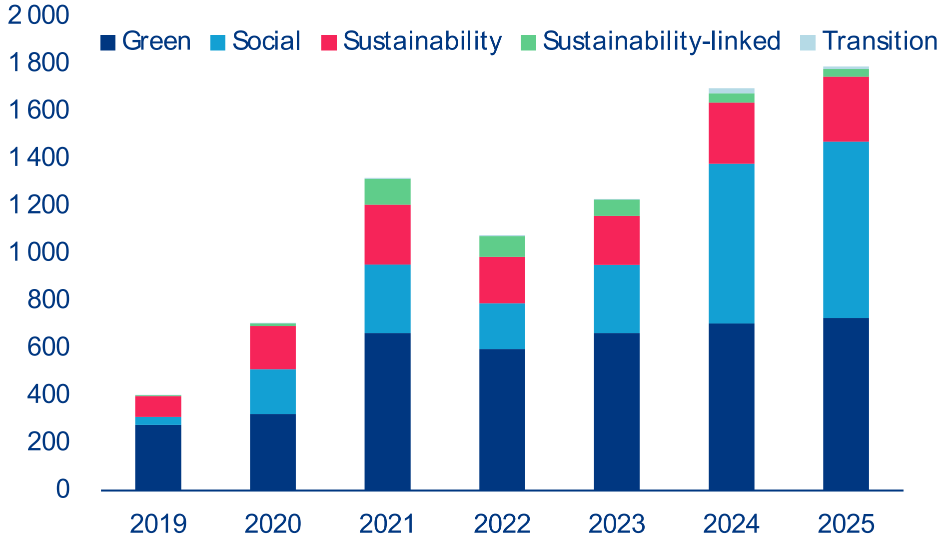
**Commercial banks should embed the floor; pricing the premium is about conviction.** The core finding for commercial banks is specific: Borrowers in the bottom sustainability tercile carry statistically higher default risk, and the 0.25pp PD reduction per sustainability decile is material in credit loss terms at portfolio scale. The practical response is to embed sustainability as a negative screen in credit due diligence: flagging and re-pricing exposure to low-sustainability borrowers, rather than rewarding high sustainability credentials with lower margins across the board. A borrower moving from moderate to strong sustainability does not generate a proportionate reduction in expected credit loss but a borrower in the bottom tercile generates elevated risk that traditional financial analysis may not fully surface. In practice, this integration means incorporating sustainability risk flags into internal ratings and pricing grids, particularly for borrowers in carbon-intensive sectors and those with governance deficiencies.

**Broad-based credit investors should consider ESG filters as downside protection, not necessarily an alpha generator.** For asset managers and credit investors, our findings support sustainability integration primarily as a downside risk mitigation tool. Indeed, the credit signal concentrates in identifying default-prone issuers, not in predicting outperformance by top-rated ones. A portfolio that systematically screens out bottom-tercile sustainability issuers captures the bulk of the credit risk reduction; however, tilting toward the highest-rated sustainability issuers does not generate proportionate improvement in expected risk-adjusted returns. This reframes the analytical objective: the primary use case is loss avoidance, not return enhancement. This framing is consistent with fund-level evidence that sustainability-tilted credit portfolios have exhibited lower spread volatility and drawdowns during stress periods – precisely because the downside-protection effect is real and concentrated at the weakest issuers. Forward-looking sustainability indicators – transition strategies, green capex trajectories, emissions commitments – are increasingly relevant as transition risk begins to crystallize in credit spreads, particularly in the three sectors where our analysis shows the largest effects. Issuer engagement focused on sustainability risk management in communication services, consumer staples and energy represents a more targeted and defensible application of the credit signal than broad portfolio sustainability tilts. Investors who can distinguish genuine sustainability improvement at the bottom of the distribution from cosmetic score improvements elsewhere will have the most informative signal as the market matures.

**Insurers are exposed to sustainability-related credit risk through both underwriting and investment activities and the sector concentration in our findings is directly relevant to where analytical effort should be focused.** The credit signal from sustainability is largest in communication services, consumer staples and energy, and weakest in materials and IT. A uniform sustainability overlay applied to a corporate bond portfolio overstates the signal in low-materiality sectors and risks underweighting it in high-materiality ones – concentrating integration on the three high-signal sectors generates a materially better signal-to-noise ratio. The analytical focus for insurers is also shifting from static ESG scores toward forward-looking indicators to distinguish issuers that are credibly managing transition risk from those accumulating deferred exposure. During phases of spread decompression, when idiosyncratic credit risk dominates and return dispersion rises, sustainability-integrated analysis is most valuable precisely because the signal concentrates at the bottom of the distribution: it helps identify which issuers are deteriorating while others hold, rather than providing a uniform quality tilt. As climate change and shifting social expectations reshape sectoral risk profiles over the medium term, sector-level sustainability views should feed directly into strategic allocation decisions – the large dispersion our analysis shows across sectors in how strongly sustainability affects credit risk is itself a portfolio signal.

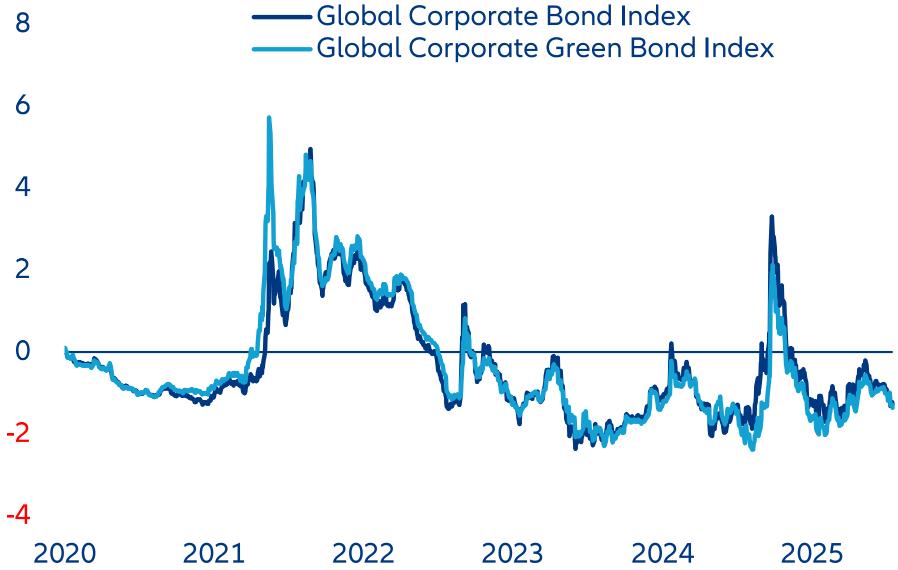
**For corporations, improving sustainability can yield tangible benefits in terms of credit quality and funding costs.** Our findings translate into a specific and financially grounded strategic priority for corporate issuers: the credit and financing benefits of sustainability concentrate at the threshold, and the penalty for falling below it is real. Companies with weak sustainability profiles (particularly weak environmental practices and poor governance) face measurably higher borrowing costs, tighter capital market access and elevated default risk. The strategic implication is to prioritize the most material sustainability risks rather than optimizing for the highest possible combined ESG score. Clear, measurable commitments to addressing material environmental and governance risks, transparent reporting and credible transition strategies are the inputs that matter for credit assessment. Companies that credibly clear the sustainability floor benefit from improved credit terms and a broader investor base; those that do not will face systematic pricing penalties as banks and investors formalize negative screen frameworks. The ESG investment that removes a credit penalty has a higher financial return than the ESG investment that earns marginal recognition at the top of a distribution where the signal is weak.

**Figure 3:** Labelled bond issuance volume (in USD bn)




Sources: Bloomberg. Includes corporate, government, supranational bonds, municipals and securitized products

**Figure 4:** 12M Normalized OAS




Source: Bloomberg

**The expanding market for sustainability-labelled bonds provides investors with targeted exposure to environmental and social objectives,** supported by use-of-proceeds and reporting frameworks that enhance transparency and allow for an assessment of issuers' sustainability trajectories. Issuance of GSSS bonds has risen markedly over the last six years, driven largely by issuers with comparatively strong sustainability profiles (see Figure 3). It remains dominated by green bonds, with activity concentrated in the financials and utilities sectors. While corporate green bonds historically traded at a premium ("greenium") due to scarcity amid strong demand, this premium has largely vanished at the aggregate index level as the market has scaled up, though sector-level dispersion persists (see Figure 4). Empirically, green bonds have exhibited lower spread volatility during market widening phases than the broader corporate bond universe, suggesting potentially more resilient risk-return characteristics, provided issuers' sustainability commitments are credible.

# Appendix: Data and methodology details

**Refinitiv Sustainability Scores – Methodology:**

Refinitiv sustainability scores are constructed using a transparent, rules-based methodology that aggregates approximately 180–200 publicly available environmental, social, and governance indicators into standardized measures of corporate sustainability performance. Raw indicators, derived from company disclosures and publicly reported information, are mapped into defined sustainability categories and themes, normalized on an industry-relative basis, and combined using pre-determined, industry-specific weights to produce Environmental, Social, and Governance pillar scores on a 0–100 scale. In addition to the standalone ESG score, Refinitiv calculates an ESG Combined (ESGC) score, which incorporates an overlay for sustainability-related controversies by adjusting the ESG score based on the severity and frequency of documented incidents. Both ESG and ESGC scores are updated as new disclosures and controversy information become available.

**Data sources and sample construction:** We compiled firm-level data from Refinitiv covering the years 2011 through 2024. The ESG data comes from Refinitiv's ESG database, which provides an ESG Combined Score (a comprehensive rating including environmental, social, governance aspects and deducting for controversies) as well as separate pillar scores for Environmental, Social, and Governance (each scaled 0-100, with higher = better ESG performance). We also collected financial data from Refinitiv for the same firms, including balance sheet and income statement items required to calculate the Altman Z-score and other financial ratios. Our initial universe included over 130,000 companies globally, but many were missing ESG or financial data. We restricted the sample to firm-year observations with non-null values for the variables of interest. The final panel comprises about 280,000 observations on ~7,400 unique firms worldwide. The regional breakdown of firms (by headquarters) was: roughly 18% North America, 20% Developed Europe, 6% Japan, 7% other developed Asia-Pacific, 34% Emerging Markets, and 15% unclassified/unknown region (the latter are cases where region wasn't specified, often smaller frontier market firms). Data coverage varied, with developed markets generally having better ESG disclosure. For instance, 67% of U.S. firm-year observations had an ESG score available, versus ~44% in Emerging Markets, as noted earlier.

**Altman Z-score calculation:** We used the **Altman Z-score** as a proxy for credit risk or default risk of a firm. The Altman Z-score is a well-known composite financial indicator developed by Edward Altman, which combines five accounting ratios using fixed weights. We applied the version for publicly traded firms (Altman's Z-score for manufacturing, since most of our firms fall in non-financial sectors). The formula is:

$$\mathrm{Z} = 1.2 \times \frac{\text{Working Capital}}{\text{Total Assets}} + 1.4 \times \frac{\text{Retained Earnings}}{\text{Total Assets}} + 3.3 \times \frac{\text{EBIT}}{\text{Total Assets}} + 0.6 \times \frac{\text{Market Value of Equity}}{\text{Total Liabilities}} + 1.0 \times \frac{\text{Sales}}{\text{Total Assets}}.$$

This formula produces a single score where higher values indicate a lower probability of bankruptcy. As mentioned, Z > 2.6 is usually considered "safe", Z < 1.1 is "distressed", and the 1.1–2.6 range is a grey zone. We winsorized extremely high or low Z-scores to avoid undue influence of outliers (e.g., a few observations had negative equity or abnormally large market capitalizations leading to very high Z).

In our dataset, not all firms had the market value data needed (especially if not publicly traded or if data was missing), which is why the Altman Z sample is smaller (~2,100 firms). Notably, financial sector firms (banks, insurance) are excluded from Altman Z analysis since the Z-score isn't designed for them and their balance sheets aren't comparable .

**Credit quality composite:** To measure credit quality more broadly (and to include more firms), we constructed a composite index reflecting a firm's earnings and cash flow capacity, which are crucial to creditworthiness. Specifically, we took three indicators: Return on Assets (ROA = Net Income / Total Assets), Leverage (Total Debt to Total Assets) and Operating Cash Flow to Total Assets (OCF/TA), which capture profitability and cash generation respectively. We standardized these within each year (to account for economic cycles) and took an average to form a composite score. This "credit quality index" is not a standard metric like Altman Z, but it serves as a general gauge of a firm's fundamental strength to service debt – high values mean the company is profitable and generating cash relative to its asset base (implying it likely can cover interest and reinvest, hence higher credit quality). We included this second measure because credit risk is multifaceted; a firm might have a low Z-score due to high market leverage but still strong cash flows to pay debts, or vice versa. The composite helps capture dimensions the Z-score might miss and greatly expands coverage to smaller firms that don't have market value data.

**Regression specifications:** We estimated panel regressions of the form:

$$\mathrm{CreditMetric}_{i,t} = \alpha + \beta \times \mathrm{ESGMetric}_{i,t-1} + \gamma \mathbf{X}_{i,t-1} + \mathrm{FE} + \varepsilon_{i,t},$$

where $\mathrm{CreditMetric}_{i,t}$ is either Altman Z or the credit quality index for firm i in year t. $\mathrm{ESGMetric}_{i,t-1}$ is the ESG variable of interest (we ran separate regressions for the Combined ESG Score, ESG Score, Environmental pillar, Social pillar, and Governance pillar) lagged one year. $\mathbf{X}_{i,t-1}$ represents control variables, which included firm size (log of total assets) and profitability (ROA) – both lagged one year as well. "FE" denotes fixed effects; for the Altman Z models, we included sector-by-year fixed effects (SYFE), meaning we account for any shocks or trends affecting a whole sector in a given year (like an oil price spike affecting all oil & gas firms' Z-scores in 2020, for example). In the credit quality models, we included both sector-year and region-year fixed effects (since that sample was much larger, we could control for regional macro conditions as well). These fixed effects absorb broad macroeconomic or industry effects, isolating the impact of ESG at the firm level. We did not include firm fixed effects because our ESG scores don't exhibit enough within-firm time variation for many firms (many have short time series), and we aimed to capture between-firm differences as well – effectively assessing if higher-ESG firms have better credit metrics than lower-ESG firms, controlling for size and ROA.

We used robust standard errors clustered at the firm level to account for heteroskedasticity and serial correlation. The statistical significance was assessed at the 10%, 5%, and 1% levels as customary. The regression tables in the main text (Table 1 and 2) summarize the coefficient estimates for the ESG variables and their significance. The R-squared (within) for the Altman Z regressions were around 0.05 (i.e., 5% of the variation in Z within sectors/years was explained by ESG and controls – not surprising as many firm-specific factors drive Z). For the credit quality regressions, the within R-squared was higher (~0.69) because profitability (ROA) is itself part of the dependent variable and explains a lot of its variation, and fixed effects soak up a lot of variance as well.

**Supplementary tables:** Below we provide a summary comparison of regional results mentioned in the text for the credit quality regressions (coefficients of ESG variables by region), to give a fuller picture:

Table A1. ESG coefficient estimates by region – Credit Quality Index regressions (signs and significance of ESG Combined Score and Pillar effects)

| Beta Coefficients of ESG vs Credit Risk | ESG Combined Score | E Pillar Score | S Pillar Score | G Pillar Score |
|---|---|---|---|---|
| US | positive & significant | positive & significant | positive | positive & significant |
| Developed Europe | positive & significant | positive & significant | positive & significant | positive & significant |
| Japan | positive & significant | positive & significant | positive | positive & significant |
| Developed America ex US | positive & significant | positive & significant | positive | positive |
| Developed Asia ex Japan | positive & significant | negative | negative | negative |
| Emerging Markets | negative | positive | positive | positive |

Sources: Refinitiv & Allianz Research

# References:

| Date | Publication Source | Authors | Title | Main Findings |
|---|---|---|---|---|
| 2011 | Journal of Banking & Finance | Goss & Roberts | The impact of corporate social responsibility on the cost of bank loans | CSR concerns linked to higher loan spreads and tighter terms; lenders price ESG-related risk, especially for weaker borrowers. |
| 2013 | Journal of Business Ethics | Attig et al. | Corporate Social Responsibility and Credit Ratings | Higher CSR performance associated with better credit ratings, robust to endogeneity concerns. |
| 2014 | The Financial Review | Oikonomou et al. | The Effects of Corporate Social Performance on the Cost of Corporate Debt and Credit Ratings | Strong social performance linked to lower bond yield spreads and higher assessed credit quality. |
| 2017 | Principles for Responsible Investment (PRI) | PRI | Why ESG factors matter in credit risk analysis | ESG affects default risk, downgrade risk, and recovery values rather than just bond pricing. |
| 2020 | Banque de France / ACPR | Billio et al. | Inside the ESG ratings: (Dis)agreement and performance | Large ESG rating disagreement implies ESG–credit results depend strongly on provider choice. |
| 2022 | Review of Finance | Berg, Kölbel & Rigobon | Aggregate Confusion: The Divergence of ESG Ratings | Low correlation across ESG ratings; measurement noise complicates credit-risk inference. |
| 2022 | Journal of Risk Finance | Barth, Hübel & Scholz | ESG and corporate credit spreads | Higher ESG scores associated with lower CDS spreads; ESG shocks are priced in credit markets. |
| 2022 | NY Fed Staff Report | Seltzer, Starks & Zhu | Climate Regulatory Risk and Corporate Bonds | Climate regulatory risk causally worsens credit ratings and increases yield spreads. |
| 2022 | World Bank | Gratcheva et al. | Credit Worthy: ESG Factors and Sovereign Credit Ratings | Clarifies how ESG factors relate to sovereign creditworthiness; warns against mechanical ESG–rating mapping. |
| 2023 | International Review of Economics and Finance | Lian et al. | How does ESG performance affect bond credit spreads? | Better ESG linked to lower bond spreads via reduced financial risk and better transparency. |
| 2025 | International Review of Economics and Finance | Chen et al. | Impact of ESG factors on credit ratings | Environmental factors positively associated with European banks' credit ratings. |

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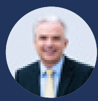

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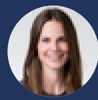

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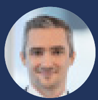

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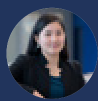

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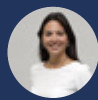

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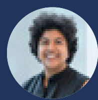

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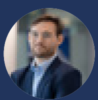

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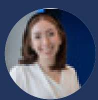

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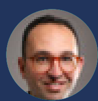

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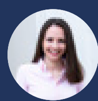

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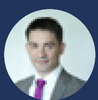

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Allianz Research encompasses Allianz Group Economic Research and the Economic Research department of Allianz Trade.

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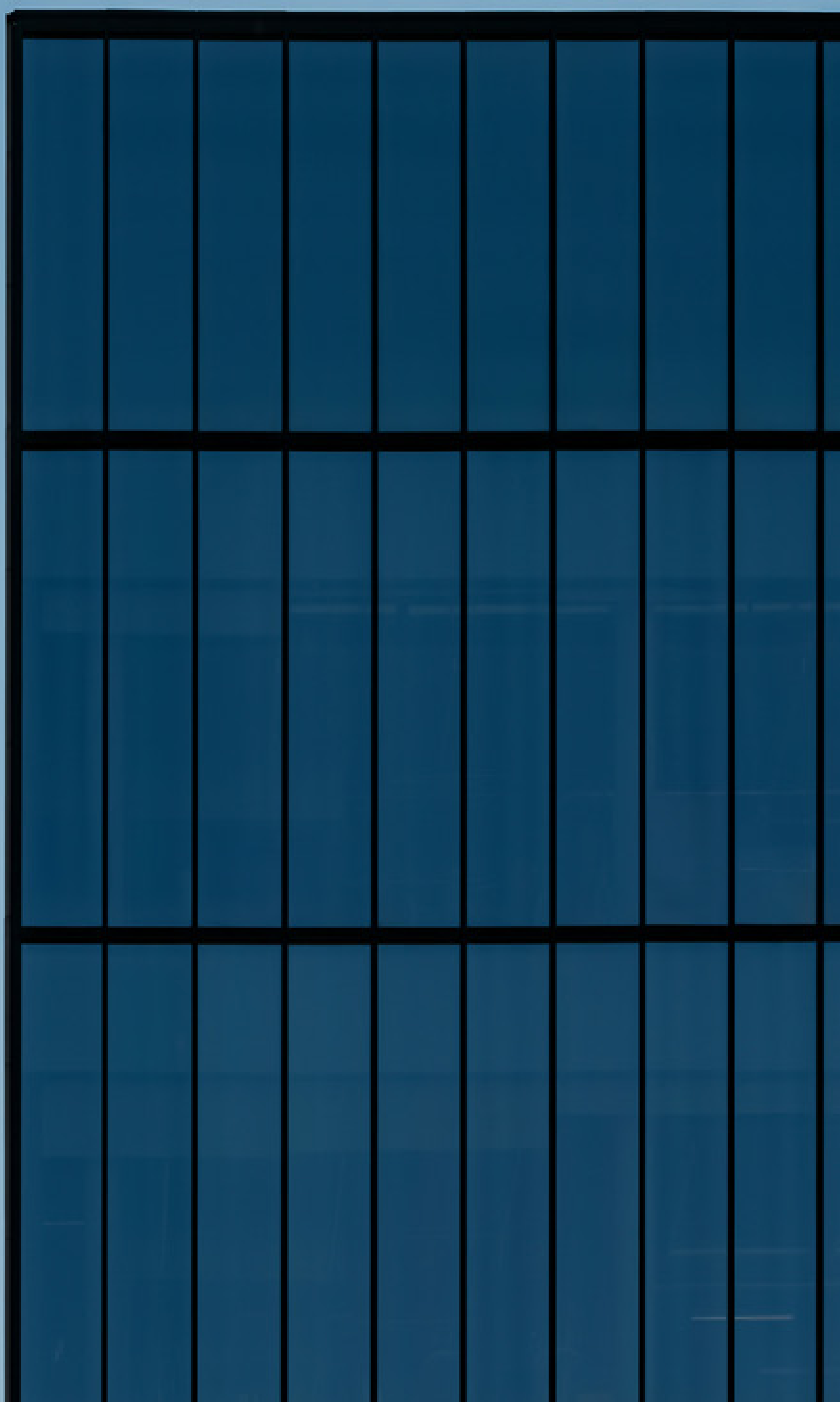
# Allianz Global Insurance Report 2026: The future of insurance in a fragmenting world

28 May 2026



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# Content

28 May 2026

# Executive Summary


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- **The insurance industry is cooling from exceptional growth, not slowing into weakness.** Global premiums rose by +7.1% to EUR6.9trn in 2025, adding EUR456bn to the global premium pool. Although growth moderated from the exceptional +9.4% recorded in 2024, it remained comfortably above the ten-year CAGR of +5.6%, confirming that the industry's growth drivers remain firmly intact. Life insurance remained the largest segment (EUR2,861bn; +EUR185bn in 2025), followed by P&C (EUR2,320bn; +EUR85bn) and health (EUR1,688bn; +EUR185bn).

- **Asia remains the structural center of growth even as advanced markets outperform cyclically**. Western Europe grew nearly twice as fast as its 2015–2025 average, while emerging market growth softened as China slowed to +7.4%, more than 3pps below its long-term average. Excluding China and Japan, however, Asia remained the industry's fastest-growing major region, with premium growth accelerating to +10.9% versus a long-term average of +4.9%.

- **Much of the industry's recent expansion reflects higher prices rather than materially broader coverage.** Global insurance penetration rose only modestly to 7.2% of GDP in 2025 and remains below levels seen a decade ago. The disconnect is most visible in P&C, where penetration has stagnated around 2.5% of GDP despite strong premium growth. Trends across business lines are diverging: health-insurance penetration rose from 1.4% of GDP in 2015 to 1.8% in 2025, while life insurance at 3.0% remains well below the levels above 4% seen two decades ago.

- T**he P&C boom is normalizing while life and health remain structurally strong.** Global P&C growth slowed to +3.8% in 2025 as pricing cycles matured and claims inflation stabilized. North America remained dominant, accounting for 52% of global P&C premiums, while Asia continued to exhibit the world's largest protection gap, with penetration of just 1.3% compared with 4.3% in North America. Life insurance grew by a still robust +6.9%, though the US annuity boom is fading. Health insurance remained the industry's strongest growth story, expanding by +12.3%, the fastest pace since 2014, driven by rising medical costs and growing demand for private healthcare protection.

- **Despite Asia's rise, global insurance remains overwhelmingly dominated by the US.** North America increased its share of global premiums from 42.5% to 46.4% over the past decade, meaning that almost every second euro written globally now originates from the region. China has emerged as a clear number two, increasing its market share from 7.5% to 10.9%. Yet, at EUR746bn in premiums, it remains less than a quarter of the size of the North America market at EUR3,191bn. Western Europe, by contrast, will continue to lose relative weight across most business lines.

- **Geopolitics and fragmentation are becoming central forces shaping the insurance industry.** The Iran war is acting as a major external supply shock, disrupting energy markets, trade flows and supply chains. In our central scenario, global GDP growth is expected to slow to +2.6% in 2026, while Eurozone growth will fall to just +0.8%. If the conflict is not resolved during the summer, we expect additional upward pressure on inflation and a materially worse global growth outlook. More broadly, geopolitical fragmentation is creating a structurally more complex operating environment, challenging assumptions around global integration, capital mobility and cross-border risk diversification. Insurers will need to adapt by building more regionally resilient operating models, integrating geopolitical analysis more directly into underwriting and capital allocation and developing products tailored to emerging risks such as cyber escalation. While fragmentation raises costs and operational complexity, it also increases demand for protection and resilience, reinforcing the strategic relevance of insurance in a more uncertain global environment.

- **Protection remains a structurally growing need in an increasingly uncertain and ageing world.** The global insurance market is expected to expand by +5.3% annually over the next decade, driven by rising demand for protection, demographic change, higher interest rates and growing health and retirement needs. Health insurance is projected to remain the fastest-growing segment at +6.7% p.a., followed by life insurance at +4.9% and P&C at +4.7%. Asia, particularly China and India, will remain the main growth engine, adding 5pps to their global market share, while North America is expected to retain its dominant global market position at roughly 46%. Europe, by contrast, is likely to continue ceding global market share over the coming decade (-4pps).

- **Asia will generate most of the industry's future growth, with India emerging as the standout opportunity.** The global premium pool is expected to expand by EUR5,260bn by 2036, with more than half of additional premiums originating in Asia. China will remain the region's dominant market, but India is expected to emerge as the fastest-growing major insurance market. Despite already ranking among the world's ten largest, India remains significantly underinsured, with penetration of just 3.8% of GDP and per capita spending of roughly EUR85. Rising incomes, demographic change and recent reforms are expected to drive annual premium growth of around +10.7% over the next decade.

- **Climate change is turning insurance affordability into a structural challenge for the industry.** Insured NatCat losses are rising by +5–7% annually in real terms, while higher premiums are increasingly colliding with weakening household purchasing power. The result is widening protection gaps, pushing lower-income households out of coverage. As affordability deteriorates, more climate losses are shifting onto already strained public balance sheets. Preserving long-term insurability will require moving beyond repricing toward stronger adaptation incentives, resilience-focused underwriting, targeted affordability mechanisms and deeper public-private risk-sharing frameworks, supported by large-scale investment in climate adaptation and resilient infrastructure.

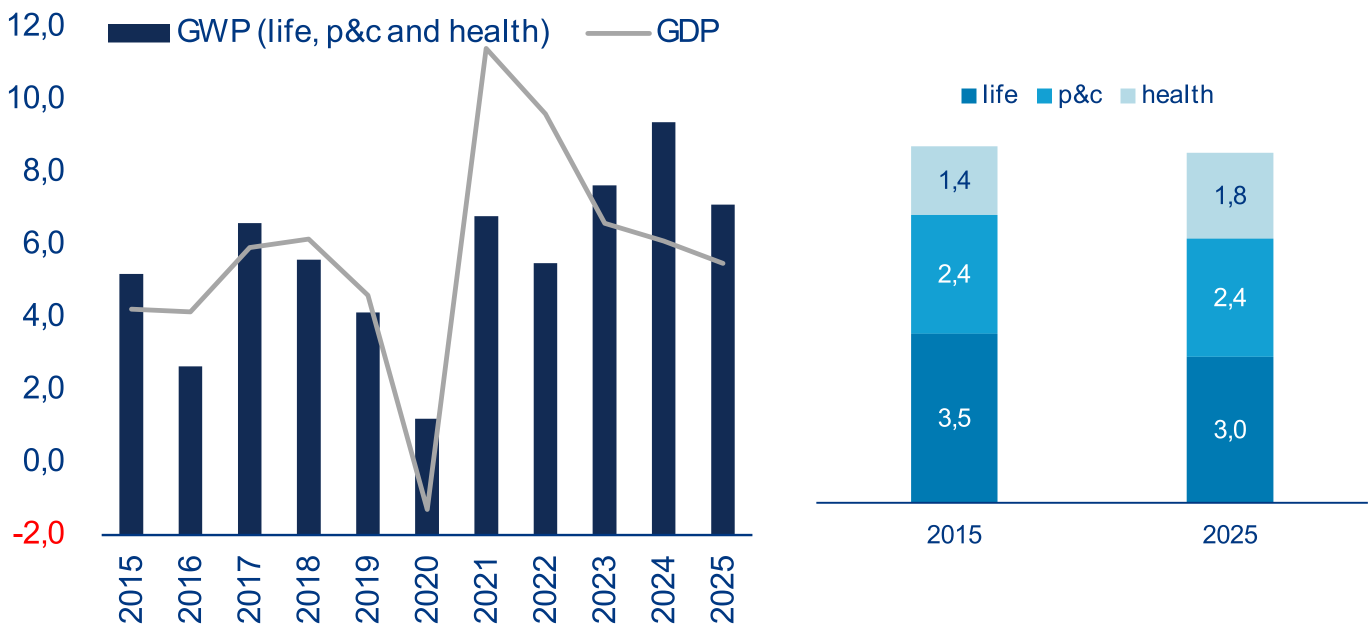
# Looking back: From boom to normalization

**The global insurance industry returned to a more sustainable growth trajectory in 2025, with total gross written premiums (GWP) rising by +7.1% to EUR6.9trn.** The expansion added EUR456bn to the global premium pool over the course of 2025. The life segment remained the largest (EUR2,861bn; +EUR185bn in 2025), followed by P&C (EUR2,320bn; +EUR85bn) and health (EUR1,688bn; +EUR185bn). However, growth decelerated from the exceptional +9.4% recorded in 2024, returning the industry toward a more sustainable trajectory after two years of super-charged growth. Crucially, even at this more moderate pace, the 2025 rate remained comfortably above the ten-year CAGR of +5.6%, confirming that the structural growth drivers that have fueled the global insurance market expansion since the mid-2010s remain in place. Over the full decade from 2015 to 2025, total GWP expanded from EUR3.97trn to EUR6.87trn – a cumulative addition of more than EUR2.9trn, driven by rising insurance penetration, expanding middle classes in emerging economies and the continued broadening of coverage across all three major segments.

**Much of the industry's recent premium growth reflects higher prices rather than a comparable deepening of insurance protection.** With consumer price inflation running at approximately 2–3% across major markets in 2025, part of the headline growth merely reflects inflationary pass-through rather than additional coverage written. Insurance penetration, measured as premiums relative to GDP, increased only modestly to 7.2% in 2025 from 7.0% in 2024 and remains below the 7.3% recorded a decade earlier. This divergence between nominal premium growth and real insurance deepening has become a defining feature of the post-2022 insurance cycle. While the industry has expanded substantially in monetary terms, the share of economic activity devoted to insurance has yet to recover to its long-run level. The disconnect is particularly visible in P&C, where higher claims costs have driven pricing increases without materially expanding insurance penetration.

**The trend in global insurance penetration is increasingly uneven across business lines, with P&C broadly stable, health steadily gaining relevance and life insurance still recovering from the ultra-low-rate era.** P&C penetration has remained broadly stable at around 2.5% of GDP, while health insurance has steadily gained importance, rising from 1.4% of GDP in 2015 to 1.8% in 2025 as demand for private healthcare protection increased. Life insurance, by contrast, continues to recover only gradually from the prolonged period of ultra-low interest rates. Although penetration has edged back up since 2022 to 3.0% of GDP in 2025, it is still below the 3.5% a decade earlier as the segment remains well below historical norms (Figure 1). The longer-term decline is even more striking: two decades ago, global life insurance penetration exceeded 4% of GDP.

**Figure 1:** Premium growth remains ahead of GDP
Global gross written premiums* and nominal GDP growth* (y/y, in %) and global gross written premiums* as % of GDP by segments

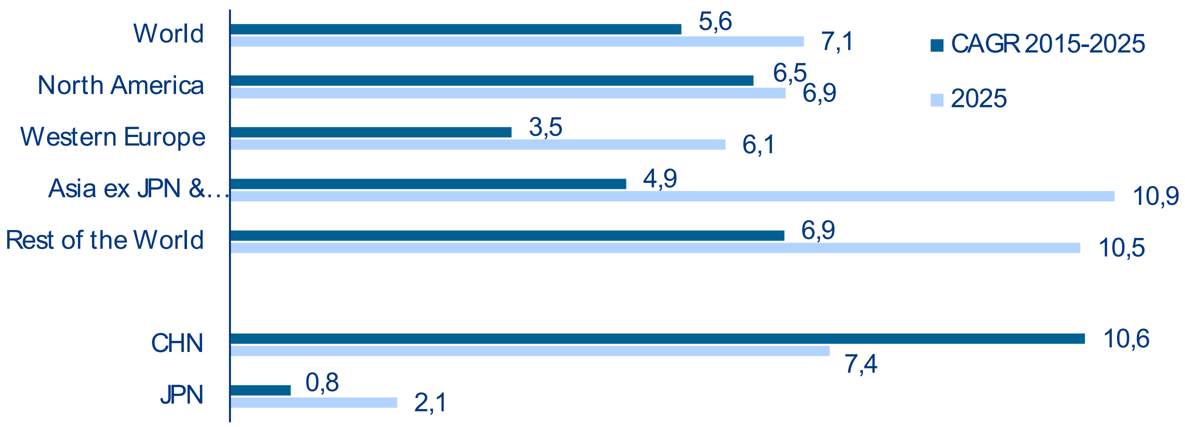



*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

**Growth remains strong globally, but the center of momentum is shifting as advanced markets outperform cyclically while Asia retains its structural growth lead.** While all major regions grew faster than their respective long-term averages in 2025, advanced markets were the main drivers of this acceleration, with Western Europe (+6.1%) expanding nearly twice as fast as its 2015-2025 average. Emerging market growth, by contrast, softened relative to the past decade, largely reflecting slower expansion in China, which accounts for almost 60% of the emerging market aggregate. Chinese premium growth slowed to +7.4% in 2025, more than 3pps below its long-term average. While this is to a large extent a function of past excellence rather than a weak current performance, nominal growth has also been held back by annual inflation registering just above 0%. Meanwhile Asia[1] (when excluding China and Japan) continues to significantly outperform Europe and North America, with premium growth reaching +10.9% in 2025 compared with a long-term average of +4.9%, reinforcing its position as the industry's key structural growth market.

[1] Throughout the report, Asia always refers to the region excluding China and Japan. If both countries are included, we use the term wider Asia.

**Figure 2:** 2025 growth outpaces long-term average
Gross written premium growth*, 2025 and CAGR 2015-2025 by region in %




*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

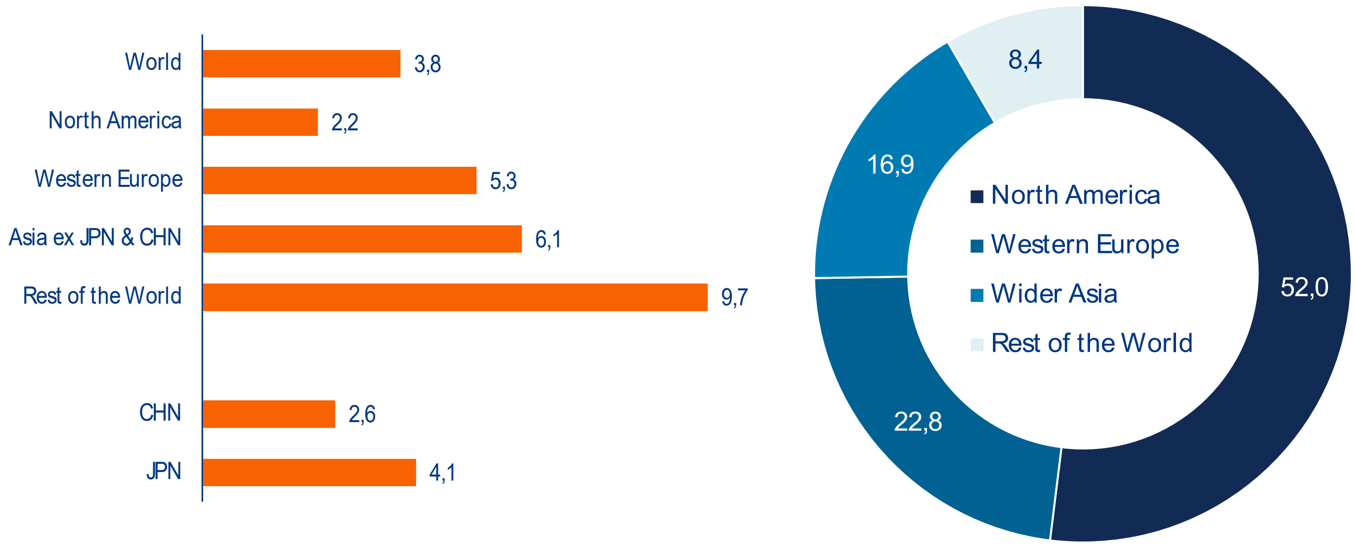
**The global P&C market is entering a phase of normalization after several years of exceptional pricing-driven growth.** The segment grew by +3.8% in 2025 to reach EUR2,320bn globally, a notably softer performance than both the overall market and the segment's own ten-year CAGR of +5.6%, as well as last year's growth of +8.5%. The moderation reflects a maturing pricing cycle. After several years of sharp rate increases across motor, property and specialty lines, pricing has started to normalize in some regions, while claims costs, although still elevated, are no longer accelerating at the pace seen in 2022 and 2023. North America[2] remains the undisputed center of global P&C activity, accounting for 52.0% of world P&C premiums, with EUR1,205bn written in 2025. However, regional growth slowed to +2.2% from +9.7% in the previous year, reflecting both a more mature rate cycle and a -6.3% decline in US accident & health premiums[3] that weighed on overall market growth (US: +2.2%); excluding A&H, however, the US P&C market is estimated to have grown by 4.1% [4](Figure 3).

**Western Europe's P&C market posted solid growth in 2025, although the exceptional post-inflation growth surge is now beginning to normalize.** Premiums grew by +5.3% to EUR529bn, a solid expansion but notably below the exceptionally strong growth rates of +8.5% in 2023 and +6.8% in 2024 as inflation pressures continued to ease. Performance across the region was increasingly uneven. Southern European markets remained particularly dynamic, with several recording high single-digit growth above +8%, while Scandinavia, the UK, Belgium and Switzerland expanded more modestly at around +1% to +4%. Among the largest Eurozone markets, growth remained comparatively robust, led by Germany at +7.7%, France at +7.0%, Spain at +6.5% and Italy at +6.4%. Wider Asia P&C premiums grew by +4.1% to EUR391bn in 2025, broadly in line with the previous year. Yet despite a population more than six times larger than Western Europe's, the region's P&C market remains significantly smaller. The gap is equally visible in insurance penetration and spending levels. P&C penetration stands at just 1.3% in wider Asia and 1.1% in China, compared with 2.5% in Western Europe and 4.3% in North America. Per capita P&C spending reached only EUR98 across Asia (China: EUR135), compared with EUR1,232 in Western Europe and EUR3,111 in North America. These disparities underline the scale of structural underinsurance and the region's long-term catch up potential.

**China is by far the largest market in the wider region, accounting for almost every second euro written.** At +2.6% China registered one of the weakest performances in recent years outside the Covid-19 period and well below the country's average growth rate of +6.6% over the past decade. The slowdown largely reflects China's relatively muted nominal growth dynamics. All other insurance markets (rest of the world) recorded growth of +9.7% in the P&C segment, driven by strong (largely inflationary) increases of above +10% in Eastern Europe and Latin America.

[2] 96% of P&C premiums in North America are written in the US; the rest is attributable to Canada.
[3] In this report, Accident & Health (A&H) premiums written by Life/A&H entities are included in the US P&C market figures, consistent with our general approach of classifying accident lines as part of the P&C market.
[4] This estimate is based on growth in net premiums written as direct written premium figures had not yet been published at the time of writing.

**Figure 3:** Europe and Wider Asia power ahead
P&C gross written premium growth* and market share by region, 2025 in %



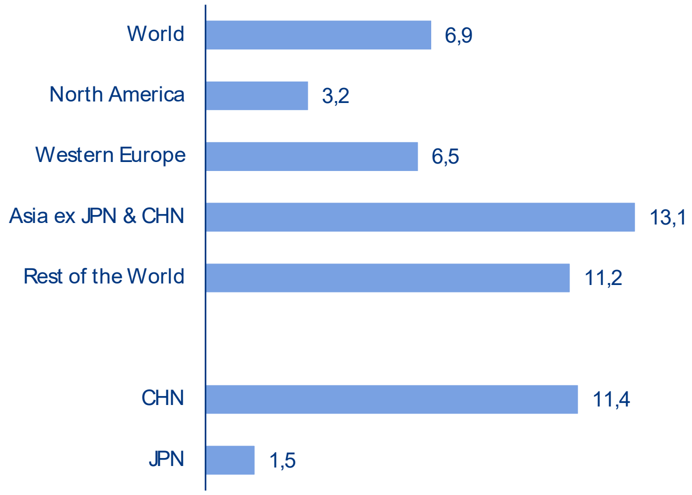

*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

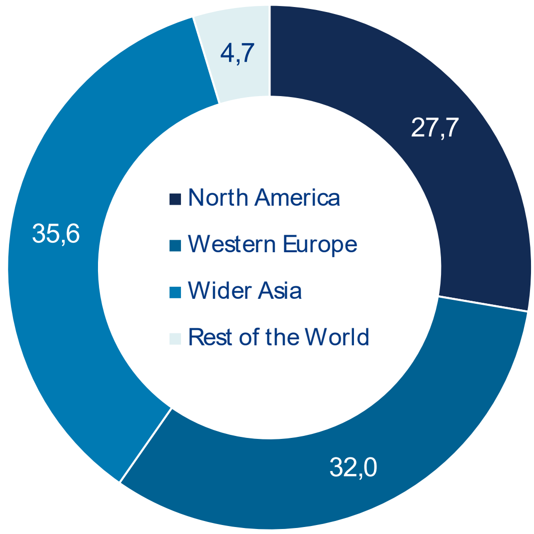
**The life insurance boom is cooling in North America, while Asia is reasserting itself as the industry's main growth engine.** Global life insurance premiums grew by +6.9% in 2025, a clear moderation from the exceptionally strong +11.3% recorded in 2024 but still a robust performance by historical standards. The slowdown was driven primarily by North America, where the annuity boom triggered by households locking in higher interest rates has begun to lose momentum. Growth leadership has now shifted back toward Asia (Figure 4).

**Higher interest rates continue to support life insurance globally, but regional momentum is increasingly diverging.** In Western Europe, life premium growth moderated to +6.5% in 2025 from +9.9% in the previous year, although higher rates continued to support savings products. Performance remained uneven across markets. France, the Netherlands, Finland and Portugal posted another year of double-digit growth, while Switzerland continued to lag. Germany returned to positive territory with growth of +5.3%. Wider Asia recorded healthy +9.9% growth in 2025, with the majority of markets posting strong double-digit gains.

**As a result, Asia as a whole remains the largest life insurance market, well ahead of North America and Europe (Figure 4),** reflecting both demographic pressures and weaker public pension systems. This is also reflected in insurance penetration, which stands at 3.3% in Asia, driven by the popularity of life products in advanced Asian markets such as South Korea, Taiwan and Singapore. But China at 2.5% and India at 2.8% already approach North American levels of 2.8%, although still below Western Europe's 4.3%. These relatively high Asian penetration rates reflect greater reliance on private retirement savings in the absence of comprehensive public pension systems. By contrast, North America's comparatively low penetration partly reflects the widespread use of occupational retirement schemes such as 401(k) plans.

**Figure 4:** Asia pulls further ahead
Life gross written premium growth* and market share by region, 2025 in %



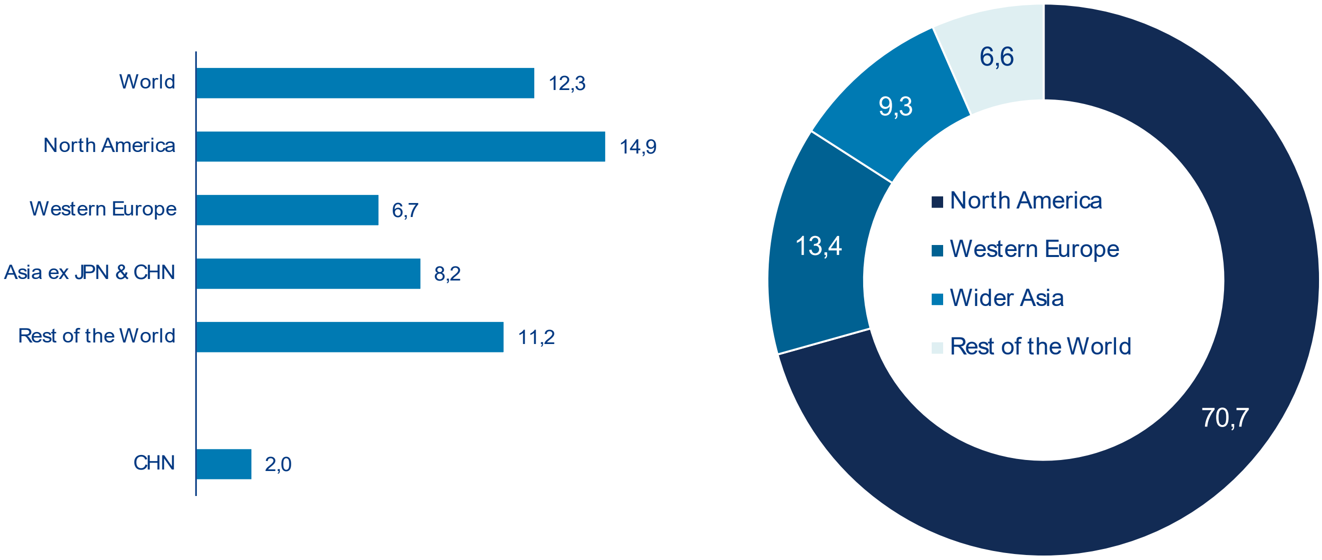

*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

**Health insurance[5] remains the industry's fastest-growing segment, driven by rising medical costs and increasing demand for private protection**. The US continues to dominate the global health market, accounting for more than 70% of worldwide health premiums (Figure 5). In many other markets, private health insurance is still a niche segment, albeit a very dynamic one: global premium growth averaged +8.5% p.a. over the last decade with particularly strong rates in recent years as Covid-19 boosted demand for additional health protection. In 2025, global health premiums increased by +12.3%, the strongest expansion since 2014, driven largely by North America where medical inflation pushed growth to +14.9%.

**Despite a recent normalization in parts of Asia, the region's low health-insurance penetration continues to point to substantial long-term growth potential.** Health-insurance growth in wider Asia slowed to +3.4% in 2025, reflecting a sharp deceleration in China, where growth dropped to just +2%. However, many markets including Indonesia, Singapore, Hong Kong, Thailand and India continued to record double-digit growth, with several accelerating further in 2025. This strong momentum partially reflects the still low penetration in the region, which is below 1% in all markets except Taiwan. Even more than in life insurance, demand is mainly driven by the status of the social security system, i.e. the coverage and quality of the public healthcare system. This explains why even in some developed markets, such as the Scandinavian markets, private health insurance is almost non-existent.

[5] In general, data visibility of health-insurance premiums is still rather low. Often, health premiums are included in other segments as health coverage is seen as part of other products and not separately reported; in Japan, for instance, health is treated as "third sector products" within life.

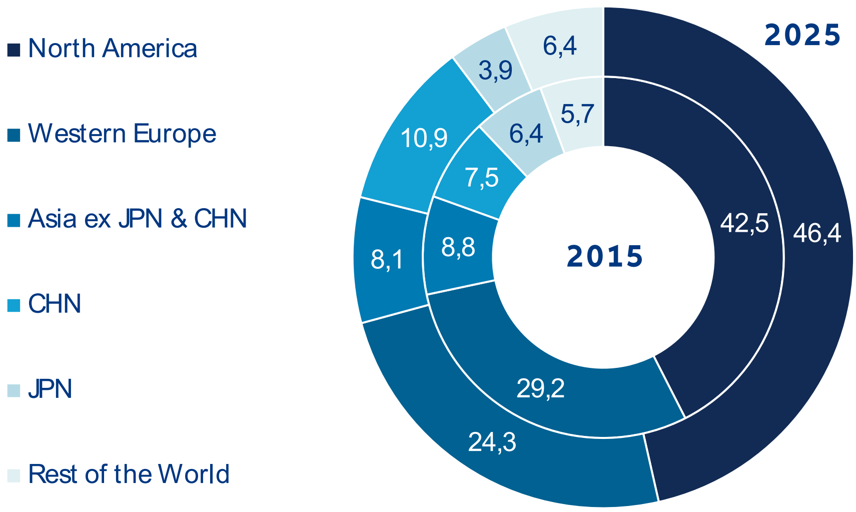
**Figure 5:** No slowdown in sight
Health gross written premium growth* and market share by region, 2025 in %




*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

**The global insurance industry remains overwhelmingly dominated by the US despite the rise of China and broader Asia.** Unlike many other industries where traditional markets are losing importance to new emerging markets, the global insurance industry is still dominated by the US. In fact, over the past decade, the North American insurance market has increased its share of global premiums from 42.5% to 46.4%, meaning that almost every second euro written globally now originates from the region. On the other hand, other „old" markets such as Western Europe (-4.9pps) and Japan (-2.5pps) behaved more or less as expected, losing market share mainly to China, which boosted its global share from 7.5% to 10.9%. However, the gap with the North America market is still huge: while the Chinese market is already the clear number two in terms of premium income (EUR746bn in 2025), it is dwarfed by the North America market, which generates more than four times as much premium income (EUR3,191bn). Given this huge discrepancy, it does not look like the US dominance in global insurance will end any time soon (Figure 6).

**Figure 6:** US dominance remains unchallenged
Total gross written premiums*, 2015 and 2025 by region in %

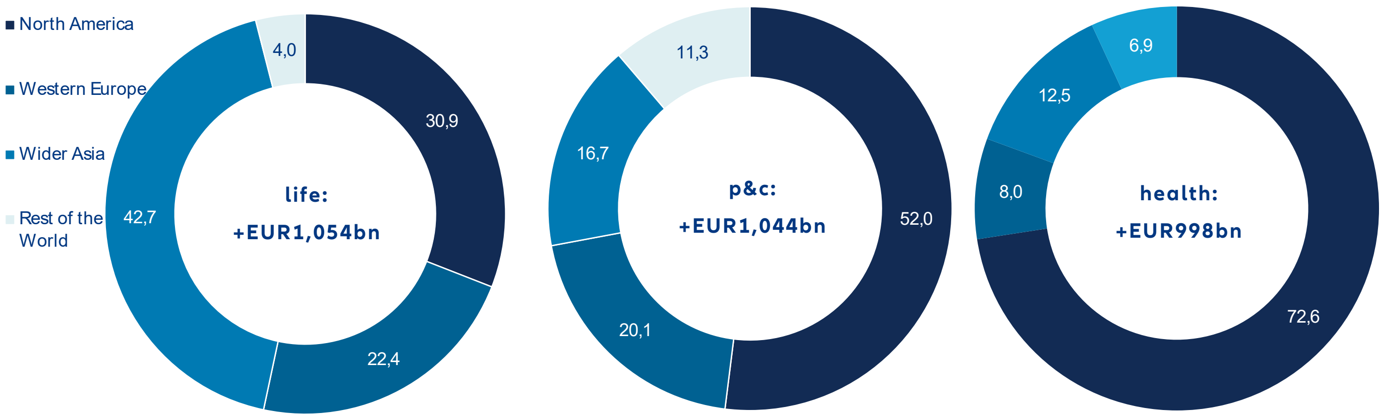



*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

**The past decade has reshaped the global insurance market unevenly across regions and business lines.** North America accounted for 52% of additional global P&C premiums and 71% of additional health premiums over the past ten years, underlining the region's overwhelming importance in both segments. In life insurance, however, the picture is more balanced. Despite the recent US annuity boom, only around one-third of additional life premiums originated in North America, while wider Asia captured by far the largest share of global growth. Western Europe, by contrast, continues to lose relative weight across all major business lines, falling behind not only the US market but increasingly also wider Asia.

**Another notable shift is the growing convergence in absolute premium growth across segments.** Although life insurance remains substantially larger in size at EUR2,861bn compared with EUR2,320bn in P&C and EUR1,688bn in health, the absolute increase in premium volumes has recently been broadly similar. While life insurance growth was constrained for years by ultra-low interest rates, health insurance has benefited strongly from the post-Covid increase in risk awareness and demand for private health protection (Figure 7).

**Figure 7:** Western Europe continues to lose ground
Share of absolute premium growth* by region, 2015 - 2025 in %




*The conversion into EUR is based on 2024 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

# The changing risk landscape

## Economic and capital market outlook

**The global economy has entered a period of heightened uncertainty as the Iran war acts as a significant external supply shock, disrupting energy markets, raising transport costs and fragmenting supply chains.** The result is renewed inflationary pressure alongside weaker growth at a time when the global outlook was already fragile. The scale and persistence of the economic impact remain highly uncertain and will depend critically on how the conflict evolves.

**In our central scenario, in which a US-Iran agreement struck during the summer months leads to a normalization of oil and gas trade before the end of the year, we forecast global GDP growth of 2.6% in 2026, revised down by 0.5pp from pre-conflict expectations.** The impact across major economies is uneven. In the US, growth is expected to remain relatively more robust at +2.0% as substantial investment in artificial-intelligence infrastructure and stronger private consumption driven by tax cuts largely offset the energy shock. China would also prove comparatively resilient at +4.7%, supported by a more diversified energy mix and solid external demand. Europe, however, would bear the heaviest burden. Eurozone growth would nearly halve to +0.7% as the region's greater dependence on energy imports adds to longstanding structural weaknesses that continue to constrain economic momentum.

**Inflation is set to rise on both sides of the Atlantic in 2026, interrupting the gradual normalization seen over the past year.** In the US, the energy shock will push inflation back toward levels last seen during the post-pandemic surge. CPI inflation is forecast to rise to 3.6% in 2026, peaking at around 4.0% around mid-year before gradually easing as energy prices stabilize and weaker activity dampens underlying price pressures. The Fed is expected to raise rates by 25bps in 2026, largely looking through the energy-driven spike as the risk of inflation expectations becoming de-anchored appears contained. After all, unlike in 2022, labor markets are softer and monetary conditions are no longer highly accommodative, limiting the scope for broad-based second-round effects. In the Eurozone, inflation dynamics would be similar in direction but more persistent in profile. Headline CPI is forecast at 3.1% for 2026 as the pass-through from higher wholesale gas prices into household utility bills keeps inflation elevated for longer than in the US. This leaves the ECB in an uncomfortable position: inflation is rising above target just as growth weakens sharply. We forecast the ECB to hike rates by 50bps in 2026, with a focus on keeping inflation expectations in check.

**Government bond markets are undergoing a sharp repricing as the energy shock pushes inflation expectations higher while simultaneously weakening the growth outlook**. Long-term sovereign yields are rising as investors demand greater compensation for inflation risk and increasingly question the sustainability of large fiscal deficits in a weaker growth environment. On balance, we expect 10-year Treasury yields to reach 4.6% by year-end, with persistent fiscal deficits of around 7% of GDP adding further upward pressure on US borrowing costs. In Europe, 10-year Bund yields are forecast at 3.0%, as weaker growth and safe-haven demand from investors diversifying away from US assets partially offset rising inflation expectations. In the periphery, however, Italian and French borrowing costs are increasing more sharply than Germany's as markets grow more concerned about the capacity of heavily indebted sovereigns to absorb energy-related fiscal pressures without compromising fiscal consolidation commitments – a dynamic likely to intensify as France approaches its 2027 election cycle.

**Risks to the outlook remain clearly skewed to the downside.** A prolonged closure of the Strait of Hormuz into fall would produce a materially more adverse outcome. In this downside scenario, the Eurozone economy would fall into technical recession, with annual growth in 2026 slowing to just +0.2%, while US growth would weaken sharply to +1.5% as equity market corrections feed through to consumer confidence and spending. Inflation would rise to 4.1% in the Eurozone and 4.2% in the US, forcing central banks into a more aggressive tightening response despite deteriorating growth conditions – a policy dilemma with few good options. Beyond the direct energy channel, a prolonged conflict would amplify second-round effects through weaker consumer confidence, rising insolvencies, tighter financial conditions and growing fiscal strain across energy-importing economies. At a time of weakening international policy coordination and increasingly constrained public finances, policymakers may have less capacity to cushion additional shocks. In such an environment, non-linear dynamics could emerge quickly, turning an initial energy shock into a broader and more persistent global downturn.

# How geopolitical fragmentation impacts insurance

**The insurance industry was built for a world of open capital flows, stable rules and predictable cross-border commerce, and that world is receding**. A series of geopolitical shocks, from the US-China trade conflict and the Covid-19 pandemic to the Russia-Ukraine war and the US-Iran conflict, has accelerated the fragmentation of trade, finance and supply chains, pushing the global economy toward a more fractured geoeconomic order. For insurers, this is not simply a challenging macroeconomic backdrop but a structural shift in the foundations of the global insurance business model.

**The question is no longer whether fragmentation will reshape global insurance, but how far and how fast.** The assumptions that supported global operating models for decades are weakening simultaneously: capital is becoming less mobile, regulatory regimes are diverging and borders are increasingly functioning as strategic barriers rather than manageable operational frictions. For insurers, this directly challenges the foundations of global risk pooling. Insurance depends on the ability to diversify exposures across geographies and move capital and reinsurance capacity freely across borders. Fragmentation weakens both. As cross-border activity becomes more constrained, volatility in claims, currencies and investment returns increases, while regional pools struggle to replicate the efficiency and resilience of global ones.

**Regional pools are structurally less efficient because fragmentation increases correlation across exposures while reducing the fungibility of capital and reinsurance.** Political shocks increasingly affect currencies, asset prices, trade flows and claims simultaneously, weakening diversification benefits precisely when they are most needed. At the same time, sanctions risk, capital controls and regulatory divergence increase the likelihood that capital becomes trapped within jurisdictions during periods of stress, forcing insurers to hold more capital against the same underlying risks. This undermines one of the key advantages of global insurance groups: efficient cross-border capital allocation and diversification. As trapped capital and duplicated solvency buffers rise, returns on capital are likely to come under structural pressure. The challenge is therefore no longer preserving the hyper-globalized insurance model of the past, but maintaining sufficient diversification and risk-sharing capacity in a more regionalized world.

**Regulatory fragmentation is reinforcing these pressures.** The decades of supervisory convergence that produced Solvency II, IAIS standards and equivalence regimes are giving way to more modular and bloc-specific systems. Cross-border reinsurance recoveries, sanctions compliance and contract enforceability can no longer be treated as stable assumptions. Meanwhile, the weakening of multilateral cooperation is amplifying the very risks insurers must absorb. Climate change, cyber threats, pandemics and financial instability all require international coordination to contain; as geopolitical rivalry undermines that cooperation, these risks become larger, more interconnected and harder to price.

**The investment consequences are equally significant. Insurers collectively manage trillions in long-duration assets and fragmentation permanently raises the cost and complexity of deploying that capital globally.** When the movement of capital increasingly requires political approval rather than responding to market logic, global integration loses efficiency by design. Currency and convertibility risks rise alongside political tensions, sovereign risk becomes harder to hedge and friendshoring creates exposures that are often less liquid and operationally more complex.

**The effects are unevenly distributed across business lines. Political-risk insurance is experiencing a structural increase in demand as multinational companies move from treating geopolitical exposure as episodic to embedding it permanently within risk-management frameworks.** In trade credit, tariffs are squeezing margins and raising insolvency risk, particularly among smaller companies exposed to disrupted supply chains. Marine and aviation insurers must navigate rerouted shipping lanes and persistent geopolitical volatility; the closure of major maritime chokepoints has shown how quickly coverage availability and pricing can deteriorate when political risk crystallizes into physical disruption. State-sponsored cyber threats are escalating, with attribution challenges straining the boundaries of standard policy language. In D&O, rising regulatory investigations and extra-territorial enforcement actions are increasing claims exposure for internationally active firms. Retail insurance faces a more indirect but persistent transmission through slower growth, higher inflation and weaker household purchasing power.

**At the same time, fragmentation is also creating new insurable risk categories that reward analytical capability and underwriting specialization.** The push for energy independence is increasing demand for specialized engineering insurance, while industrial policy initiatives including the CHIPS Act, European sovereignty investment programs and rising defense spending are generating substantial new property and infrastructure exposures. Geopolitics is on balance a net headwind for the sector's long-term growth prospects, but rising investment in defense, energy security, cyber resilience and strategic infrastructure is creating offsetting opportunities for insurance, risk transfer and long-duration capital deployment.

**The fact that the insurance sector has remained broadly stable through successive geopolitical shocks so far is no guarantee of future success.** Capital positions have remained healthy, combined ratios have held and capacity has not materially decreased. This is testament to the resilience of the insurance industry. However, each shock raises the structural baseline, which is not fully reversed during recovery periods. This means that the industry has to reprice for a genuinely higher level of structural risk. This makes strategically adapting to a fragmenting world a matter of current urgency.

**The industry's strategic response must operate on several levels simultaneously.**

• **First, operating models must become less centralized and more regionally resilient.** In a more fragmented world, resilience and adaptability matter more than pure scale efficiency. Capital must be increasingly pre-positioned within jurisdictions rather than assumed freely transferable, and entities must be capable of operating with greater autonomy during periods of geopolitical disruption. Scenario planning must also replace narrow point forecasts. Tail risks such as severe bloc bifurcation or financial decoupling are no longer remote enough to ignore in strategic planning, even if the most likely trajectory remains one of escalating but contained protectionism.

• **Second, portfolio and underwriting decisions must incorporate explicit geopolitical assessment as a systematic input rather than a qualitative overlay.** Insurers should classify markets along a clear spectrum from core to optional to potential exit candidates, combining underwriting analysis with regular geopolitical, sanctions and affordability assessments rather than making decisions reactively under pressure. Stress testing must extend beyond financial variables to include sanctions regimes, reinsurance enforceability and diversification failure under fragmentation scenarios. The industry cannot assume that mechanisms designed for a globalized world will continue to function under sustained geopolitical strain.

• **Third, product strategy must evolve to address the new exposures fragmentation is creating rather than relying primarily on exclusions and silent coverage.** Recent conflicts have exposed significant gaps in affirmative cover for war risk, political violence, trade disruption, cyber escalation and supply-chain interruption. Insurers that develop explicit and well-priced solutions for these exposures will benefit from a structural increase in corporate demand. Those that rely primarily on exclusions risk marginalization from some of the fastest-growing segments of the commercial market.

**Fragmentation will reshape how the global insurance industry operates.** A more regionalized world implies higher capital requirements, weaker diversification benefits and greater operational complexity. Yet the same forces that reduce efficiency are also increasing demand for protection, resilience and specialized risk transfer. Insurance may become structurally more complex and capital-intensive, but also strategically more important.

# Looking ahead: Insurance remains a growth industry

**Insurance will remain a structural growth industry over the next decade, although growth will become more regionally uneven.** We expect the global insurance market to expand by +5.3% annually over the next ten years, broadly in line with last year's long-term outlook. Beneath this stable headline picture, however, regional dynamics are shifting meaningfully.

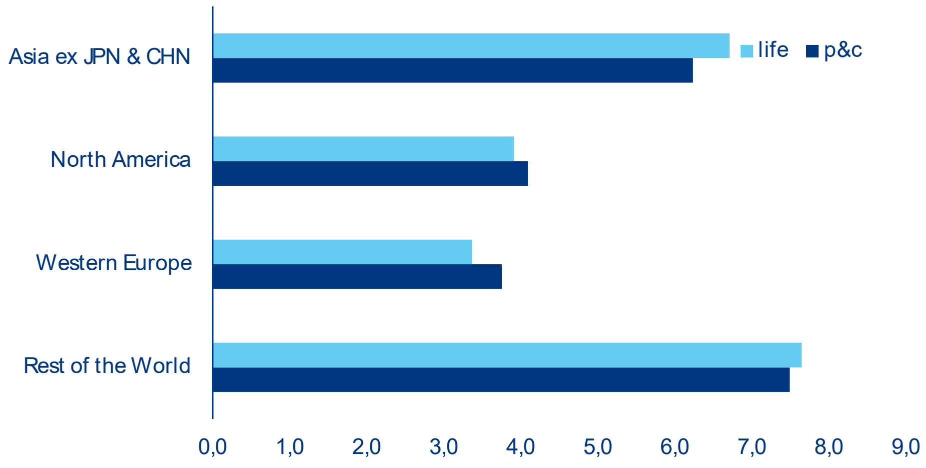
**North America is likely to remain the industry's dominant profit and growth pool.** Compared to last year, our outlook for the region has become more optimistic across all three business lines, particularly in P&C. Three forces are driving this shift: the expected AI investment boom and associated productivity gains, continued increases in household spending on protection – especially in health insurance – and elevated interest rates supporting life and annuity products.

**Europe faces a structurally weaker outlook.** Our view on Western Europe has become more cautious compared with last year. The region is likely to continue struggling with the fallout from higher energy costs triggered by the escalation in the Middle East (see Chapter 3 for more details), slower than expected progress on strategic sovereignty and both weaker AI investment and slower AI adoption than the US. As a result, we expect European P&C growth to lag the US over the next decade at +3.8% versus +4.1%. Although rising natural-catastrophe losses should continue to support premium growth, we now expect only a limited increase in penetration rates, rising by just +0.1pp to 2.6% by 2036.

**Asia will remain the industry's principal long-term growth engine despite rising geopolitical headwinds.** We expect P&C growth in Asia to slow modestly relative to previous forecasts as trade tensions and supply-chain restructuring weigh on economic momentum. Even so, projected annual growth of +6.2% in wider Asia and +6.3% in China remains comfortably above expected growth rates in Europe and North America. Overall, we expect global P&C premiums to expand by +4.7% annually through 2036 as demand for protection continues to rise worldwide.

**Life and health insurance will remain the industry's key structural growth segments.** We expect life insurance to grow at a CAGR of +4.9% over the next decade, benefiting from elevated interest rates and rising retirement needs. Asia and China should remain the principal growth engines as demographic ageing accelerates and pension systems continue to deepen. Health insurance is expected to remain the industry's fastest growing segment at +6.7% annually, supported by rising healthcare costs and still substantial unmet demand, particularly across Asia (Figure 8).

**Figure 8:** Solid growth
Gross written premium* growth, CAGR 2026 -2036 by region and segment in %

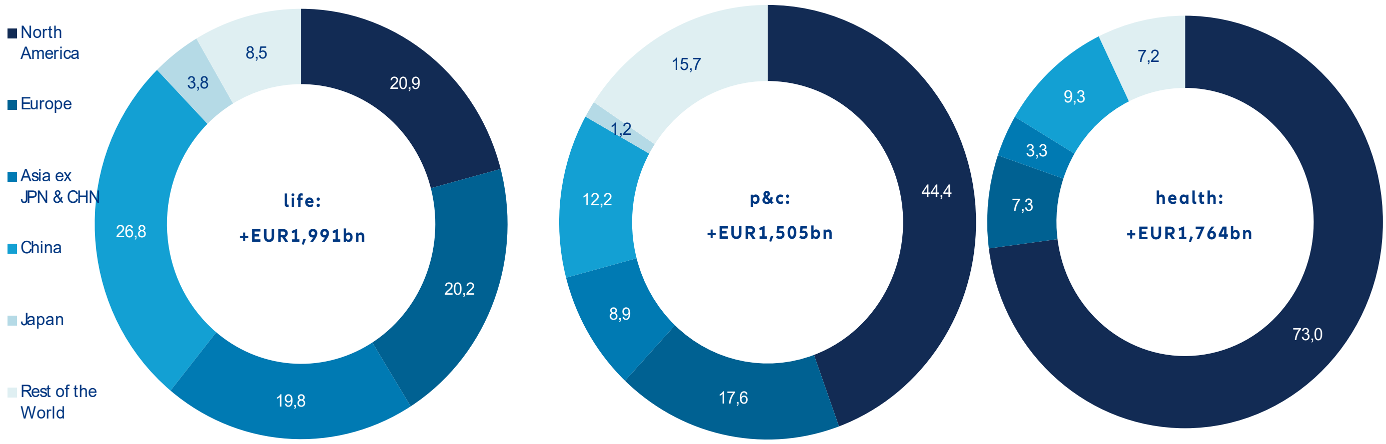



*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research..

**Asia will generate most of the industry's additional premium growth.** The global premium pool is expected to expand by EUR5,260bn by 2036 (Figure 9), with life insurance contributing the largest share at EUR1,991bn. More than half of all additional premiums will originate in Asia including China, exceeding the combined contribution of North America and Europe. While China will remain the region's dominant market in absolute terms with growth of +7.6% annually, India is expected to emerge as the fastest-growing major market at +10.5% annually, positioning it to challenge Japan as Asia's second-largest life insurance market by 2036.

**Despite Asia's rise, the US insurance market will remain unrivaled in scale.** Although North America's share of incremental P&C growth is expected to decline relative to the previous decade, the US market will continue to dominate globally. By 2036, US P&C premiums are projected to exceed EUR1,800bn, almost five times larger than China's market and more than twice the size of Europe's. In health insurance, US dominance is even more pronounced, with more than two-thirds of expected global premium growth projected to originate from the US market.

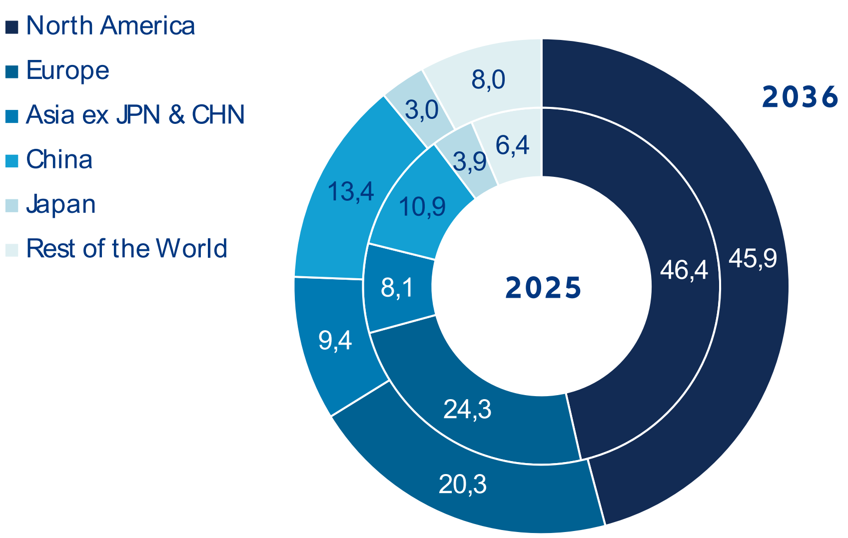
**Figure 9:** Life remains on top
Share of additional gross written premiums* by 2036, by region in %




*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

**The global insurance map will continue shifting eastward, albeit gradually.** North America is expected to retain a global market share of roughly 46% through 2036, surrendering only marginal ground over the next decade (-0.5pp). India and China, by contrast, are expected to continue gaining relevance, together adding almost 4pps of global market share. Western Europe will continue to lose relative weight. A glimmer of hope for the Old Continent: while it lost 5.3pps of market share in the last decade, it may lose „only" 4pps in the next decade (Figure 10).

**Figure 10:** North America, a slightly smaller giant
Total gross written premiums*, 2025 and 2036 by region in %

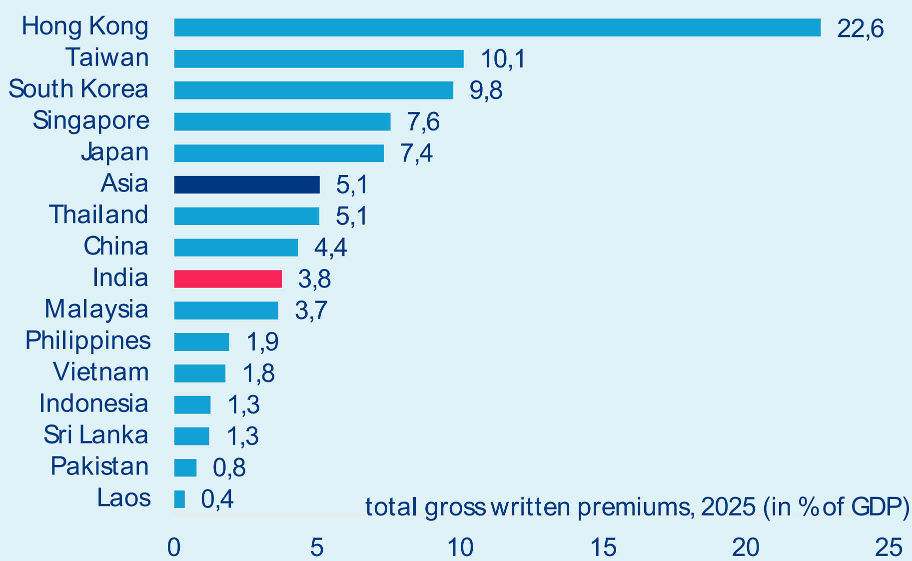



*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

# Indian insurance market deep dive – turning scale into depth

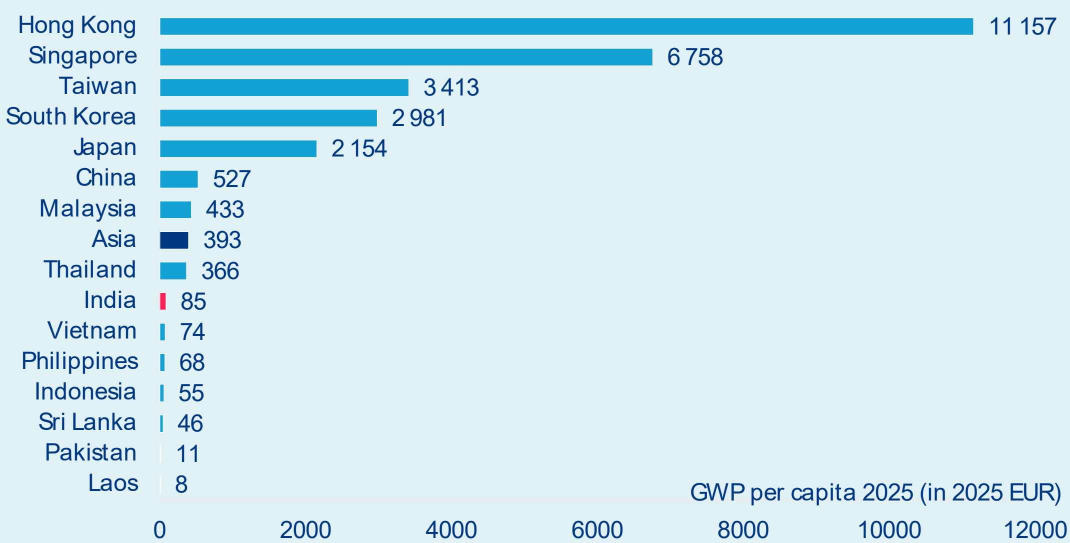
**India ranks among the world's ten largest insurance markets.** But by almost every measure of depth, it remains significantly underdeveloped. Despite double-digit growth rates and total gross written premium income of EUR124bn in 2025, insurance penetration and density still indicate substantial pent-up demand. In 2025, total gross written premium income corresponded to 3.8% of GDP compared to 4.4% in China or 5.1% in Thailand. And while per-capita spending for insurance amounted to the equivalent of around EUR527 in China and EUR366 in Thailand, it was merely EUR85 in India (Figure 11 and 12).

**Figure 11:** Insurance penetration, selected countries (gross written premiums in percent of GDP)




Source: National financial supervisory authorities, insurance associations and statistical offices..

**Figure 12:** Insurance density, selected countries (gross written premiums per capita, in 2025 EUR)

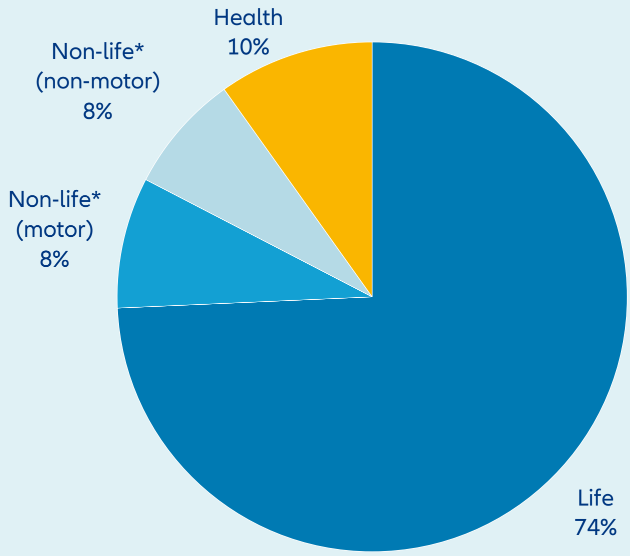



Source: National financial supervisory authorities, insurance associations and statistical offices.

**Life insurance dominates the market, but the non-life and health segments reveal where the structural gaps and the growth potential are most acute.** India's insurance market is dominated by the life insurance segment, which accounted for 74% of total gross written premium income in 2024. Non-life insurance premiums, including personal accident and travel insurance, accounted for 16% and health insurance accounted for 10% of total gross written premiums (see Figure 13).

**Mandatory motor coverage is the backbone of non-life insurance yet, half of all vehicles on Indian roads remain uninsured, illustrating the gap between legal requirements and market reality.** As in most countries, motor insurance was the most important business line in the non-life segment (excluding health), with a share of 52% in fiscal year 2024–2025. Third-party liability business made up more than half of the motor segment as such coverage is mandatory for all vehicles driven on public roads. Nevertheless, it is estimated that over 50% of vehicles remain uninsured[6]. Given the importance of the agricultural sector for the Indian economy, crop insurance accounted for 16% of general premium income. However, there were significant differences in the composition of premium income across individual states, largely reflecting their respective economic structures (see Figure 14).

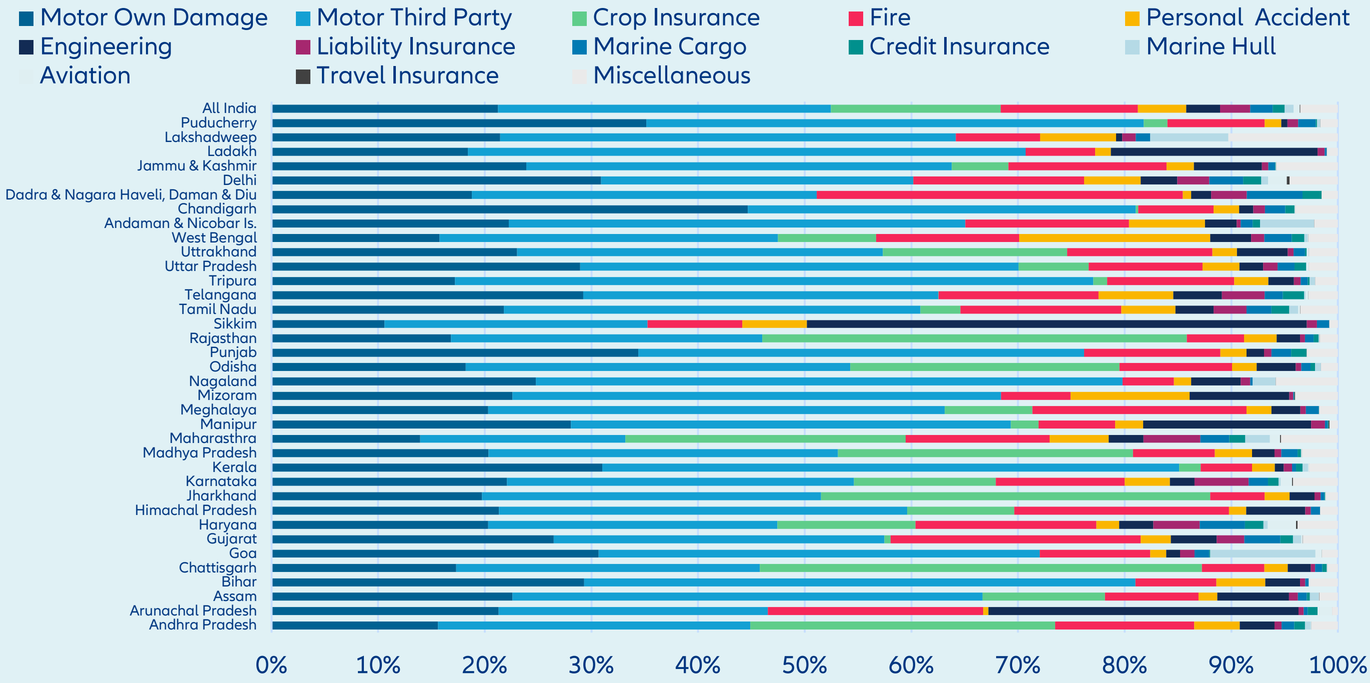
**Figure 13:** Gross written premiums by business line (in percent)




*including personal accident and travel insurance premiums
Source: IRDAI, Allianz Research.

[6] See Axco (2025): Insurance market reports (non-life). India (as of March 2026).

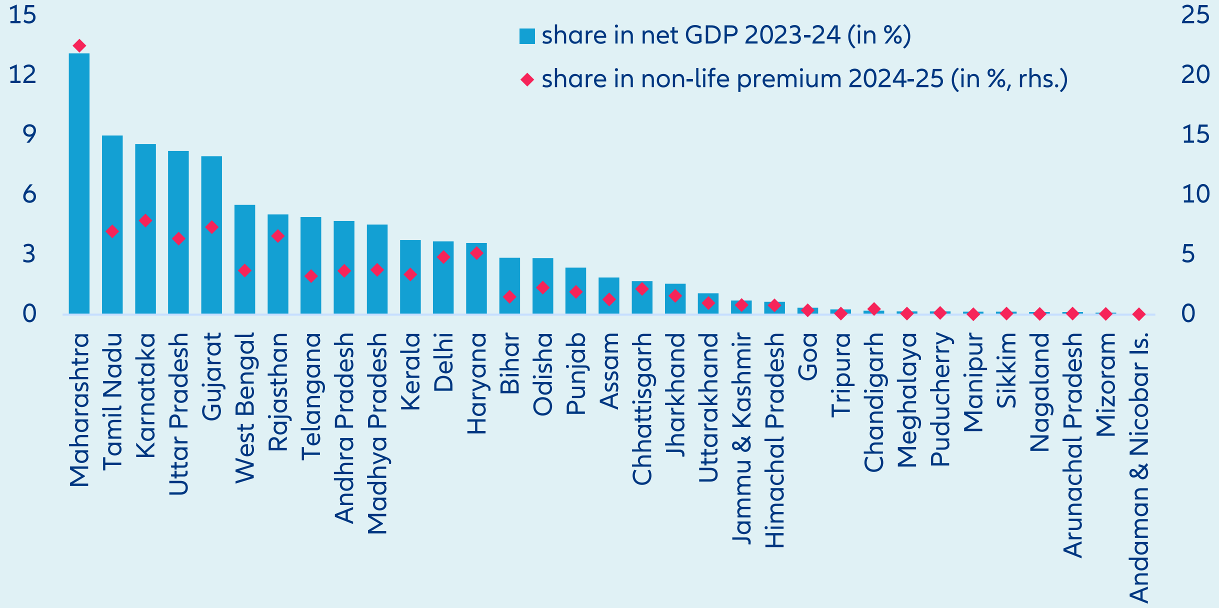
**Figure 14:** Non-life insurance premium by line of business and state (in percent)




Source: IRDAI

**Premium income is as unevenly distributed as economic activity itself, and product design that ignores this geography will systematically miss the largest underserved populations.** Five states – Gujarat, Karnataka, Maharashtra, Rajasthan and Uttar Pradesh – accounted for 48% of general premium income, with Maharashtra alone accounting for 20% (see Figure 15). In health insurance, Karnataka, Maharashtra and Tamil Nadu accounted for 49.2% of premium income in 2023–2024, with 22.5% generated in Maharashtra. Despite signs of economic convergence, marked regional differences persist. In 2023–2024, per-capita net state domestic product ranged between EUR570 in Bihar, representing around a third of the national average of EUR1,790, to EUR5,568 in Sikkim. These differences need to be taken into account when it comes to the design and distribution of insurance products.

**Figure 15:** Non-life premium income by state




Source: IRDAI, Reserve Bank of India

**The most powerful long-term growth driver for India's insurance market, alongside rising household wealth, is the still incomplete and uneven reach of the country's social protection system**. Market expansion is therefore expected to be driven primarily by pent-up demand and the growing need for supplementary private risk protection, as public social security remains at a relatively early stage of development. According to the latest available ILO data, only 64.3% of India's population was covered by at least one branch of the social security system at all. Effective pension coverage is even more limited: the ILO estimates that only 45.8% of the working-age population was covered by the pension system, compared with close to universal coverage in most industrialized countries. Without targeted government initiatives such as Atal Pension Yojana and the e-Shram portal aimed at informal workers, as well as the supplementary state-level pension schemes, this share would be significantly lower. This is particularly relevant given that only around 12% of India's workforce is employed in the formal sector and therefore benefits from standard pension coverage. Against the backdrop of rising life expectancy, these structural gaps point to a growing need for supplementary, capital-funded private pension solutions to secure retirement income. A similar dynamic is evident in healthcare. Despite the successful rollout of Pradhan Mantri Jan Arogya Yojana (PM-JAY), out-of-pocket spending still accounts for 43.9% of total health expenditure - high by international standards. This underscores both the need for further reforms to broaden public healthcare coverage and the resulting demand for supplementary private health insurance. In this context, medical inflation in India, which is among the highest in Asia, poses a significant financial challenge for households and health insurance providers alike.

**Regulatory reform and an ambitious universal coverage target are creating both the policy tailwind and the competitive catalyst for the next growth phase.** Recent insurance market reforms, notably the opening up to foreign insurance participation, will increase the efficiency of the insurance sector and together with the IRDAI's „Insurance for All by 2047" program are expected to spur double-digit insurance market growth of an estimated average 10.5% in life insurance, 10.2% in non-life and 12.5% in health insurance per annum until 2036.

**India's insurance sector will expand not just in scale but in structural sophistication, shifting from a peripheral to a systemically important pillar of the financial system.** Rising incomes and a growing middle class will lift demand for protection and savings products, driving higher penetration and density. Demand is expected to shift from basic, often mandatory covers such as motor and simple life towards more complex products including health, pensions and wealth-linked insurance. Insurers will increasingly act as institutional investors, supporting capital market development and long-term financing, particularly for infrastructure. Higher incomes and improving financial literacy would also help reduce underinsurance and encourage more proactive risk management by households and firms. Together with demographic trends, notably ageing, this would further boost demand for private health and retirement solutions. India's still-developing social security system implies a greater reliance on private provision compared to other markets such as China. Overall, sustained growth is likely to transform the sector into a more mature and systemically important pillar of the financial system.

# The next frontier: Climate risk, affordability and the future of Insurability

**Affordability is emerging as a key structural challenge facing the insurance industry.** In some markets and lines of business, the cost of adequate coverage is beginning to outpace households' capacity to pay, driven by claims severity, inflation and broader economic pressures. Where this gap becomes persistent, it translates into structural underinsurance, concentrating uninsured risk where financial resilience is weakest. Climate risk is where this dynamic is already most visible and most advanced.

**Global insured NatCat losses are rising by 5–7%[7] annually in real terms, underlining that climate risk is now a structural underwriting issue rather than a cyclical volatility event.** Climate change is an undisputed driver, but observed loss trends are also shaped by socioeconomic factors, including exposure growth in hazard-prone areas and inflation rather than by climate effects alone[8] . Rising premiums are increasingly colliding with falling household purchasing power. Climate shocks alongside other socioeconomic pressures such as demographic change, wars and geopolitical instability are driving both insurance costs and greater financial stress. This dynamic is already visible in household insurance markets: In the US, homeowners' insurance premiums rose by +38% between 2019 and 2024, roughly twice the rate of inflation. The same climate shocks driving up premiums also erode the capacity to pay: droughts reduce crop yields and push up food prices, while heat stress lowers labor productivity and wages. Global food prices could be around 13% above baseline by 2050 in a +2°C warming scenario, with Asia-Pacific facing increases of up to 27%, while the most exposed European economies could see cumulative GDP losses of 5–7% over five years[9] . Households are therefore being squeezed from both directions: rising insurance costs and weakening income resilience.

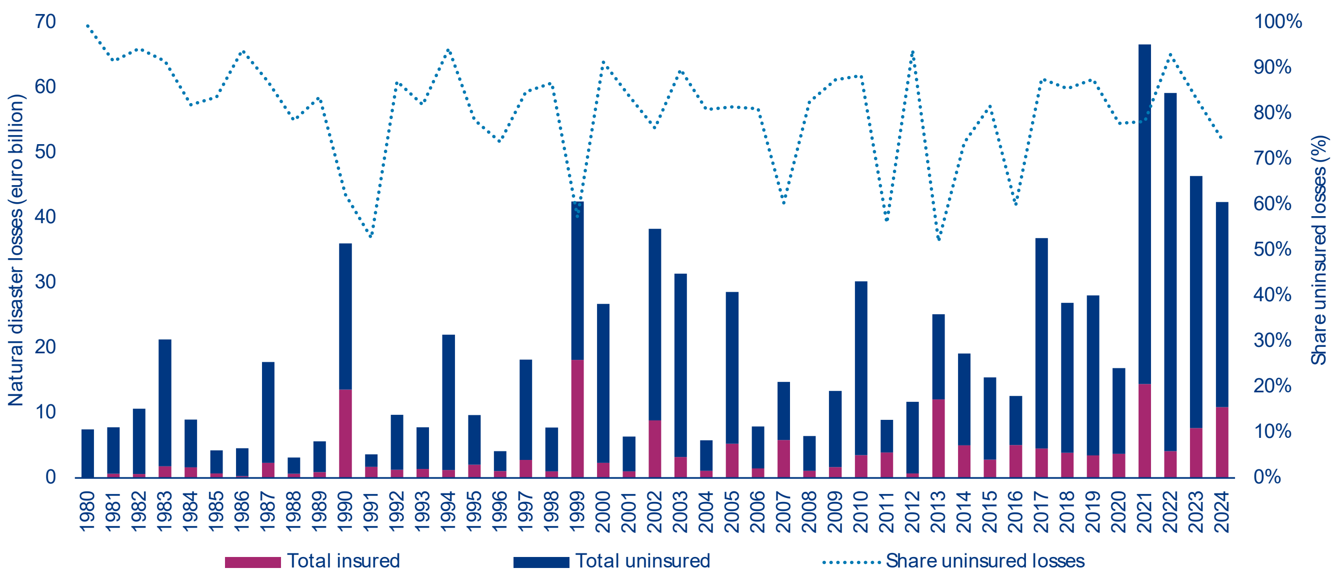
**Lower-income households face a self-reinforcing vulnerability trap as higher premiums and weaker incomes push them out of insurance coverage.** The poorest and most climate-exposed households are hit hardest, creating a cycle in which rising risks deepen both poverty and underinsurance. Evidence from developing countries shows that the poorest 40% experience climate-driven income losses 70% larger than the average[10], while a 1°C increase in temperature raises poverty rates by 0.9 to 2.3pps[11]. Similar patterns are emerging in developed

[7] Swiss RE (Sigma 1/2025)
[8] The primary driver to date has been exposure growth (urbanization, asset accumulation and rising construction costs) which accounts for over 80% of the long-term increase in weather-related insured losses. Climate change effects have so far been a smaller but growing contributor, and Swiss Re cautions that for certain perils and regions, such as wildfire in North America and severe convective storms in Europe, hazard intensification is already accelerating losses well beyond what exposure alone can explain.
[9] Feeding a warming world: Securing food and economic stability in a changing climate | Allianz
[10] Climate change through a poverty lens | Nature Climate Change
[11] World Bank Document

economies, where climate change also drives chronic rather than temporary poverty, making exclusion from insurance more persistent. As premiums rise, disaster insurance risks becoming a product accessible only for wealthier households. Evidence from Florida shows that without public investment in risk reduction and affordability mechanisms, both traditional and parametric insurance products become inaccessible for low- and middle-income households[12].

**As private insurance affordability declines, climate losses increasingly shift onto public balance sheets that are themselves being weakened by climate shocks.** In the EU, uninsured direct disaster losses averaged EUR59bn annually between 2021 and 2023 (Figure 16), highlighting how a large share of climate losses continues to sit outside formal insurance coverage and is ultimately absorbed by households, firms and governments[13]. This gap is significantly demand-driven: EIOPA research highlights[14] that low risk awareness, expectations of government compensation and lack of product clarity mean many households and businesses simply do not purchase available NatCat coverage, a challenge distinct from the supply-side retreat seen in parts of the US market. This is also visible in the rapid expansion of state-backed insurance schemes. Heat stress alone could worsen European fiscal balances by around 0.5% of GDP annually. France faces additional fiscal pressure of 2.2% of GDP, while Italy and Spain face 1.9% and 1.7% respectively. The governments most exposed to climate risk are often those with the least fiscal space to subsidize insurance or invest in adaptation, creating a dangerous feedback loop of rising losses, weaker coverage and declining fiscal resilience. In the US, California's FAIR Plan increased insured exposure from USD220bn in September 2022 to USD750bn by early 2026[15] , and following the Los Angeles wildfires required a USD1bn emergency levy on insurers to maintain solvency. Florida's Citizens scheme peaked at USD618bn of insured exposure in 2023, showing how quickly public contingent liabilities can grow when private insurers retreat[16].

**Figure 16:** Development of NatCat losses in Europe and share of uninsured losses




Source: CATDAT

[12] Insurance and climate risks: Policy lessons from three bounding scenarios | PNAS
[13] Climate and nature risks fuel multi-billion insurance protection gaps | WWF
[14] Climate insurance protection gaps: a demand-side challenge - European Insurance and Occupational Pensions Authority
[15] California FAIR Plan Association (March 2026)
[16] Citizens Property Insurance Corporation (Policies in Force)

**Europe is increasingly moving toward mandatory and state-backed NatCat systems, but these solutions must avoid creating long-term moral hazard.** Public-private NatCat systems are becoming a structural feature to adapt to extreme weather events. France already operates a mandatory state-backed NatCat reinsurance scheme funded by a surcharge on all property insurance policies, and after repeated climate-related losses that charge was nearly doubled last year to 20%. Since 2025, all businesses in Italy have been required to carry NatCat insurance, with the government covering up to half of the exposure through its state reinsurer. Germany is about to introduce a mandatory scheme, and the German Insurance Association has proposed a new private-sector pool for high-risk residential properties backed by a government stop-loss mechanism for catastrophe losses above EUR30bn. At the EU level, supervisors are exploring a Europe-wide risk-sharing system combining a premium-financed pool with a public loan-funded mechanism. These schemes can preserve insurability, but if public support becomes too strong, pricing signals could weaken and incentives for resilience investment, relocation and land-use discipline could be reduced. Equally, even well-designed public-private schemes face structural limits if investment in climate-change adaptation, from enforced flood defenses and resilient infrastructure to smart agriculture and water management systems, does not scale fast enough to reduce costs of the rising loss trends. Without combined strategies, the fiscal and insurance capacity underpinning the public-private NatCat schemes will be progressively eroded.

**The deeper issue is that climate risk is exposing the limits of pricing alone as a long-term solution to maintaining insurability.** Insurance pricing continues to rely primarily on claims experience, probabilistic NatCat models and increasingly granular hazard maps that capture location-specific risk differentials. These tools remain broadly effective for pricing incremental climate risks and preserving underwriting discipline. However, annual repricing becomes increasingly difficult when physical risks and claims severity rise faster than households' ability to absorb higher premiums. For insurers, the challenge is no longer simply to reprice risk, but to preserve insurability in economies where affordability is structurally weakening.

**Preserving long-term insurability requires stronger adaptation incentives, efficiency gains, targeted affordability measures and public-private solutions.** This requires moving beyond pure risk transfer toward stronger risk reduction and resilience incentives. Insurance will increasingly need to incentivize adaptation through pricing, deductibles and prevention management, while efficiency gains become essential to offset structurally rising claims severity driven by technological complexity, specialized repair services and legal costs. Automation, AI and digital claims management already show significant potential to reduce operational costs without compromising coverage quality. Critically, however, affordability cannot be preserved through artificially suppressed premiums. Underpricing weakens incentives for risk reduction and ultimately accelerates rather than closes protection gaps. Where lower-income households are being priced out of coverage, targeted support mechanisms such as means-tested vouchers or public reinsurance are likely to prove more effective and transparent than below-risk pricing. In areas where physical risks have shifted permanently, managed relocation may ultimately be more sustainable than maintaining coverage at unsustainably low prices. More broadly, protection gaps cannot be solved by private insurance alone. Public-private partnerships, sovereign backstops and targeted affordability mechanisms will become increasingly necessary where risks become privately uninsurable, but must always be combined with credible adaptation investment at scale. Without enforced building codes, flood defenses and resilient infrastructure, public-private schemes absorb rising losses without addressing their source, progressively eroding the fiscal and insurance capacity that underpins them.

# Appendix

| **Appendix A** **Insurance markets 2025[2] KPIs** | Total premium income (in EUR bn)[1] | | | Penetration (as % of GDP) | | | Density (per capita, in EUR)[1] | | |
|---|---|---|---|---|---|---|---|---|---|
| | p&c | life | health | p&c | life | health | p&c | life | health |
| Argentina | 11,8 | 1,5 | 0,03 | 2,4 | 0,3 | 0,01 | 258 | 33 | 0,6 |
| Australia | 40,5 | 14,0 | 18,1 | 2,5 | 0,9 | 1,1 | 1 500 | 517 | 670,6 |
| Austria | 15,6 | 5,2 | 3,4 | 3,0 | 1,0 | 0,7 | 1 710 | 568 | 376,8 |
| Bahrain | *0,4* | *0,1* | *0,2* | *1,1* | *0,2* | *0,5* | *260* | *41* | *134,6* |
| Belgium | *13,3* | *17,8* | *2,5* | *2,1* | *2,8* | *0,4* | *1 127* | *1 516* | *214,7* |
| Brazil | 24,0 | 10,5 | 53,6 | 1,2 | 0,5 | 2,7 | 113 | 49 | 251,6 |
| Bulgaria | *2,2* | *0,3* | 0,2 | *3,7* | *0,5* | 0,3 | *326* | *48* | 28,6 |
| Canada | *49,9* | *66,6* | *25,4* | *2,5* | *3,3* | *1,3* | *1 244* | *1 659* | *633,0* |
| Chile | 4,9 | 9,1 | 1,1 | 1,5 | 2,9 | 0,4 | 247 | 460 | 57,6 |
| China | 190,9 | 433,2 | 121,5 | 1,1 | 2,5 | 0,7 | 135 | 306 | 85,8 |
| Colombia | 5,8 | 6,8 | 1,1 | 1,4 | 1,6 | 0,3 | 109 | 127 | 21,0 |
| Croatia | 1,6 | 0,3 | 0,2 | 1,7 | 0,4 | 0,2 | 411 | 87 | 39,4 |
| Czech Republic | 7,3 | 2,1 | 0,6 | 2,1 | 0,6 | 0,2 | 684 | 196 | 60,2 |
| Denmark | *12,8* | *36,2* | *0,5* | *3,1* | *8,8* | *0,1* | *2 134* | *6 034* | *88,7* |
| Egypt | *0,9* | *1,1* | *0,3* | *0,3* | *0,3* | *0,1* | *8* | *9* | *2,6* |
| Finland | 4,8 | 6,0 | 0,8 | 1,7 | 2,2 | 0,3 | 850 | 1 075 | 141,3 |
| France | *98,2* | 192,1 | 48,4 | *3,3* | 6,4 | 1,6 | *1 474* | 2 882 | 726,7 |
| Germany | 103,6 | 95,0 | 54,1 | 2,3 | 2,1 | 1,2 | 1 232 | 1 130 | 643,9 |
| Greece | 2,7 | 2,8 | 0,6 | 1,1 | 1,1 | 0,2 | 267 | 283 | 55,6 |
| Hong Kong | *3,3* | *77,0* | *2,3* | *0,9* | *21,1* | *0,6* | *441* | *10 411* | *305,0* |
| Hungary | 3,0 | 2,1 | 0,1 | 1,3 | 0,9 | 0,1 | 312 | 214 | 13,4 |
| India | 19,7 | *92,1* | 12,4 | 0,6 | *2,8* | 0,4 | 13 | *63* | 8,5 |
| Indonesia | 6,2 | 8,2 | 1,4 | 0,5 | 0,7 | 0,1 | 22 | 29 | 5,0 |
| Ireland | *4,6* | *24,1* | *3,9* | *0,7* | *3,8* | *0,6* | *874* | *4 539* | *741,0* |
| Italy | *48,0* | *122,6* | 5,0 | *2,1* | *5,4* | 0,2 | *812* | *2 073* | 84,0 |
| Japan | 58,5 | 206,7 | n/a | 1,6 | 5,7 | n/a | 475 | 1 679 | n/a |
| Kazakhstan | 1,2 | 1,8 | 0,1 | 0,4 | 0,7 | 0,1 | 55 | 86 | 6,6 |
| Kenya | *0,9* | *1,4* | *0,6* | *0,8* | *1,2* | *0,5* | *16* | *25* | *10,9* |
| Laos | *0,1* | *0,01* | n/a | *0,3* | *0,1* | n/a | *6* | *1* | n/a |
| Malaysia | *4,7* | 10,6 | *0,3* | *1,1* | 2,5 | *0,1* | *130* | 296 | *7,4* |
| Mexico | 18,1 | 22,0 | 8,7 | 1,1 | 1,3 | 0,5 | 137 | 167 | 65,6 |
| Morocco | 2,8 | 2,7 | 0,5 | 1,7 | 1,7 | 0,3 | 72 | 72 | 12,6 |
| Netherlands | *16,7* | *15,7* | 66,5 | *1,4* | *1,3* | 5,6 | 912 | 855 | 3 624,3 |
| Nigeria | *0,9* | *0,4* | n/a | *0,4* | *0,2* | n/a | *4* | *2* | n/a |
| New Zealand | *5,5* | *1,6* | n/a | *2,5* | *0,7* | n/a | *1 039* | *303* | n/a |
| Norway | 10,7 | 15,2 | 0,5 | 2,3 | 3,3 | 0,1 | 1 910 | 2 708 | 87,5 |
| Pakistan | *1,5* | *1,2* | n/a | *0,4* | *0,4* | n/a | *6* | *5* | n/a |
| Peru | *2,1* | 3,5 | *0,4* | *0,7* | 1,2 | *0,1* | *61* | 103 | *12,5* |
| Philippines | 2,1 | 5,6 | *0,2* | 0,5 | 1,4 | *0,1* | 18 | 48 | *1,8* |
| Poland | *14,5* | *3,6* | 1,7 | *1,6* | *0,4* | 0,2 | *381* | *95* | 43,8 |
| Portugal | 6,2 | 8,1 | 1,8 | 2,0 | 2,7 | 0,6 | 600 | 783 | 171,1 |
| Romania | 3,5 | 0,8 | *0,2* | 0,9 | 0,2 | *0,1* | 183 | 44 | *11,7* |
| Saudi Arabia | *5,8* | *2,8* | *10,7* | *0,5* | *0,3* | *1,0* | *169* | *81* | *308,4* |
| Singapore | 3,7 | 33,5 | 2,5 | 0,7 | 6,4 | 0,5 | 625 | 5 705 | 428,3 |
| Slovakia | 1,4 | 0,6 | 0,2 | 1,0 | 0,5 | 0,1 | 253 | 115 | 36,1 |
| South Africa | 11,8 | 40,2 | n/a | 3,0 | 10,2 | n/a | 183 | 622 | n/a |
| South Korea | 79,8 | 74,2 | n/a | 5,1 | 4,7 | n/a | 1 545 | 1 437 | n/a |
| Spain | 36,5 | 35,9 | 13,5 | 2,2 | 2,1 | 0,8 | 763 | 750 | 280,9 |
| Sri Lanka | 0,4 | 0,6 | 0,1 | 0,4 | 0,7 | 0,1 | 16 | 27 | 2,7 |
| Sweden | 10,8 | 39,4 | n/a | 1,8 | 6,5 | n/a | 1 017 | 3 697 | n/a |
| Switzerland | 22,1 | 24,3 | 14,4 | 2,4 | 2,6 | 1,5 | 2 462 | 2 710 | 1 609,0 |
| Taiwan | 9,8 | 56,1 | 13,0 | 1,3 | 7,2 | 1,7 | 424 | 2 428 | 560,3 |
| Thailand | 8,0 | 14,0 | 4,2 | 1,6 | 2,7 | 0,8 | 112 | 196 | 58,2 |
| Türkiye | 16,5 | 3,5 | 4,2 | 1,3 | 0,3 | 0,3 | 188 | 40 | 47,9 |
| United Arab Emirates | 7,3 | 1,9 | 8,2 | 1,5 | 0,4 | 1,7 | 642 | 170 | 726,3 |
| United Kingdom | *122,0* | *275,3* | *10,4* | *3,5* | *7,9* | *0,3* | *1 754* | *3 958* | *150,1* |
| United States | *1155,5* | 725,6 | 1167,6 | *4,4* | 2,8 | 4,5 | *3 327* | 2 089 | 3 362,1 |
| Vietnam | *2,7* | *4,9* | n/a | *0,6* | *1,2* | n/a | *26* | *48* | n/a |

[1] 2025 exchange rates.

[2] *Estimates are shown in italics.*

## Appendix B

**Insurance markets**
**Long-term development**

| | CAGR 2015-2025 (in %) | | | CAGR 2026-2036 (in %) | | | Total premium income 2036 (in EUR bn)[1] | | |
|---|---|---|---|---|---|---|---|---|---|
| | p&c | life | health | p&c | life | health | p&c | life | health |
| Argentina | 61,0 | 54,4 | 61,0 | 17,5 | 18,8 | 22,8 | 69,8 | 10,1 | 0,3 |
| Australia | 6,5 | -8,1 | 4,3 | 4,7 | 3,1 | 4,2 | 67,1 | 19,5 | 28,5 |
| Austria | 3,8 | -2,3 | 5,6 | 3,1 | 1,2 | 5,0 | 21,8 | 5,9 | 5,9 |
| Bahrain | 1,2 | -6,3 | 6,7 | 3,4 | 3,7 | 5,6 | 0,6 | 0,1 | 0,4 |
| Belgium | 3,2 | 0,9 | 5,2 | 3,0 | 4,0 | 6,1 | 18,4 | 27,4 | 4,8 |
| Brazil | 7,1 | 11,6 | 9,2 | 5,5 | 11,4 | 5,8 | 43,0 | 34,5 | 99,9 |
| Bulgaria | 10,5 | 7,2 | 16,9 | 7,2 | 7,4 | 15,4 | 4,7 | 0,7 | 0,9 |
| Canada | 4,1 | 5,3 | 4,8 | 3,7 | 4,1 | 4,7 | 74,8 | 104,0 | 42,1 |
| Chile | 8,2 | 9,3 | 11,1 | 6,4 | 7,3 | 6,3 | 9,7 | 19,9 | 2,2 |
| China | 6,6 | 11,3 | 18,2 | 6,3 | 7,6 | 8,1 | 374,3 | 966,7 | 285,6 |
| Colombia | 9,1 | 12,4 | 14,6 | 6,7 | 8,0 | 11,5 | 11,9 | 15,8 | 3,7 |
| Croatia | 7,0 | -0,4 | 13,7 | 4,4 | 2,6 | 12,2 | 2,5 | 0,4 | 0,5 |
| Czech Republic | 6,6 | -2,7 | 8,9 | 4,7 | 2,4 | 8,6 | 12,0 | 2,7 | 1,6 |
| Denmark | 2,8 | 6,2 | 7,2 | 3,0 | 5,6 | 8,1 | 17,8 | 66,0 | 1,3 |
| Egypt | 20,4 | 20,2 | 30,5 | 11,4 | 9,8 | 20,9 | 3,0 | 3,0 | 2,5 |
| Finland | 1,8 | 0,2 | 9,0 | 3,3 | 4,5 | 9,8 | 6,8 | 9,8 | 2,2 |
| France | 4,5 | 3,7 | 2,8 | 4,1 | 3,5 | 3,2 | 153,5 | 280,8 | 68,7 |
| Germany | 4,6 | 0,5 | 3,7 | 4,3 | 2,5 | 3,9 | 165,3 | 124,3 | 82,7 |
| Greece | 2,3 | 4,5 | 14,4 | 5,4 | 3,6 | 6,8 | 4,7 | 4,2 | 1,1 |
| Hong Kong | 2,3 | 8,2 | 6,8 | 4,1 | 5,2 | 5,9 | 5,1 | 135,1 | 4,2 |
| Hungary | 10,7 | 5,1 | 16,1 | 6,8 | 4,5 | 12,6 | 6,2 | 3,3 | 0,5 |
| India | 10,9 | 10,4 | 18,3 | 10,2 | 10,5 | 12,5 | 57,3 | 277,0 | 45,2 |
| Indonesia | 8,0 | 3,7 | 13,1 | 7,3 | 7,9 | 12,9 | 13,3 | 18,8 | 5,4 |
| Ireland | 5,1 | 8,0 | 4,9 | 3,5 | 3,4 | 4,9 | 6,8 | 34,8 | 6,7 |
| Italy | 2,9 | 0,5 | 8,3 | 3,6 | 2,8 | 7,6 | 70,6 | 167,0 | 11,1 |
| Japan | 1,9 | 0,5 | n/a | 2,5 | 2,9 | n/a | 77,2 | 282,9 | n/a |
| Kazakhstan | 12,0 | 32,8 | 12,7 | 7,1 | 11,8 | 12,1 | 2,4 | 6,1 | 0,5 |
| Kenya | 5,5 | 12,9 | 12,7 | 7,8 | 12,1 | 12,5 | 2,1 | 5,0 | 2,3 |
| Laos | 11,3 | 35,0 | n/a | 10,4 | 15,0 | n/a | 0,2 | 0,0 | n/a |
| Malaysia | 3,3 | 6,1 | 2,1 | 5,6 | 5,2 | 6,7 | 8,5 | 18,6 | 0,5 |
| Mexico | 9,2 | 9,8 | 12,2 | 8,2 | 8,5 | 9,7 | 43,0 | 53,8 | 24,0 |
| Morocco | 5,8 | 10,8 | 5,2 | 5,9 | 7,1 | 6,8 | 5,2 | 5,8 | 1,0 |
| Netherlands | 2,5 | 0,0 | 4,4 | 3,2 | 2,7 | 4,3 | 23,8 | 21,1 | 105,7 |
| Nigeria | 21,6 | 21,7 | n/a | 15,7 | 20,0 | n/a | 4,7 | 3,3 | n/a |
| New Zealand | 7,1 | 3,9 | n/a | 3,6 | 3,6 | n/a | 8,0 | 2,3 | n/a |
| Norway | 7,1 | 6,2 | 12,8 | 5,0 | 4,4 | 8,7 | 18,3 | 24,4 | 1,2 |
| Pakistan | 27,7 | 11,5 | n/a | 13,8 | 13,8 | n/a | 6,4 | 5,2 | n/a |
| Peru | 5,0 | 10,7 | 8,3 | 5,2 | 8,0 | 5,2 | 3,7 | 8,3 | 0,8 |
| Philippines | 7,6 | 8,7 | n/a | 8,5 | 10,0 | n/a | 5,1 | 16,0 | n/a |
| Poland | 6,9 | -3,8 | 24,1 | 4,2 | 3,6 | 8,7 | 22,9 | 5,3 | 4,2 |
| Portugal | 6,1 | -2,2 | 11,2 | 4,0 | 3,9 | 8,1 | 9,7 | 12,5 | 4,2 |
| Romania | 9,6 | 11,1 | 27,8 | 3,2 | 3,3 | 21,6 | 4,9 | 1,2 | 1,9 |
| Saudi Arabia | 5,8 | 26,8 | 10,5 | 6,2 | 11,7 | 8,2 | 11,3 | 9,4 | 25,3 |
| Singapore | 4,2 | 7,5 | 23,9 | 5,6 | 5,5 | 8,4 | 6,7 | 60,2 | 6,1 |
| Slovakia | 3,6 | -5,4 | n/a | 4,0 | 3,5 | 10,5 | 2,1 | 0,9 | 0,6 |
| South Africa | 7,0 | 5,0 | n/a | 5,6 | 5,6 | n/a | 21,6 | 73,0 | n/a |
| South Korea | 5,4 | 0,8 | n/a | 4,9 | 4,4 | n/a | 135,6 | 119,0 | n/a |
| Spain | 4,1 | 3,4 | 5,8 | 3,4 | 3,6 | 6,0 | 52,9 | 52,8 | 25,6 |
| Sri Lanka | 8,8 | 15,9 | 10,6 | 10,6 | 12,8 | 8,3 | 1,1 | 2,4 | 0,2 |
| Sweden | 4,8 | 5,4 | n/a | 3,6 | 4,1 | n/a | 16,1 | 61,2 | n/a |
| Switzerland | 2,0 | -3,3 | 3,1 | 2,9 | 1,6 | 2,4 | 30,4 | 28,9 | 18,8 |
| Taiwan | 5,9 | -1,3 | 4,1 | 4,8 | 4,3 | 4,8 | 16,4 | 89,2 | 21,6 |
| Thailand | 2,5 | 0,5 | 9,7 | 4,5 | 3,6 | 8,6 | 12,9 | 20,8 | 10,4 |
| Türkiye | 40,5 | 43,8 | 48,1 | 11,8 | 14,4 | 15,3 | 56,3 | 15,6 | 20,0 |
| United Arab Emirates | 7,7 | -0,4 | 11,2 | 4,7 | 5,3 | 6,3 | 12,0 | 3,4 | 16,1 |
| United Kingdom | 7,8 | 4,5 | 4,8 | 3,4 | 3,4 | 3,3 | 176,1 | 397,2 | 14,9 |
| United States | 5,7 | 4,9 | 9,0 | 4,1 | 3,9 | 6,9 | 1 799,6 | 1 103,9 | 2 437,4 |
| Vietnam | 10,6 | 16,4 | n/a | 9,5 | 7,0 | n/a | 7,3 | 10,2 | n/a |

[1] 2025 exchange rates.

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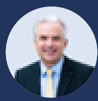

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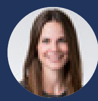

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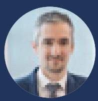

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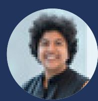

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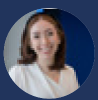

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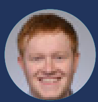

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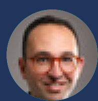

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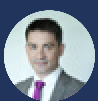

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Allianz Trade

# Too hot to grow

## The economic costs of extreme heat

28 May 2026

Allianz Research

# Content

# Executive Summary

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- **Extreme heat is emerging as a structural economic risk, with Europe highly exposed.** Heat stress events have multiplied sevenfold since the 1980s while the average death toll per event has risen fivefold. That share partly reflects measurement: vital registration and excess-mortality surveillance are far more developed in Europe than in much of Africa and South Asia, where heat deaths go largely uncounted. But genuine structural vulnerability is also at work – ageing populations, dense urban building stock designed to retain warmth and severely underdeveloped cooling infrastructure, with air-conditioning penetration averaging 19% across Europe against roughly 90% in the US.

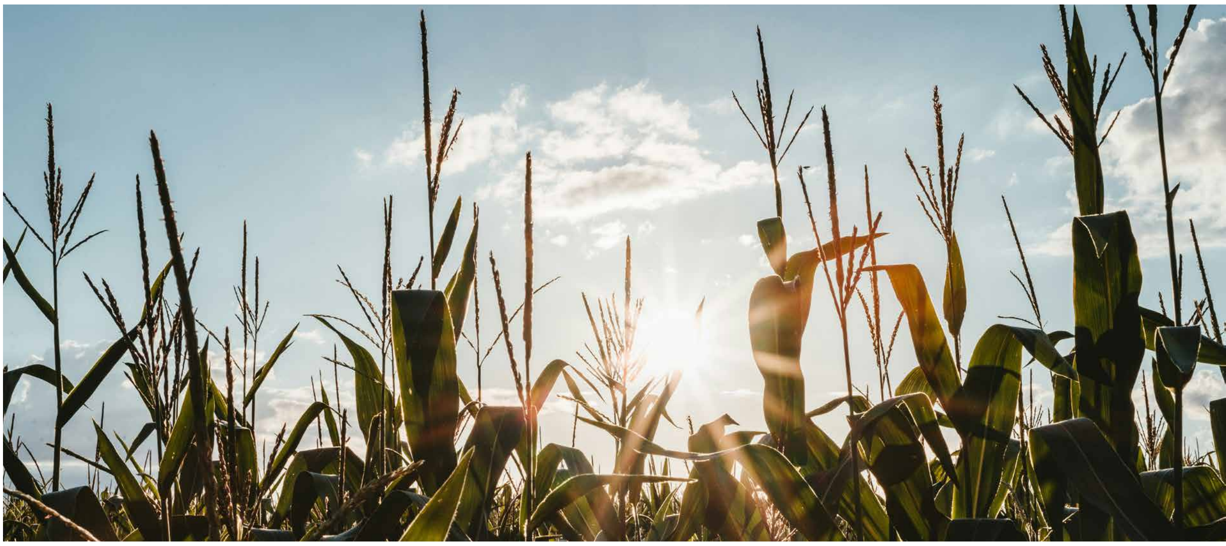
- **The economic transmission of heat stress is non-linear, with a critical threshold around 30°C beyond which productivity losses intensify sharply.** Below this level, warming reduces heating costs and is associated with modest productivity gains. Above this level, the relationship reverses and both channels worsen with each additional degree. The dominant effect operates through labor: output per hour declines by approximately USD1.3 (constant PPP, ~3% of mean hourly output in our 2014-2024 sample) for every degree across the 30-35°C range. Wage adjustments follow productivity with a lag, so the short-run cost falls disproportionately on firm profitability before gradually transmitting to household income and consumption. A second, smaller channel runs through energy: consumption rises by around 1.2% per degree, raising firms' input costs at exactly the temperatures where labor productivity is falling.

- **To gauge the macroeconomic stakes, we construct a stress scenario in which the five hottest years observed in each country between 2014 and 2024 are replayed in ascending order over 2026–2030 – the fifth-hottest year in 2026, the fourth in 2027 and so on, culminating in the country's hottest year on record in 2030.** Under this trajectory, cumulative implied GDP losses (2026 – 2030) could reach 5–7% for the most exposed economies: USD240bn for France, USD354bn for Japan, USD147bn for Italy, USD131bn for Germany and USD120bn for Spain. More consequentially for long-run growth, in such a scenario the decline in fixed capital formation systematically exceeds consumption losses, reaching 8% on average across affected countries: As heat compresses expected returns on capital, investment falls, reducing future productive capacity in a self-reinforcing drag. Moreover, stagflationary dynamics should be expected, with rising prices alongside rising unemployment, placing monetary authorities in a binding trade-off that is especially acute in the Eurozone, where a single policy rate must serve economies with sharply diverging climate exposures.

- **The fiscal consequences fall most heavily on the economies least able to absorb them.** The loss of economic output due to heat reduces tax revenues: Estimated annual losses would reach 1.8% in France, 1.3% in Italy and Spain and 0.7% in Germany – partly because progressive tax systems mean revenues tend to fall faster than output itself, amplifying the fiscal drag beyond the headline GDP loss. Simultaneously inflation-indexed transfers, healthcare costs and emergency infrastructure repair raises public expenditure. Fiscal balances deteriorate by around 0.5% of GDP annually on average. Italy and Spain risk breaching the Maastricht deficit ceiling (again) once heat-related pressures are incorporated. France, already carrying a projected deficit of –4.9% of GDP, faces additional heat-related pressure of 2.2%.

- **Insured losses remain a small fraction of total damages, reflecting a structural mismatch between what heat destroys and what conventional insurance was designed to cover.** In 2022, total climatological losses in Europe reached EUR46bn while the insured share rose only marginally. Most heat damage accumulates through excess mortality, lost working hours, healthcare-system pressure and infrastructure stress – channels that indemnity contracts are not designed to handle. This makes extreme heat harder to insure than other climate risks because losses are widespread and are often indirect – like lower productivity or health impacts – making them difficult to measure and price. Closing the protection gap is therefore as much a product-design challenge as a capacity one, and the insurance toolkit is already evolving in response: parametric instruments that pay out on objective temperature or duration thresholds, public–private risk-sharing arrangements for systemic exposures and dedicated public backstops where private capacity cannot reasonably reach.

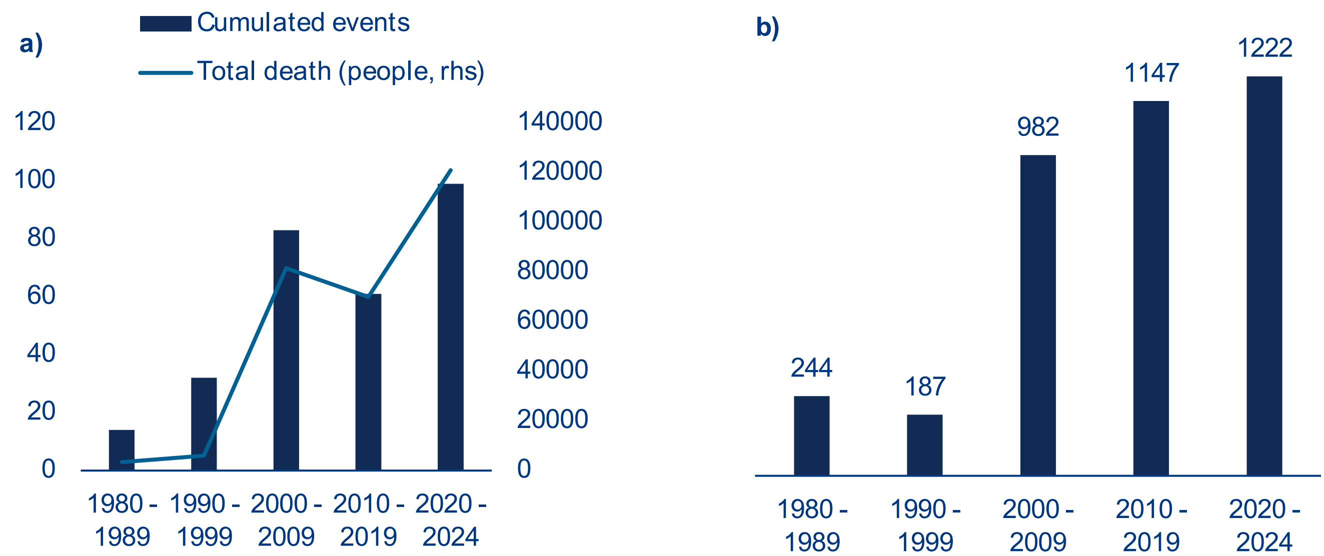
- **Heat-adaptation policy in Europe is built mainly to compensate for losses rather than prevent them.** Following the multi-actor frame of IPCC, closing the gap in Europe requires coordinated action on four fronts: labor regulation, buildings, public finance and households. A workable occupational regime needs binding temperature thresholds, automatic work restrictions when those thresholds are crossed, paid compensation for lost hours and coverage that reaches fixed-term, seasonal and platform workers. No major European economy has all four, and the gap is concentrated on the last: protections were designed around standard contracts and leave the workers most exposed to heat largely outside the regime. Prevention itself is also underused, with shifted hours, partial mechanization and indoor cooling still rare. For buildings, four pieces have to fit together: overheating standards in new construction, mandatory passive cooling in renovation, cooling access for vulnerable households as a social entitlement and grid-adequacy planning that accounts for coincident summer cooling demand and the thermal derating of generators. The revised EU Energy Performance of Buildings Directive delivers the first; the gap is the other three, where indoor temperatures, mortality and peak electricity demand are actually held down. On fiscal architecture, every major European economy has a national adaptation strategy but almost none has translated it into a multi-year budget envelope, so the response defaults to ad-hoc emergency packages – and each episode quietly consumes the fiscal space that ex-ante adaptation would have used to lower the next. The missing layer is households. EU households hold almost EUR40trn in financial assets, including a very large stock of deposits, while much of Europe's housing stock remains poorly adapted to hotter summers. Mobilizing even a small, well-targeted share – through incentives for retrofit, passive cooling and affordable parametric cover – could close part of the gap that public budgets alone cannot reach. But this is not a private-finance solution: the households most exposed are often not those with the greatest liquid savings, so public guarantees, subsidies and distributional safeguards are what turn household wealth into resilience rather than into inequality.



# The economic transmission of heat stress

**From one decade to another, the data are recording more frequent and more lethal heat stress events.** Between 1980 and 1989, only 14 heat-wave events[1] were catalogued globally; the figure more than doubled in the 1990s, jumped to 83 in the 2000s and has already reached 99 in the truncated 2020–2024 period, meaning five years have nearly matched what entire decades produced two cycles earlier (Figure 1a). Mortality has followed an even steeper trajectory: from around 3,400 deaths in the 1980s to more than 121,000 in the first half of the 2020s alone. The most analytically significant metric is the average death toll per event, which has risen fivefold from 244 in the 1980s to 1,222 in 2020–2024. This decoupling between event frequency and per-event lethality indicates that the intensity and duration of individual episodes are increasing faster than societies can adapt (Figure 1b).

**Figure 1:** Increasing exposure to heat stress: a) Evolution of number of events and total death (1980 – 2024); b) Average number of death per heat stress event (1980 – 2024)



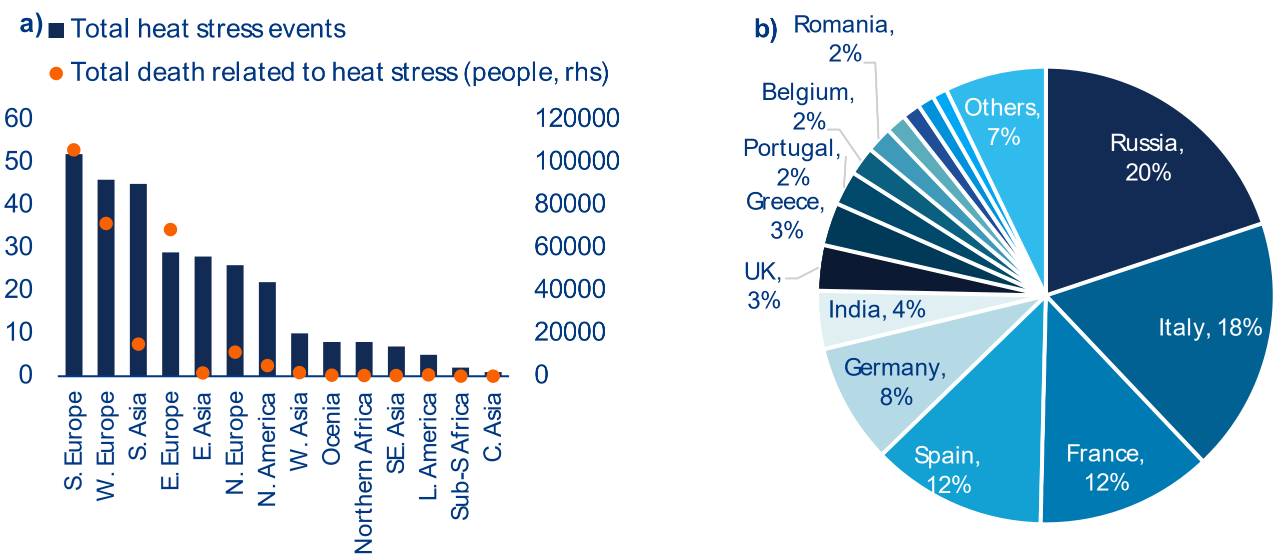

Sources: EM-DAT, Allianz Research

[1] EM-DAT defines a heat wave as a period of abnormally hot and/or humid weather lasting typically two or more days, with location-specific thresholds. Only events with ≥10 deaths, ≥100 people affected, a declared emergency, or an international assistance call are catalogued.

**The cross-sectional distribution of heat stress events and mortality adds a second layer of insight, showing that exposure and impacts do not fully align geographically (Figure 2a).** At the regional level, Southern and Western Europe concentrate the highest number of recorded events (52 and 46 respectively), closely followed by South Asia (45). Yet, this apparent convergence in frequency masks a profound divergence in outcomes. Southern Europe alone accounts for more than 105,000 deaths, while Western and Eastern Europe contribute a further 71,000 and 68,500 respectively. By contrast, South Asia, despite a comparable number of events, records fewer than 15,000 deaths over the same period. This asymmetry becomes even more pronounced when examined at the country level. A limited set of European countries dominates the global mortality profile: Russia (20%), Italy (18%), France (12%), Spain (12%) and Germany (8%) together account for 68% of total deaths. The concentration is too strong to be explained by hazard occurrence alone. Recent valuations suggest that heat-related mortality in European cities already entails welfare losses of EUR192–314 per adult per year, on par with air pollution, with avoided mortality alone worth more than EUR150bn annually.[2] This is only one channel of impact. On the productivity side, heatwaves cost 0.3–0.5% of European GDP in anomalously hot years such as 2003, 2010, 2015 and 2018, and more than 1% in the most exposed southern regions. Losses are projected to rise fivefold by 2060 in the absence of further mitigation or adaptation.[3]

**The pattern in Figure 2b needs to be read carefully: EM-DAT counts reflect reporting practices as much as biophysical exposure.** In hot regions such as the Gulf, India and interior Australia, populations have adapted over time through acclimatization, building design and work practices, so many high-temperature episodes do not register as disaster events, even though heat continues to kill outdoor workers, the poor and the elderly. Europe sits at the opposite end, combining ageing populations with impaired thermoregulation, dense historic urban form, uneven air-conditioning access and stronger vital registration. The European mortality concentration therefore reflects exposure meeting structural vulnerability and more complete death recording, not exceptional hazard frequency.

**Figure 2:** Global distribution of heat stress events: a) Number of events and total death across regions (1980 – 2024); b) Heat stress related death by country (1980 – 2024)



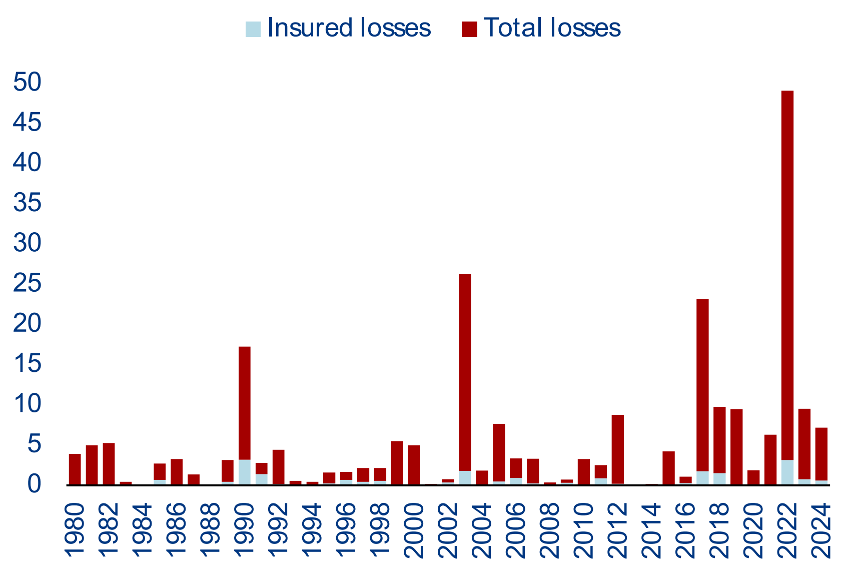

Sources: EM-DAT, Allianz Research

[2] Economic valuation of temperature-related mortality attributed to urban heat islands in European cities | Nature Communications
[3] Current and projected regional economic impacts of heatwaves in Europe | Nature Communications

**Insured and uninsured losses tell a parallel story.** Total losses from climatological hazards in Europe have risen across 1980–2024, with sharp peaks in 2003 and 2022, while the insured share has stayed limited (Figure 3). In 2022, total losses reached around EUR46bn, of which insurance covered only a small fraction. The gap is not simply a matter of market lag; it also reflects what heat actually damages. Compared with storms or floods, extreme heat less often shows up as direct damage to insured assets. Most of its cost appears through excess mortality, lost working hours, healthcare-system pressure and the disruption that follows when grids, railways and water systems are pushed past their design limits. The more conventionally insurable parts, such as direct property damage, heat-amplified wildfire losses, some crop and equipment failures etc., make up only part of the total. EEA data for 1980–2022[4] show that only around 10% of economic losses from heatwaves, droughts, wildfires and other climatological events in Europe were insured, against more than a third for storms, hail and wind. These figures cover direct asset damage only and exclude mortality, healthcare costs and lost productivity.

**Heat is also hard to insure through standard indemnity contracts.** Heatwaves can be defined meteorologically, but turning a temperature episode into a compensable loss is diffuse, lagged and hard to attribute. Losses tend to be highly correlated across large regional portfolios, which limits diversification. And the largest impacts – deaths, morbidity, lost output, public-service stress – are not the kind of damage private indemnity contracts were written for. Heat therefore sits at the boundary of conventional insurability: some asset-based losses can be transferred, but the largest welfare and productivity costs are hard to transfer at scale. Closing the gap will require new instruments, such as parametric cover, public–private risk-sharing and dedicated public backstops, without which much of the cost will keep falling on households, firms and public budgets.

**Figure 3:** Climatological (including heat waves) insured and uninsured losses in Europe (1980 – 2024)




Sources: EEA, Allianz Research

4 Economic losses from weather- and climate-related extremes in Europe | Indicators | European Environment Agency (EEA)

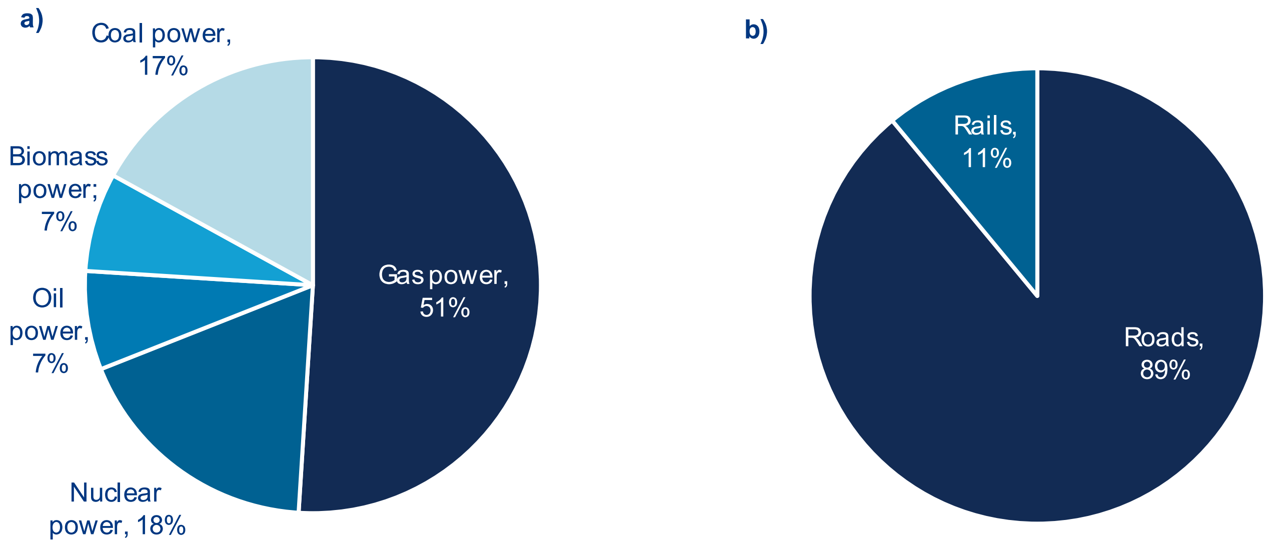
**Heat translates into measurable macroeconomic cost through two channels: additional energy demand and lower labor productivity.**[5] Both relationships are sharply nonlinear, with a critical threshold around 30°C beyond which damages intensify. The remainder of this section estimates the magnitude of each channel using panel regressions covering 35–49 countries over three decades, and quantifies what those estimates imply when each country experiences its recent peak-heat conditions relative to a 1991–2010 climatological baseline. The output of these two channels feeds directly into the global macroeconomic scenario presented in Section 2.

## Energy demand

**The relationship between temperature and energy demand is inherently non-linear, and the structure of Europe's power mix amplifies the risk on both demand and supply sides.** At moderate temperatures, milder conditions reduce space-heating needs and consumption falls; above a threshold, cooling dominates and consumption rises sharply, producing a convex U-shaped response (Auffhammer and Mansur, 2014)[6]. The supply side magnifies the demand-side pressure. A large share of European generation capacity exposed to heat stress relies on thermoelectric technologies – gas (51%), nuclear (18%) and coal (17%) – all of which depend on water availability and cooling efficiency. Elevated air and river temperatures directly constrain these systems, forcing output reductions precisely when cooling-driven electricity demand surges (Figure 4a).

**Recent episodes illustrate the systemic nature of the resulting grid stress, and analogous pressures affect transport infrastructure.** During the August 2020 California heatwave, electricity demand peaked near 47,000 MW while generation capacity was simultaneously constrained, producing the first rolling blackouts in nearly two decades. During the 2019 French heatwave, nuclear output was reduced due to cooling constraints, tightening supply and triggering sharp electricity-price spikes. Heat also degrades transmission networks, lowering the carrying capacity of generators, transformers and power lines, while even non-thermal renewables, in particular solar PV, experience efficiency losses at high temperatures. Transport infrastructure faces analogous physical pressures: road networks account for nearly 89% of heat-exposed transport assets and rail for 11% (Figure 4b), with asphalt softening and rutting under sustained heat and rail tracks prone to thermal expansion, illustrated by the UK in July 2022, when temperatures first exceeded 40°C and rail operators imposed strict speed limits or suspended services entirely.

**Figure 4:** Exposure of critical European infrastructure to heat stress : a) Energy infrastructure; b) Transport infrastructure



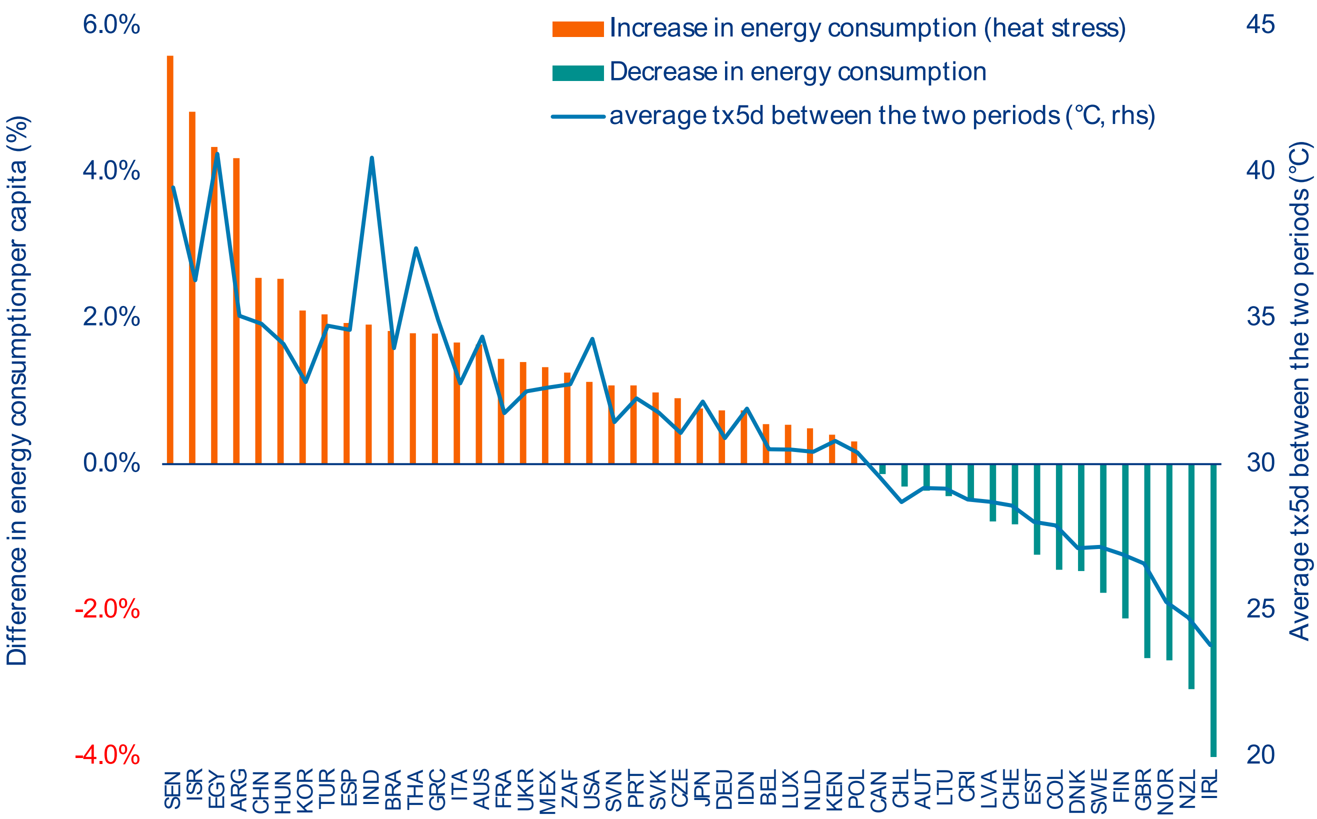

Sources: Forzieri et al. (2018)[7], Allianz Research

[5] What to watch I July 01, 2025 | Allianz
[6] Measuring climatic impacts on energy consumption: A review of the empirical literature - ScienceDirect
[7] Escalating impacts of climate extremes on critical infrastructures in Europe - ScienceDirect

**Our empirical results confirm the non-linear structure and identify a threshold close to 30°C (Appendix 1).** Using the Tx5d indicator, which is the mean daily maximum temperature over the hottest five consecutive days of the year, on an unbalanced panel of 49 countries over 1991–2023, the estimated marginal effect of heat stress on per capita energy consumption is sign-switching and temperature-dependent. At 25°C, a one-unit increase in Tx5d is associated with an approximately 0.6% decline in per capita energy consumption. At 35°C, the relationship reverses: the same marginal increase implies a 1.2% rise. The twofold differential illustrates how disproportionately energy-system stress intensifies once thermal thresholds are crossed.

**The country-level counterfactual reveals a sharply asymmetric pattern, with heat already operating as a structural driver of energy demand across much of the G20 (Figure 5).** Comparing per capita energy consumption during each country's most intense post-2014 heat episode (maximum Tx5d) with its 1991–2010 baseline average, countries already operating at high temperature levels (Senegal, Egypt, China, South Korea, Spain and Brazil) register increases above 4–5%, pushed further into the convex segment of the response. Almost all G20 economies in the tropics see rising energy demand across sectors, while only Europe and Oceania post aggregate reductions, driven by lower residential heating needs in cooler latitudes. The cooling gains are structurally fragile and likely to erode as temperatures continue to rise.

**Figure 5:** Per capita energy demand under peak heat stress (maximum Tx5d after 2014) vs. the 1991–2010 average baseline

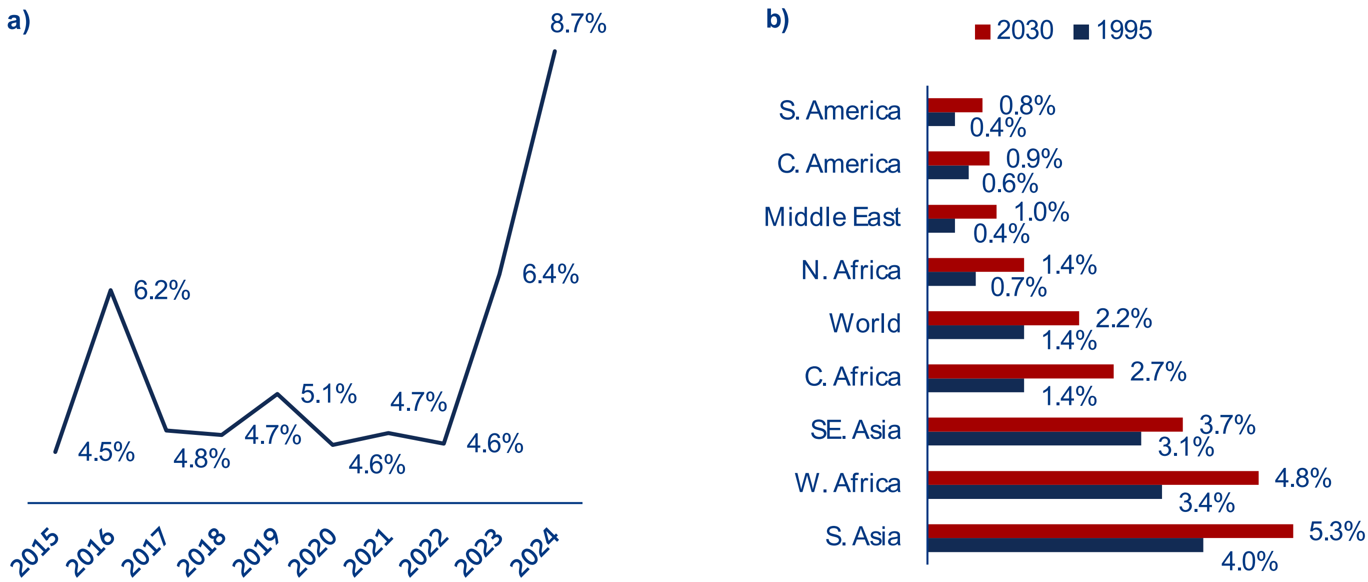



Sources: China National Customs, Allianz Research

## Labor productivity and wages

**Heat stress reduces labor output through physiological, cognitive and sleep-related channels that extend well beyond outdoor work (Figure 6).** At the worker level, sustained exposure above the thermoneutral zone induces cardiovascular strain, dehydration and cognitive impairment, with field studies consistently documenting productivity losses across sectors (Ebi et al., 2021)[8]. The share of global working hours lost to heat stress is projected to rise from 1.4% in 1995 to 2.2% by 2030, with much higher impacts in already-warm regions – 5.3% in South Asia, 4.8% in West Africa – and a near doubling even in traditionally less exposed regions (Figure 6b). Indoor workers are not insulated: night-time temperatures, which are rising faster than daytime averages in many regions, increased total sleep time lost by around 6% over 2020–2024 relative to the 1986–2005 baseline, reaching a record 8.7% in 2024, with up to 12 additional hours of sleep loss per person per year in the most affected locations (Figure 6a). Degraded sleep duration and quality has measurable next-day consequences for cognition, cardiovascular health and decision-making, transmitting heat impacts to indoor as well as outdoor labor.

**Figure 6:** Heat stress and productivity : a) Loss of sleeping time for the period 2015 – 2024, compared to a baseline 1986 – 2005 (%); b) Decline in working hours (%)




Sources: The Lancet Countdown (2025), ILO (2019), Allianz Research

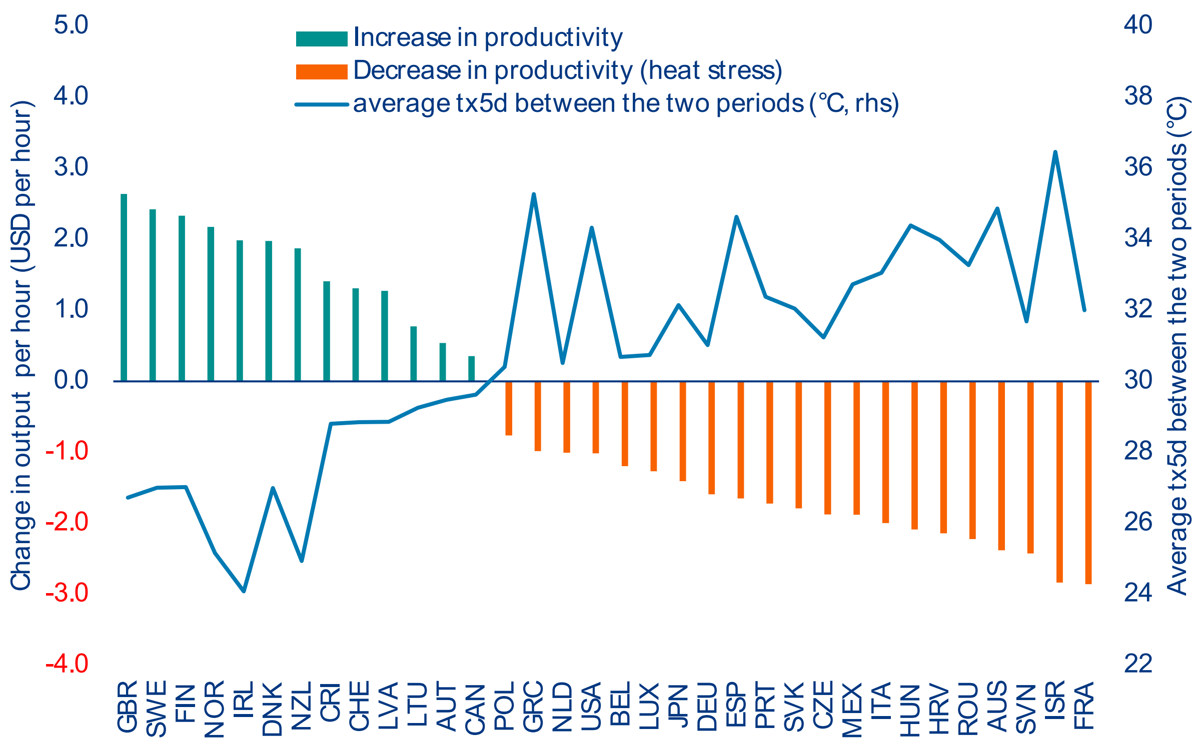
**Our regressions on output per hour reveal a clear threshold effect at 30°C, consistent with the empirical consensus (Appendix 2).** Estimated on a panel of 35 countries over 1997–2023, the coefficient on Tx5d below 30°C is positive and marginally significant (+0.59), consistent with mild productivity gains at moderate temperatures in temperate economies. Above 30°C, the interaction coefficient turns negative and statistically significant (–1.30), implying that each additional degree of heat beyond the threshold reduces output per hour by approximately USD1.30 (constant PPP, representing 3% of average output per hour in our sample between 2014 and 2024). Applied to the country-level counterfactual, where each country experiencing its maximum Tx5d observed over 2014–2024, against its 1991–2010 average, France, Slovenia, Australia and Italy emerge as the most affected, with output per hour declining by up to USD 1–3 in the most exposed cases. The UK, Finland, Ireland, New Zealand and Canada, all below the 30°C threshold, exhibit modest productivity gains (Figure 7a).

**The wage response materializes with a lag, reflecting institutional rigidities in labor markets (Appendix 3).** Using a one-year lag specification on a panel of 33 countries over 1995–2023, the relationship follows an inverted U-shape: below 30°C, higher heat stress in year t is associated with a modest acceleration of wage growth in year t+1 (approximately +0.2% at 25°C); above 30°C, the relationship reverses (approximately –0.2% at 35°C). The lag is consistent with the combination of collective-bargaining frictions, nominal wage rigidities and minimum-wage floors, all of which delay the transmission of productivity losses into labor compensation. The counterfactual maps the same threshold logic onto observed peak heat: Israel, Bulgaria, Australia, Hungary, Romania and Spain register wage declines of approximately 0.7–1.3%, while Ireland, Norway, the UK, Sweden and Finland post wage gains of 0.3–0.7% (Figure 7b).

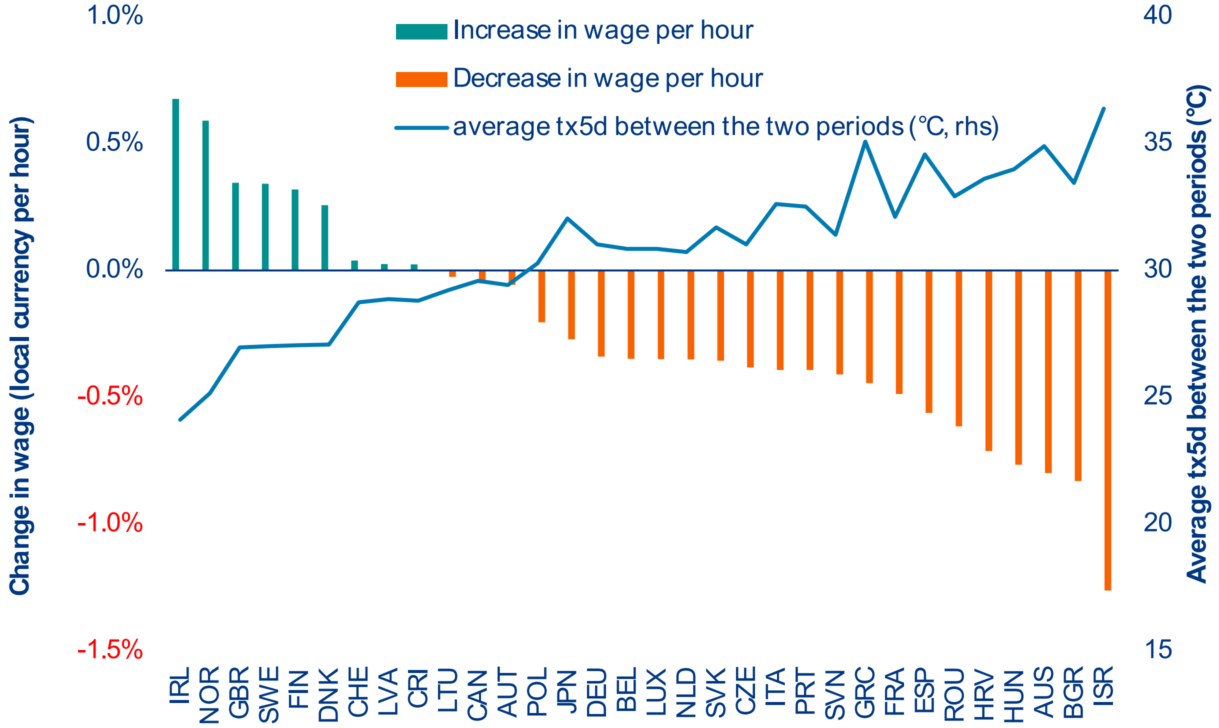
**The distributional implication is structurally important and motivates the macroeconomic scenario that follows.** In the short run, the cost of heat stress falls disproportionately on firm profitability as wage rigidities prevent compensation from adjusting in line with productivity. Over time, however, that wedge closes: real wages decline, household purchasing power erodes and consumption weakens, precisely in the economies already most exposed to rising temperatures. The two channels estimated in this section, additional energy demand and lower labor productivity, transmitted partially and with a lag into wages, are the inputs to the global macroeconomic simulation in the next section, which quantifies the general-equilibrium consequences in terms of GDP, prices, employment and fiscal balances.

**Figure 7:** The change in labor productivity under peak heat stress (maximum Tx5d post-2014) relative to the 1991–2010 baseline, measured as (a) output per hour in USD and (b) wage per hour (%).

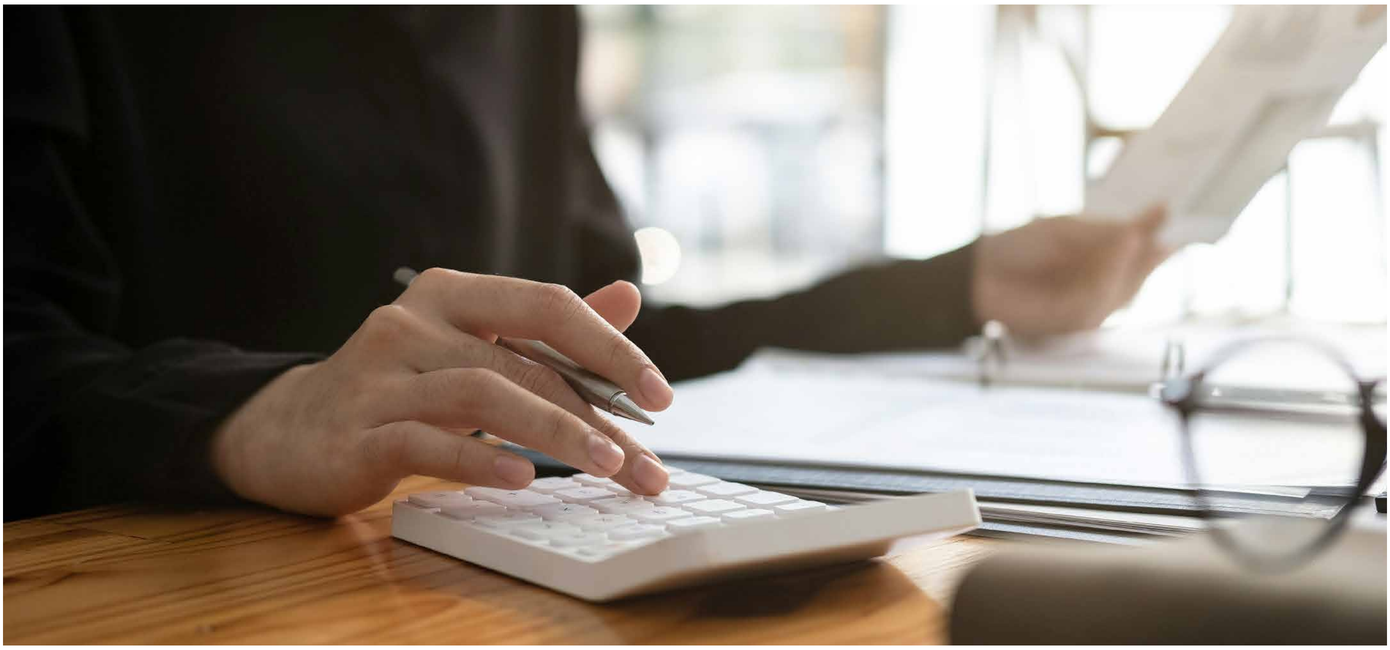
**a)**



b)




Source: Allianz Research



# The macroeconomic implications of heat stress

**To quantify the macroeconomic cost of these effects, we translate the empirical estimates into a structural scenario using the Oxford Economics Global Economic Model.** The counterfactual exercises described above provide country-specific measures of the productivity and energy demand impact of heat stress, but they remain partial-equilibrium estimates, as they capture the direct effect of temperature on each variable in isolation, without accounting for the general-equilibrium feedbacks that propagate through the broader economy. As the recent climate-macroeconomics literature has increasingly recognized, reduced-form empirical estimates of climate damages, while essential for identification, cannot capture the full macroeconomic cost because they omit the general-equilibrium feedbacks, through trade, prices, investment and fiscal balances, that amplify the initial shock.[9]

**To capture these interdependencies, we impose three simultaneous shocks on the model, each calibrated directly from our regression results.** The first is a shock to potential output[10], reflecting the estimated decline in output per hour worked under extreme heat conditions. Entering the productivity loss through potential output ensures that the shock operates on the supply side of the economy, reducing the economy's productive capacity. The second shock targets wages and salaries per employee[11], capturing the lagged and partial transmission of productivity losses to labor compensation that our analysis identifies. The exogenous wage shock therefore captures the residual labor market adjustment that our regressions document beyond the model's endogenous response. The third shock targets energy demand[12], reflecting the additional per capita energy consumption induced by heat stress. This channel translates higher cooling needs into increased household and firm expenditure, higher energy imports for net-importing economies and upward pressure on consumer prices.

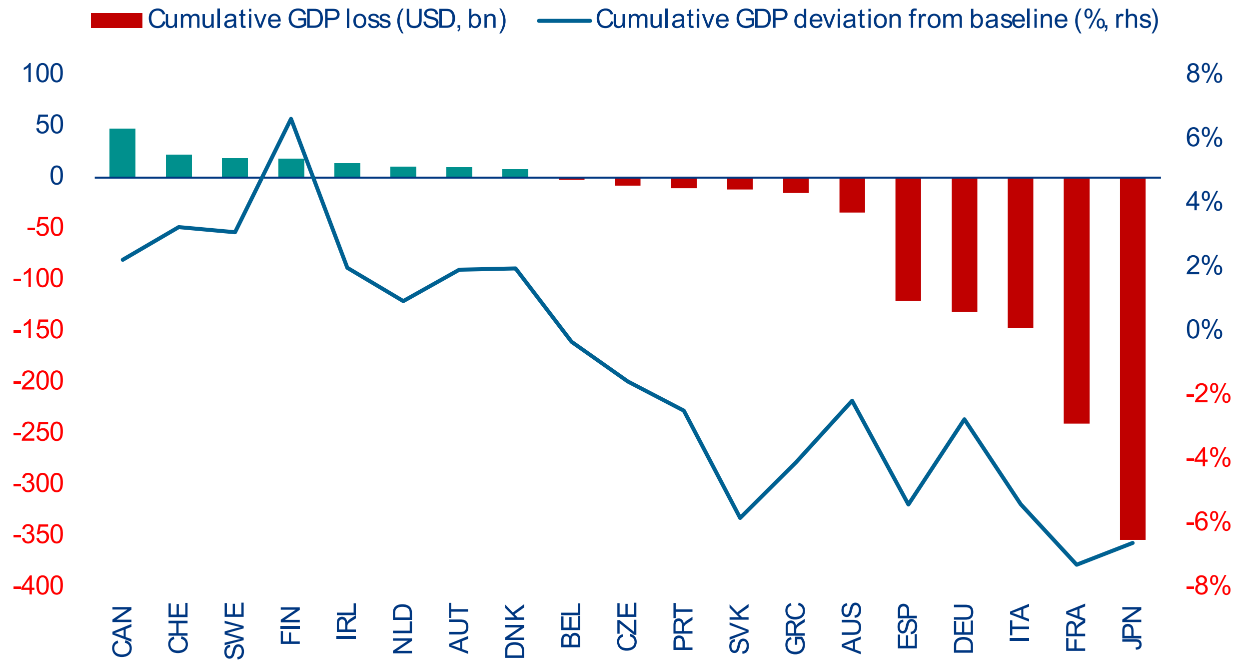
**To assess the macroeconomic cost of heat stress, we construct a counterfactual scenario and compare it against a baseline projection.** The baseline simulates economic outcomes over the period 2026–2030 under climatological normal conditions, defined as the average Tx5d per country observed over the 1991–2010 reference period. This baseline represents the economic trajectory each country would follow if heat stress remained at its historical norm. Against this baseline, we impose a stress scenario in which each country experiences,

[9] Climate Change through the Lens of Macroeconomic Modeling | NBER
[10] YHAT in Oxford Economics
[11] PEWFP in Oxford economics
[12] DENERGY in Oxford economics

consecutively from 2026 to 2030, five years with the most important heat waves recorded during the 2014–2024 period, ranked by Tx5d. This is not a climate scenario based analysis but a historically grounded exercise: it asks what the macroeconomic consequences would be if the extreme heat conditions already observed in the recent past were to recur over a sustained five-year window.

**The results reveal a clear divide across economies (Figure 8).** Countries where average Tx5d remains below the 30°C threshold under the stress scenario – Canada, Switzerland, Sweden and Ireland – experience cumulative GDP gains of 1% to 3% relative to the historical climate baseline, leading to absolute gains ranging from USD10bn to USD47bn.[13] These gains reflect the net balance of reduced heating costs and marginally improved working conditions, benefits that are structurally fragile and likely to erode as temperatures continue to rise. Beyond this group, the transition into net losses is sharp. France, Japan, Slovakia, Spain and Italy register the highest cumulative GDP losses of 5% to 7% below baseline, reflecting the direct productivity drag that our shocks impose. The largest absolute losses, however, are concentrated in the biggest economies in the sample: Spain (USD120bn), Germany (USD131bn), Italy (USD147bn), France (USD240bn) and Japan (USD354bn).

**Figure 8:** Cumulative GDP loss from sustained extreme heat stress (2026–2030) relative to historical climate baseline



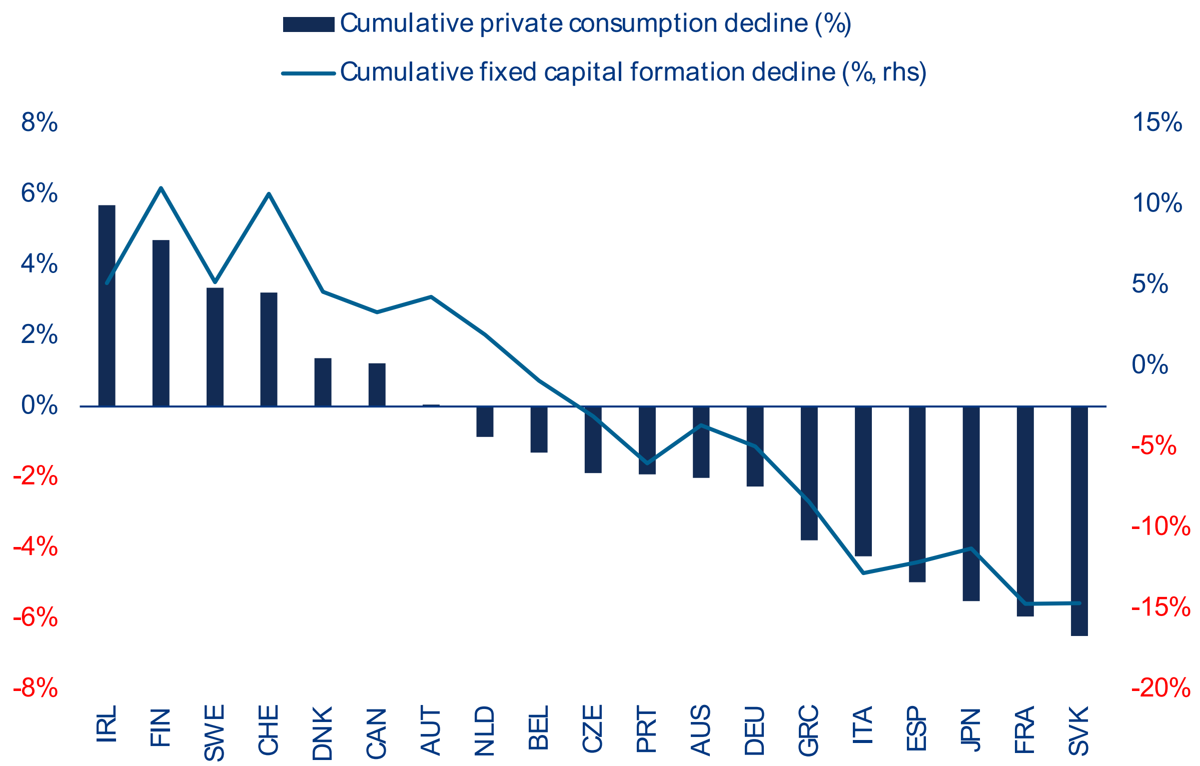

Sources: Oxford economics, Allianz Research

[13] Finland is an outlier, with a potential cumulative GDP gain of 7% under sustained extreme heat stress

**Decomposing the GDP impact into its demand-side components reveals that the macroeconomic cost of heat stress operates through two reinforcing channels: weaker household consumption and lower investment leading to a long-run decline in fixed capital formation (Figure 9).** In countries where moderate warming produces net economic benefits – Ireland, Finland, Sweden, Switzerland and Denmark – both private consumption and fixed capital formation rise relative to baseline, with investment gains generally exceeding consumption gains. The pattern inverts sharply for heat-exposed economies, and here the most striking feature of the results is the systematic amplification through investment. In virtually every country where the heat stress scenario produces economic losses, the decline in fixed capital formation exceeds the decline in private consumption, often by a wide margin. Portugal sees a consumption decline of 1.9% but an investment decline of 6.0%. Italy registers consumption losses of 4.2% against an investment collapse of 12.8%. France and the Slovak Republic, the two most affected economies, experience investment declines of 14.7% alongside consumption losses of 5.9% and 6.5% respectively. This wedge between the consumption and investment response reflects a powerful amplification mechanism: as heat stress reduces potential output and compresses firm margins, the expected return on new capital falls, discouraging investment. Lower investment in turn reduces future productive capacity, reinforcing the initial productivity shock and creating a self-reinforcing drag on the economy's growth trajectory. This asymmetry between the investment and consumption response is consistent with Casey et al. (2022)[14], who show that standard climate-economy models systematically understate investment damages because climate change disproportionately affects capital-intensive sectors exposed to outdoor heat.

**The consumption response, while smaller in percentage terms, is nonetheless consequential.** It captures the combined effect of lower real wages, which our wage shock imposes directly, and higher energy expenditure, which erodes household purchasing power. In the most affected economies, cumulative consumption declines of 4% to 6.5% compared to the baseline, representing a material reduction in living standards over a five-year horizon, with implications for domestic demand, retail activity and ultimately tax revenues.

**Figure 9:** Impact of sustained extreme heat stress on private consumption and fixed capital formation, 2026–2030 (cumulative deviation from baseline)




Sources: Oxford economics, Allianz Research

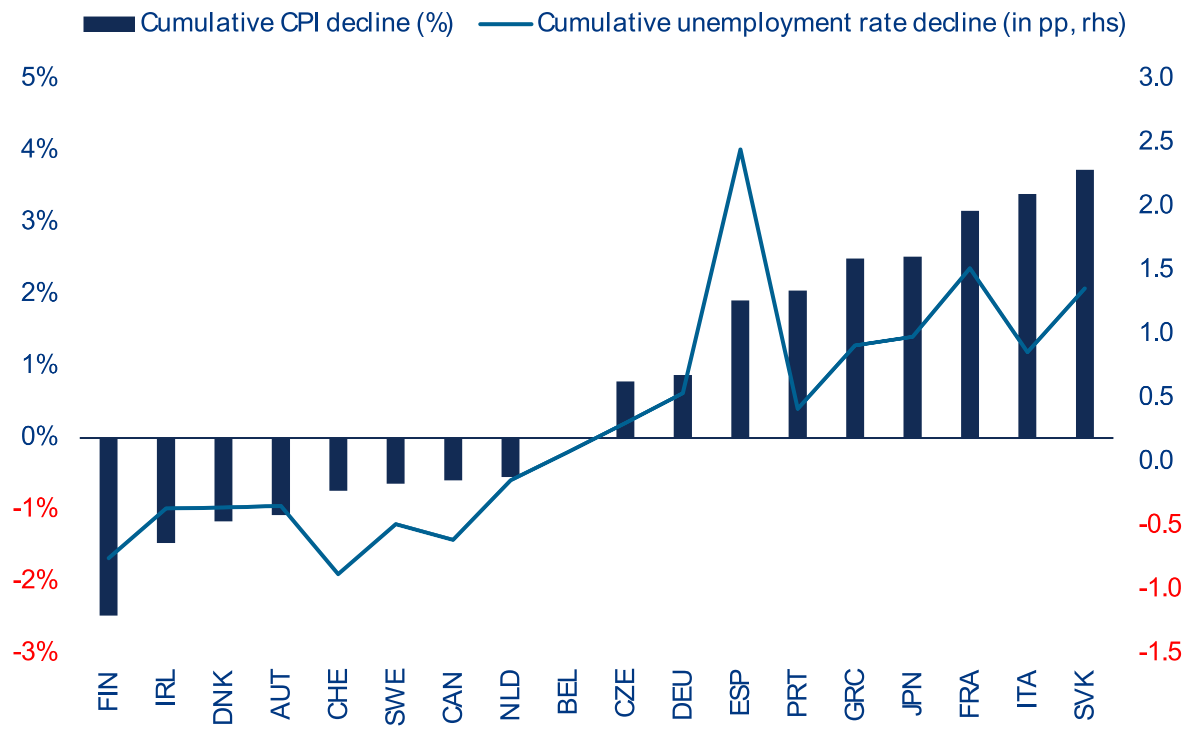
[14] Understanding climate damages: Consumption versus investment - ScienceDirect

**The joint behavior of consumer prices (CPI) and the unemployment rate under the stress scenario reveals the stagflationary nature of the heat stress shock, a finding with direct implications for monetary and fiscal policy.** In cooler economies – Finland, Ireland, Denmark, Austria and Switzerland – both indicators improve relative to the baseline: prices fall by 0.5% to 2.5% cumulatively, while unemployment declines by 0.14 to 0.88pp. In these countries, the modest productivity gains associated with below-threshold heat stress expand supply, easing price pressures and strengthening labor demand simultaneously.

**The picture reverses as heat exposure rises.** For hotter countries in our samples – the Czech republic, Germany, Spain, Portugal, Greece, Japan, France, Italy and Slovakia – both prices and unemployment move in the same adverse direction, which is a defining characteristic of a negative supply-side disturbance. Spain registers a cumulative CPI increase of 1.9% alongside a 2.45pp rise in unemployment. France, Italy and Slovakia experience price increases of 3.2% to 3.7% combined with unemployment increases of 0.86pp to 1.51pp. This co-movement of rising inflation and rising unemployment is the hallmark of stagflation, and it stands in sharp contrast to the pattern produced by a standard demand shock, where the two variables move in opposite directions along the Phillips curve. In the New Keynesian framework, Blanchard and Galí (2007)[15] show that such a trade-off arises when real imperfections, such as wage rigidities, prevent the economy from adjusting smoothly to supply-side shocks. The heat stress shock in our simulation operates through precisely this type of supply-side channel: reduced potential output pushes up unit costs and prices, while simultaneously contracting wages and salaries.

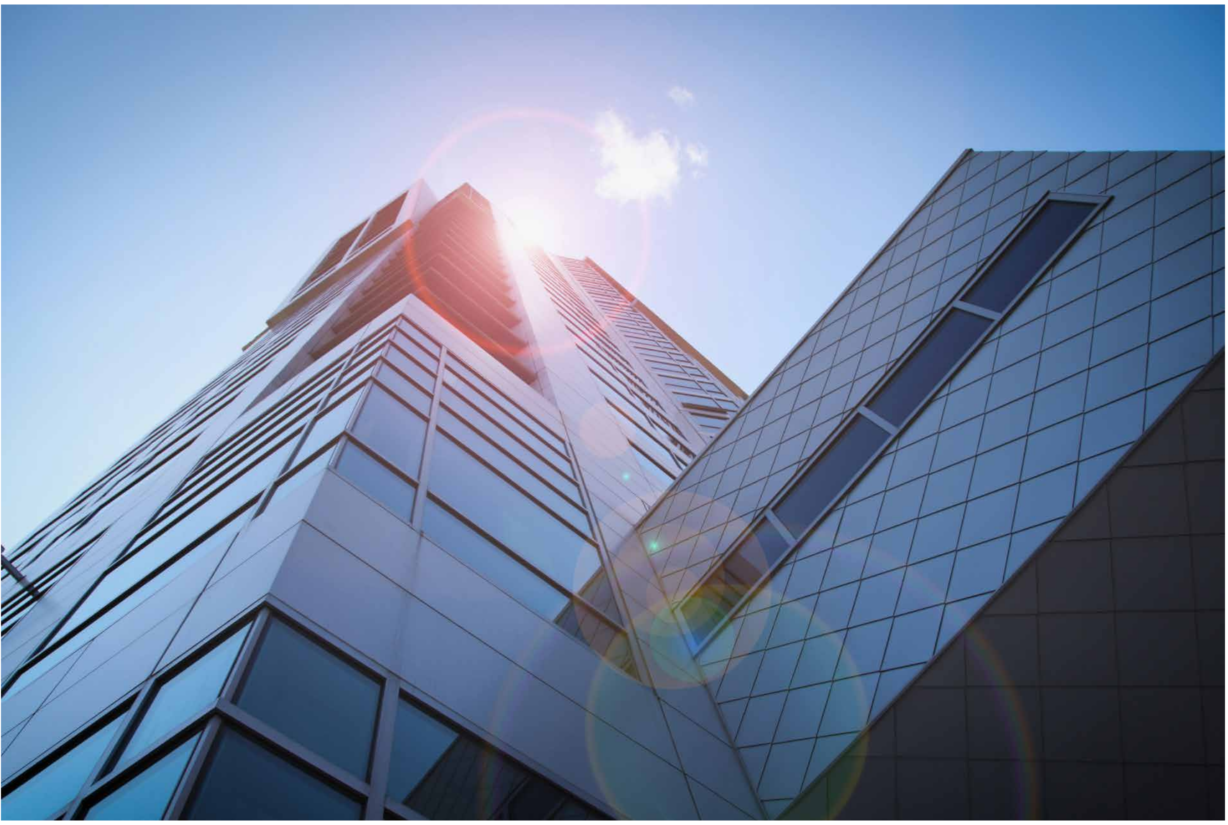
**The result of such heat stress scenario is a policy environment in which conventional stabilization tools face a binding tradeoff: tightening monetary policy to contain inflation would deepen the employment losses, while loosening policy to support the labor market would accommodate further price increases.** This dilemma is especially acute in the Eurozone context, where a single monetary policy must simultaneously address economies experiencing positive supply effects from moderate warming, such as Finland and Austria, and economies facing stagflationary pressures from extreme heat, such as Germany, France, Italy and Spain, a form of climate-induced macroeconomic divergence that existing policy frameworks are not designed to manage.

**Figure 10:** Consumer prices and labor market response to sustained extreme heat stress, 2026–2030 (cumulative deviation from baseline)




Sources: World Bank, Microsoft, Allianz Research

[15] Real Wage Rigidities and the New Keynesian Model - BLANCHARD - 2007 - Journal of Money, Credit and Banking - Wiley Online Library

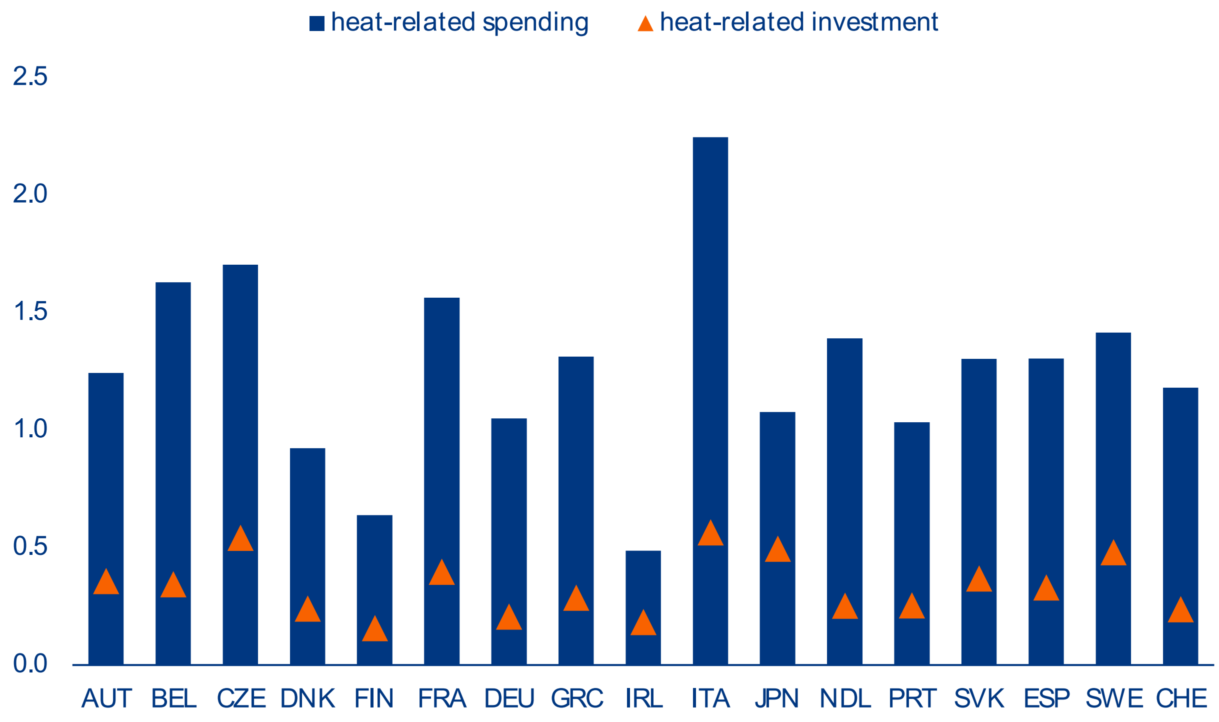


# The fiscal burden

**Even though it already receives about 50% of climate-related investments, extreme heat puts a strain on public finances by cutting revenues and raising adaption costs.** Heat reduces GDP and productivity, which lowers tax collection, while also disrupting supply chains and energy systems, which further weakens revenues. Estimated annual revenue losses are -1.8% in France, -1.3% in Italy and Spain, -1.0% in Greece and -0.7% in Germany – higher than estimated GDP losses because progressive tax systems mean revenues tend to fall faster than output itself, amplifying the fiscal drag beyond the headline GDP loss. At the same time, public spending rises. Governments face higher inflation-linked costs (e.g. wages, pensions and social benefits) as well as heat-specific expenditures (e.g. healthcare, energy subsidies, disaster response, infrastructure repair and social protection). Demand for emergency investments in cooling infrastructure also increases. In Europe, heat-related spending accounts for around 50% of climate-related public expenditure (2.1% of GDP), but this figure varies widely: 40% in Austria, 51% in Germany, 58% in France, 63% in Spain and 65% in Italy. Flooding accounts for about 35%, with the remainder going to wildfires, drought and other areas like biodiversity. As a percentage of GDP, heat-related spending ranges from 2.2% in Italy to 1.6% in France, 1.3% in Spain, 1.1% in Germany and 0.5% in Ireland (Figure 11). Adaptation to heatwaves accounts for a significant proportion of more than a quarter of heat-related budgets, primarily for cooling and urban greening measures. However, overall investment in heat resilience is still lacking to provide for a heat proof society.[16]

[16] From invisible to investible: An investment taxonomy for climate adaptation | Allianz

**Figure 11:** Heat-related spending and investment, in % of GDP 2023




Sources: OECD, Allianz Research

**On average, the fiscal balances of European countries could deteriorate by around 0.5% of GDP annually due to heat stress, creating a structural tension between the fiscal damage caused by heat and the investment required to adapt to it.** A rapidly growing literature has established that climatic risks impose economically significant costs on public finances[17], operating through both the erosion of tax revenues as climate damages reduce economic output and the increase in public expenditure as governments face rising costs for existing commitments and new demands for adaptation. Our results confirm both channels operating simultaneously. On the revenue side, the productivity and wage shocks documented in our simulations translate directly into lower tax receipts: economies producing less and paying lower real wages generate less income tax, social security contributions and value-added tax revenue. On the expenditure side, the stagflationary character of the heat shock through rising prices combined with contracting output, inflates the cost of government programs indexed to prices or wages. The share of public spending subject to such indexation varies considerably across our sample, from approximately 20% in Ireland to 58% in Finland, creating heterogeneous fiscal exposure to the same inflationary impulse.

**We identify three groups of countries based on the interaction between heat-related fiscal pressure and existing budgetary constraints.** The first group comprises countries where this tension is most acute: heat-related fiscal pressure is significant and public finances are already strained exceeding the Maastricht deficit criterion of –3%. France is the most exposed case. Heat stress could worsen an already high expected fiscal deficit of –4.9% (average 2025-2030) by an additional 2.2% of GDP. At this level, fiscal space for discretionary adaptation spending is severely compromised. Germany, with an estimated deficit of –3.6% and additional heat-related pressure of 0.9%, faces a tighter version of the same constraint. Italy's fiscal deficit of -3.1% could worsen by an additional 1.9% of GDP while Belgium, where 0.2% of additional pressure meet a deficit of –4.0%, completes this group

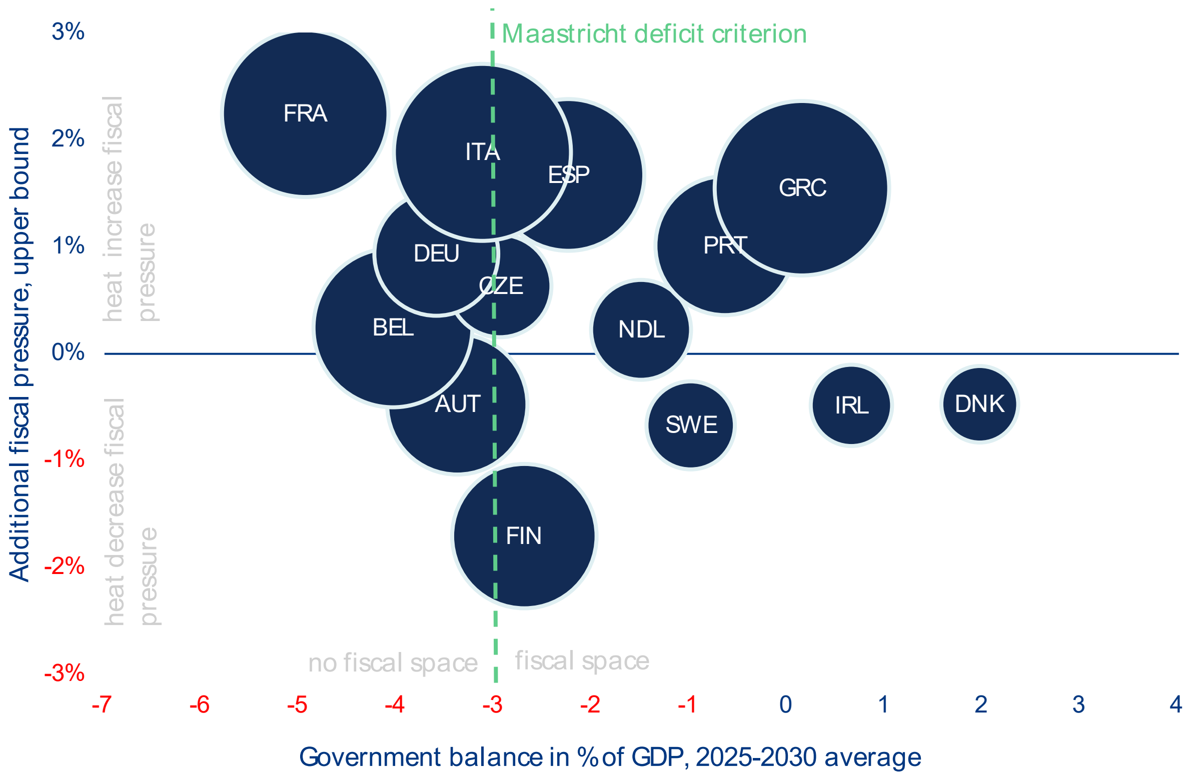
[17] Climate Change Impacts on Public Finances Around the World | Annual Reviews

in the upper quadrant (Figure 12). For these economies, the fiscal damage from heat is actively eroding the budgetary room from which the adaptation response must be financed. Considering in addition the projected debt burden of a country also makes clear that the heat-related fiscal pressure on countries like France (121% of GDP on average between 2025 and 2030, bubble size), Italy (136% of GDP) or Belgium (109% of GDP) is more worrying than for countries like Germany with a projected debt burden of about 66%.

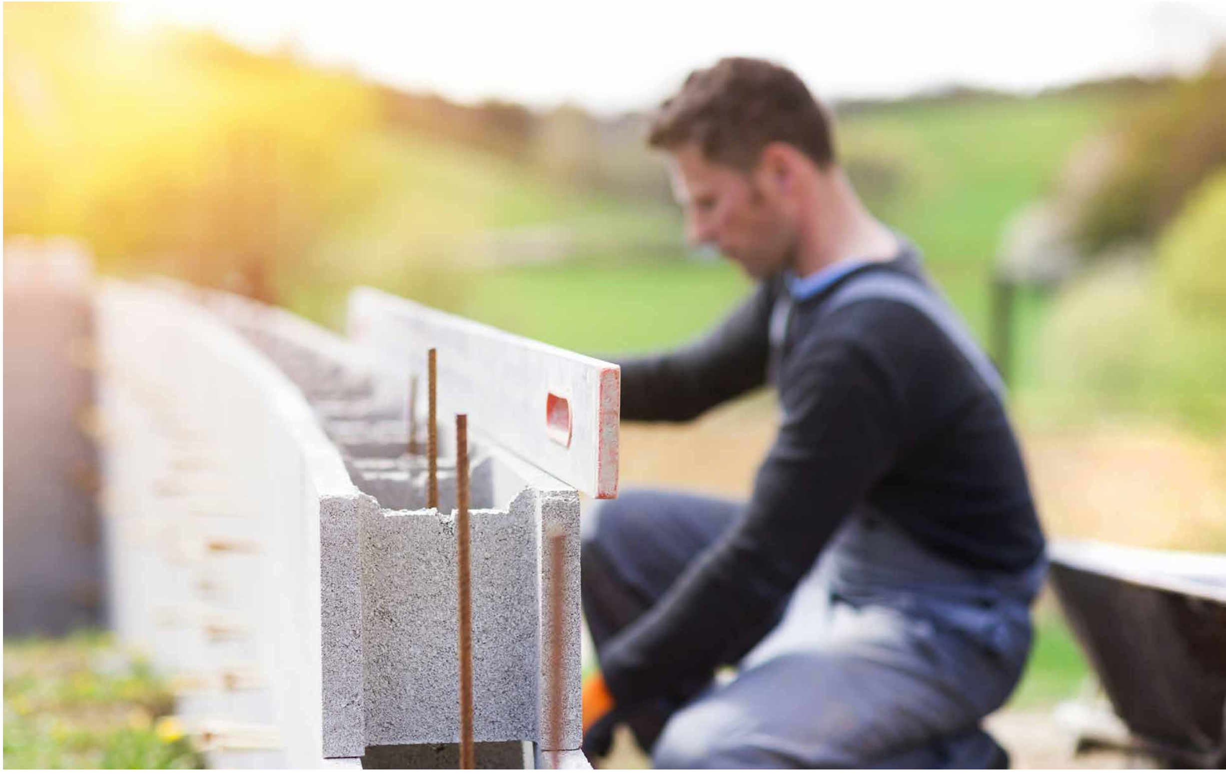
**The second group includes countries where the fiscal deficit still provides some room but where heat-related pressures from lower growth, higher inflation and adaptation investment needs are large enough to threaten the Maastricht threshold.** Italy and Spain, with estimated deficits of –2.8% and –2.4% respectively, could clearly breach the –3% criterion once heat-related pressures of 1.9% and 1.7% of GDP are factored in. The Czech Republic faces a similar dynamic, with 0.6% of heat pressure meeting a deficit hovering just above –3%. Other countries in this group, such as Greece (1.5% of heat pressure), Portugal (1.0%) and the Netherlands (0.2%), have more fiscal room and are thus less immediately constrained, but the trajectory might point toward increasing pressure.

**The third group, which contains cooler regions such as the Nordic countries, may experience an improvement in their fiscal positions as moderate warming boosts economic activity through reduced heating costs and marginally improved working conditions.** For these countries, the fiscal space created by a temporarily favorable climate position should be used for preventive adaptation investment, including updating building codes, establishing heat-health action plans, introducing workplace heat protocols, before the temperature distribution shifts further. Preventive action taken now, when fiscal conditions permit, yields substantially higher returns than remedial action taken later under fiscal constraint.

**Figure 12:** Average fiscal pressure due to heat in % of GDP (annually between 2026 and 2030) and government balance in % of GDP (2025-2030 average)




Sources: OECD, COFAS, Oxford economics, Allianz Research. Note: Bubble size indicates the projected debt burden by country 2025-2030 on average. These figures take into account the average change in inflation due to heat between 2026 and 2030, the share of indexed spending by country and heat-specific expenditure as a share of total climate-related public expenditure



# From exposure to resilience: the three-pillar adaptation plan

## Pillar 1. Protecting workers and productivity

**The labor-productivity exposure shows that above 30°C, each additional degree of Tx5d reduces output per worker-hour by approximately USD1.30.** Globally, without a tailored adaptation strategy, working hours lost to heat stress are projected to rise from 1.4% in 1995 to 2.2% by 2030 (ILO 2019). Four regulatory mechanisms shape how much of this will be mitigated: quantitative temperature thresholds, automatic work-restriction or suspension triggers, compensation for hours lost and contractual extension to fixed-term, seasonal, self-employed and platform workers.

**No major European economy currently pulls all four, and the open cells cluster on the dimensions most relevant to outdoor and non-standard workers (Table 1).** Spain comes closest, combining binding indoor thresholds (Royal Decree 486/1997) with up to four days of paid adverse-weather leave (Royal Decree-Law 8/2024), though the indoor focus and standard-contract design leave outdoor and non-standard workers partially exposed. France introduced procedural employer obligations from July 2025 but stopped short of quantitative thresholds or compensation. Germany's ASR A3.5 sets graduated thresholds at 26°C, 30°C and 35°C with a recognized compliance route under workplace-safety law, but does not by itself trigger automatic work restrictions. Finland regulates exposure time above 28°C through binding work-rest limits. The UK and Ireland rely on general workplace-safety duties.

**Table 1:** Coverage matrix — occupational heat-stress regulation across selected economies, scored on four mechanism dimensions.

| Country / Economy | Quantitative threshold | Automatic work restriction / suspension trigger | Compensation for lost time | Extension to non-standard workers |
|---|---|---|---|---|
| **Spain** | ● indoor | ◐ adverse-weather trigger | ● RDL 8/2024 | ◐ standard contract bias |
| **France** | ○ | ◐ procedural only | ○ | ○ |
| **Germany** | ● ASR A3.5 graduated | ○ compliance route only | ○ | ○ |
| **Finland** | ● 28°C / 33°C | ● work-rest limits | ○ | ○ |
| **Italy** | ○ | ◐ protocols | ◐ wage support for heat stoppages | ○ |
| **UK & Ireland** | ○ | ○ | ○ | ○ |
| **Slovenia** | ◐ indoor reported | ○ | ○ | ○ |
| **Australia** | ○ no national heat-specific mechanism | ○ | ○ | ○ |
| Legend: ● fully covered, ◐ partial or conditional, ○ not covered | | | | |

Source: Allianz Research

**Read in this format, the gap is not that most countries are silent, it is that no country closes all four cells, and the open cells concentrate where exposure is highest.** Of the four economies most affected in the output-per-worker-hour counterfactual (France, Italy, Slovenia and Australia) none appears to combine comprehensive binding heat thresholds, automatic work-restriction rules and broad compensation coverage. France has recently strengthened procedural obligations; Italy provides some wage-support channels during extreme heat, recognized for temperatures above 35°C and, in some cases, below it depending on perceived temperature and work conditions; Australia relies mainly on general work-health-and-safety duties rather than a national heat threshold and Slovenia's coverage should be treated separately depending on whether the analysis focuses on indoor thresholds or outdoor exposure. Several EU countries have no explicit heat-specific workplace provision. The most consistent open cell across all groups is contractual: compensation provisions, where they exist, are built around standard employment relationships; they may cover fixed-term and seasonal employees in principle, but access is weaker in practice, while self-employed and platform workers are often excluded or only indirectly covered. The cost of adaptation therefore falls disproportionately on the workers least able to bear it.

## Pillar 2. Cooling buildings and stabilizing the grid

**The exposure is two-sided and must be addressed simultaneously: poorly adapted buildings on the demand side, where indoor temperatures rise and cooling demand and heat mortality follow, and thermoelectric generation on the supply side, which derates under heat stress precisely when cooling demand peaks.** Four regulatory levers shape how much of this is mitigated: overheating thresholds in building codes, mandatory passive cooling in renovation, cooling access as social policy and summer adequacy planning that models coincident cooling demand and supply derating. No major European economy currently pulls all four, and the gaps cluster on renovation, social-policy access and adequacy planning (Table 2). The revised EPBD (Directive 2024/1275), with a transposition deadline of 29 May 2026, brings overheating and passive cooling into the energy-performance framework but is silent on cooling access and grid resilience. At the national level, France's RE2020 is the most advanced overheating regime, integrating a degree-hour indicator calibrated on a 2003-heatwave reference; England's Approved Document O and the Netherlands' TOjuli indicator play comparable new-build roles. Spain and Germany sit a tier below – Spain capping cooling energy demand without a dedicated overheating definition and Germany addressing summer thermal protection through DIN 4108-2 rather than a dedicated regime.

**Table 2:** Coverage matrix — building heat resilience and summer peak-demand management across selected economies, scored on four mechanism dimensions.

| Country / Economy | Quantitative overheating threshold (new build) | Mandatory passive cooling in renovation | Cooling access as social policy | Heat-adjusted summer adequacy & distribution planning |
|---|---|---|---|---|
| **EU EPBD 2024/1275 (from May 2026)** | ◐ EPC reflection, passive cooling in min. standards | ◐ depends on transposition | ○ out of scope | ○ out of scope |
| **France** | • RE2020 DH indicator | ◐ new-build focus | ○ | ◐ RTE summer adequacy assessment |
| **Germany** | ◐ DIN 4108-2 (indirect) | ○ | ○ | ○ winter-oriented |
| **Spain** | ◐ cooling-energy cap (CTE) | ◐ partial | ○ | ○ no documented mechanism |
| **United Kingdom (England)** | • Approved Doc O | ○ new-build only | ○ | ○ winter-peak focus |
| **Italy** | ○ no dedicated national overheating indicator | ○ | ○ | ○ |
| **Netherlands** | • TOjuli indicator | ◐ partial | ○ | ○ |

Legend: • fully covered, ◐ partial or conditional, ○ not covered

Source: Allianz Research

**Read in this format, the gap is twofold: cooling access remains far less institutionalized than winter heating support, and even the strongest national codes apply mainly to new build rather than to the existing stock where most exposure sits.** The forthcoming EU floor in May 2026 will raise the minimum on overheating and passive cooling but does not address cooling access or grid resilience, and its bite depends entirely on transposition. The most consistent open cell is cooling access: no major European economy has developed a systematic summer counterpart to the winter heating-support architecture, and the gap falls on the groups that drive the European mortality concentration – older people, low-income households and tenants in the least-insulated dwellings. A workable national plan therefore rests on closing the four open cells together: binding overheating thresholds extended from new build to deep renovation; mandatory passive cooling triggered by a transparent renovation threshold; cooling access as a coordinated package – passive retrofits and efficient-appliance subsidies first, targeted bill support and medically prioritized access second, paired with a regulated right-to-install subject to structural and heritage constraints and heat-adjusted summer adequacy planning that explicitly models coincident cooling demand and supply derating, paired with residential demand response.

## Pillar 3. Income protection and adaptation finance

**Pillar 3 addresses the income-transmission channel of heat risk: when productivity losses become wage losses, the burden falls unevenly across workers, households and public budgets.** The wage compression documented in previous sections lands first on outdoor workers, fixed-term, seasonal and platform workers and lower-income households. Existing stabilizers, such as unemployment insurance, short-time work, indexed transfers and energy-poverty support, absorb part of the shock, but with a structural twist: the same indexation and triggering that protect household purchasing power also raise public expenditure or compress tax receipts, lifting the fiscal cost of each heat episode.

**This is the fiscal-stabilizer paradox at the heart of Pillar 3.** The more effectively a welfare state cushions households from heat-driven income and price shocks, the more it shifts the climate burden onto public budgets. If that burden is met through ex-post compensation rather than ex-ante adaptation, it consumes the fiscal space needed to lower future exposure. The paradox has two legs: a household-income leg, where existing social protection determines who absorbs the shock, and a fiscal-architecture leg, where public finance determines whether the resulting cost goes into prevention or compensation.

**Only Spain (Royal Decree-Law 8/2024 paid adverse-weather leave, including heatwave situations) and partially Italy (CIG/CIGS wage supplementation activated for extreme heat) have operational heat-specific income-protection mechanisms.** The remaining major economies – France, Germany, the UK, the Netherlands and most other member states – have general schemes that could in principle absorb heat-driven income loss but require ad-hoc activation each time, with no automatic heat trigger in legislation. Parametric risk-transfer for residual income volatility remains essentially absent at the national level in Europe; the European Commission's Climate Resilience Dialogue (2024) classifies it as an evolving tool requiring further analysis.[18] National adaptation strategies are now in place across all major European economies but remain strategic documents rather than multi-year budget envelopes. At EU level, several frameworks enable national action without amounting to a European income-protection or adaptation-budget regime.

**To reduce the gap, income protection, residual risk-transfer, budget planning and investment appraisal have to be redesigned as one fiscal architecture, working on both sides of the problem.** Existing social-protection instruments should carry explicit heat triggers conditional on workplace adaptation by employers. Parametric instruments calibrated to occupational heat thresholds can complement social insurance for the residual income volatility it does not absorb, with prioritized access for outdoor and non-standard workers and a public-backstop component for affordability. Multi-year adaptation budget envelopes, set out within medium-term expenditure frameworks, should replace reliance on ex-post emergency packages, paired with climate-risk screening across public investment management and coordinated oversight bringing together the health, housing, environment, energy, transport and labor ministries.

[18] Climate Resilience Dialogue - Final Report - Zurich Climate Resilience Alliance

# Appendix

## A1 – The impact of heat stress on energy demand

**To empirically assess the relationship between heat stress and energy demand, we construct a panel regression framework that captures how rising heat stress translates into higher energy consumption – primarily through the increased use of cooling technologies such as air conditioning and refrigeration.** As climate change drives more frequent and intense heat extremes, this mechanism is expected to become an increasingly important driver of energy demand, with significant implications for grid stability, energy security and the pace of the low-carbon transition.

**Our analysis draws on two data sources.** Energy consumption data are sourced from the IEA's World Energy Balances 2025, expressed in per capita megajoules (MJ) to allow for cross-country comparability. To capture heat stress, we use the Tx5d indicator from the ERA5 reanalysis dataset, defined as the mean daily maximum temperature over the hottest five consecutive days of a given year. This metric is particularly well-suited to our analysis, as it reflects the kind of sustained heat episodes most likely to trigger a behavioral and technological response in energy use, rather than isolated daily peaks.

**The empirical model is estimated on an unbalanced panel covering 49 countries over the period 1991–2023.** It provides both broad geographic coverage and sufficient temporal depth to identify meaningful climate-energy relationships. We employ a two-way fixed effects specification, controlling for both country-level heterogeneity and common time trends, with robust standard errors clustered at the regional level to account for potential spatial correlation in residuals. The regression model reads

$$\ln(\mathrm{energy}) = \beta_1 \mathrm{tx5d} + \beta_2 \mathrm{tx5d}^2 + \mathrm{C(ISO3)} + \mathrm{C(Year)};\ \mathbf{Eq.}\ 1$$

**The relationship between heat stress and energy consumption is nonlinear, following a convex U-shaped curve (Table A1), confirming previous literature results.**[19] At moderate temperature levels, energy demand declines as milder conditions reduce the need for space heating and other thermal energy services. This downward trend reverses once temperatures exceed approximately 30°C, a commonly used threshold for heat stress[20], at which point cooling demand begins to dominate, driven primarily by the widespread use of air conditioning.

**This non-linearity is reflected in the regression results, which show that the marginal effect of heat stress on energy consumption is both temperature-dependent and sign-switching.** At 25°C, the effect is negative: a one-unit increase in the Tx5d indicator is associated with an approximately 0.6% decrease in per capita energy consumption. At 35°C, the relationship reverses: the same marginal increase in heat stress is associated with a 1.2% rise in energy consumption. This twofold difference in magnitude illustrates how the energy implications of heat stress intensify disproportionately at higher temperature levels. A dynamic with important consequences for energy planning and infrastructure resilience in a warming climate.

[19] Electricity Consumption and Temperature: Evidence from Satellite Data, WP/21/22, February 2021
[20] Keeping cool in a hotter world is using more energy, making efficiency more important than ever – Analysis - IEA

Table A1: Regression results: The impact of heat stress on energy consumption

| Variable | Coefficient | p-value |
|---|---|---|
| **Intercept** | 10.3390 | 0.000 |
| **tx5d** | -0.0514 | 0.005 |
| **$tx5d^2$** | 0.0009 | 0.008 |

Source: Allianz Research

## A2 – The impact of heat stress on productivity

**The adverse effects of heat stress on labor productivity are well established in the academic literature.** Growing evidence from both micro and macro studies shows that exposure to extreme temperatures affects health, labor supply, and labor productivity in the short run.[21] At the physiological level, heat stress affects workers through physiological and behavioral responses, in turn reducing labor supply, productivity and overall labor capacity, with the relationship largely non-linear, declining sharply beyond peak temperature thresholds.[22] Despite considerable heterogeneity across work sectors and study designs, nearly all field studies consistently find a reduction in productivity due to occupational heat exposure.[23]

**Building on this evidence, we adopt the same empirical framework used for our energy consumption analysis to derive an econometric estimate of heat stress impacts on productivity.** Heat stress is measured using the Tx5d index, as defined previously. For productivity, we draw on two complementary indicators from the OECD database[24], output (gross value added) per hour worked, expressed in USD, and wages per hour worked, which together capture both the economic output and the compensation dimension of labor performance. The models reads as follow

$$\mathrm{GVA}_{\mathrm{per}_{\mathrm{hour}}} = \mathrm{tx5d} + \ \mathrm{tx5d}_{\geq 30} + \mathrm{C(ISO3)} + \mathrm{C(Year)}; \mathbf{Eq.}\ 2$$

$$\Delta \ln(\mathrm{wage}_{\mathrm{per}_{\mathrm{hour}}}) = \mathrm{tx5d}_{(\mathrm{t}-1)} + \ \mathrm{tx5d}_{(\mathrm{t}-1)}{}^{2} + \mathrm{C(ISO3)} + \mathrm{C(Year)}; \mathbf{Eq.}\ 3$$

**Equations 2 and 3 depart from the baseline specification of Equation 1 in two important ways, each motivated by the distinct channels through which heat stress affects labor productivity.** Equation 2 models gross value added per hour as a function of both the Tx5d index and an interaction term, Tx5d when heat stress exceeds 30°C, to account for the non-linear intensification of heat impacts at higher temperature levels. Equation 3, which models the log change in wages per hour, follows the same structure as the energy consumption model (Equation 1), but introduces a one-year lag specification for both the linear and quadratic Tx5d terms. This lagged structure reflects the hypothesis that wage adjustments to heat stress occur with a delay, as labor market responses, such as renegotiated contracts or sectoral reallocation, tend to materialize more slowly than changes in physical output.[25]

[21] Reflections—Temperature Stress and the Direct Impact of Climate Change: A Review of an Emerging Literature | Review of Environmental Economics and Policy: Vol 10, No 2
[22] Heat stress and the labour force | Nature Reviews Earth & Environment
[23] Frontiers | Occupational heat stress, heat-related effects and the related social and economic loss: a scoping literature review
[24] OECD Data Explorer • Productivity database
[25] Economic impact of labor productivity losses induced by heat stress: an agent-based macroeconomic approach | Climatic Change | Springer Nature Link

**The output per hour regression (Eq. 2) is estimated on a panel of 35 countries over the period 1997–2023 (Table A2).** The results reveal a clear threshold effect consistent with the non-linear relationship hypothesized in our framework. Below 30°C, the coefficient on Tx5d is positive and marginally significant at the 10% level (0.59), suggesting that moderate heat is associated with a slight increase in output per hour – potentially reflecting seasonal productivity gains in temperate economies. Above 30°C, however, the picture changes markedly: the coefficient on the interaction term is negative and statistically significant at the 5% level (–1.30), indicating that each additional degree beyond this threshold is associated with a reduction in output per hour of approximately 1.30 USD. These findings underscore the non-linearity of heat's impact on labor productivity, where the results point to 30°C as a critical inflection point beyond which heat stress begins to impose measurable economic costs on labor output.

Table A2: Regression results: The impact of heat stress on output per hour

| Variable | Coefficient | p-value |
|---|---|---|
| **Intercept** | 13.7401 | 0.142 |
| **tx5d** | 0.5900 | 0.072 |
| **$tx5d_{\geq 30}$** | -1.3039 | 0.042 |

Source: Allianz Research

**The wage regression (Eq. 3) is estimated on a panel of 33 countries over the period 1995–2023 (Table A3).** The results are broadly consistent with those obtained for output per hour, with a turning point again emerging around 30°C. The use of a one-year lag structure reflects the hypothesis – supported by the data – that labor compensation adjusts to heat stress with a delay, as wage-setting mechanisms respond more slowly to productivity shocks than output itself.

**The relationship between heat stress and wage growth follows an inverted U-shape.** In the range below 30°C, higher heat stress in year t is associated with a modest acceleration in wage growth in year t+1, consistent with the productivity gains observed at moderate temperatures in the output per hour specification. Above 30°C, this relationship reverses: additional heat stress dampens wage growth the following year, reflecting the transmission of heat-induced productivity losses into labor compensation over time.

**The marginal effect of heat stress on wage growth is both temperature-dependent and sign-switching.** At 25°C, the effect is positive, with a one-unit increase in Tx5d associated with an approximate 0.2% increase in wage growth. At 35°C, the relationship reverses, with the same marginal increase in heat stress associated with a 0.2% decline in wage growth. While the magnitude of the effect may appear modest, its consistency with the output per hour results and its statistical significance lend confidence to the finding that heat stress above 30°C exerts a meaningful drag on labor compensation, with implications for household income, consumer demand and the broader macroeconomic costs of climate change.

Table A3: Regression results: The impact of heat stress on per hour wage growth

| Variable | Coefficient | p-value |
|---|---|---|
| **Intercept** | -0.0756 | 0.364 |
| **$tx5d_{(t-1)}$** | 0.0120 | 0.027 |
| **$tx5d_{(t-1)^2}$** | -0.0002 | 0.013 |

Source: Allianz Research

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# How AI is rewiring global trade

## Concentrating Power, Dependencies, and Supply Chains

21 May 2026

Allianz Research

# Content

# Executive Summary


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AI and trade are no longer separate policy domains. AI growth depends on globalized supply chains for semiconductors, computing infrastructure and digital services while trade is increasingly shaped by who controls AI infrastructure, data flows and cloud capacity.

- **Trade openness is a structural precondition for AI-driven productivity gains.** Open economies benefit disproportionately from cheaper inputs, faster innovation diffusion and AI adoption spillovers. Trade openness accounts for 23% of the variation in AI adoption across countries with highly open economies, such as Singapore, UAE and Ireland, leading in diffusion. However, while AI can significantly boost growth, its benefits are unlikely to be evenly distributed.

- **Export volumes of AI-enabling goods have surged from USD1trn in 2014 to USD3.8trn in 2025 (+280%), accounting for 15% of global trade and far outpacing the 40% growth in goods trade overall.** Asia dominates the supply side, accounting for 65% of global AI-related exports and seven of the top ten exporters, led by China (18% of AI-related exports), Taiwan (12%) and Hong Kong (11%). The composition remains concentrated in intermediate inputs (76%) and equipment (23%), reflecting deep dependence on semiconductors and data-center infrastructure. On the demand side, the US has tripled its AI-related imports since 2023, underpinned by 5,427 operational data centers, 45% of the global total. Europe's import growth of just +40% underscores a widening infrastructure gap.

- **Services imbalances could scale with AI.** ICT services trade reached USD900bn in 2024 (11% of global services trade), with Ireland alone exporting USD173bn, exceeding both the rest of the EU (USD142bn) and the US (USD108bn), and reflecting its role as a billing hub for US multinationals. However, it masks structural dependence, with the EU excluding Ireland running a USD45bn ICT services deficit (mainly with Ireland), highlighting its reliance on US digital ecosystems. And things could get worse. Hypothetically, if the AI services subscription penetration rate of US providers such as ChatGTP Plus and Claude Pro increases from 3% to 50% in a high adoption scenario in the Eurozone, annual payments to US providers could reach EUR34bn (currently EUR2.7bn), equivalent to 20% of the current EU–US goods services deficit. AI thus acts as a scalable, recurring channel that exacerbates structural EU–US digital imbalances and has the potential to lower the US overall trade deficit.

**Beyond the headline data, three structural dynamics are reshaping the global AI trade order:**

**1. Supply chain concentration as single points of failure**
The AI supply chain is concentrating rapidly at the technological frontier. Taiwan, South Korea, the US and the Netherlands dominate the production of advanced chips, high-bandwidth memory and semiconductor equipment, creating critical single points of failure with no near-term redundancy. Unlike broader goods trade, which has partially reoriented along geopolitical lines, AI-related goods remain largely unfragmented across geopolitical blocs, with semiconductors and key components still flowing globally except at the technological frontier (notably advanced US export controls) due to deep interdependencies, challenging the narrative of technological decoupling.

**2. Infrastructure as geopolitical power**
Control over data centers, cloud infrastructure and computing capacity is emerging as a primary source of geopolitical leverage, distinct from traditional goods trade dynamics. Europe's strategic exposure is acute. With less than 10GW of operational data-center capacity, four times below the US (60GW), and a development pipeline six times smaller, the gap is not expected to narrow. US hyperscalers already control 35% of European computing capacity and account for nearly half of the upcoming pipeline, consolidating a 70% cloud market share. Structural constraints, fragmented regulation, complex permitting processes, grid connection delays, no domestic hyperscaler and limited VC or state-backed funding reinforce this dependency. Finally, reliance on Asian hardware inputs complements the US dependence that will increase without sufficient EU investment in the area. Under that backdrop, Europe is permanently under the threat of a US "kill switch" on cloud data. In the meantime, this effective expansion of this key infrastructure is dependent on inputs sourced in Asia, notably in China (14% import ratio for AI datacenter components) and Taiwan for semiconductor, and as a result exposed to current supply-chain disruption risks along new geopolitical tensions in Middle East. Regaining control and strengthening sovereignty on AI good production but also service delivery is critical for the region to avoid suffering a twin external dependence.

**3. Industrial policy: from subsidies to protectionism**
Global tariffs on AI-enabling goods have fallen from 5.6% in 2015 to 2.8% in 2025, well below the 7.8% average for manufacturing overall, but non-tariff measures have surged, driven primarily by the US and China, reflecting a strategic shift from financial support toward technology protectionism. Over 3,600 AI subsidy measures targeting critical materials, semiconductors, GPUs, computing equipment and optical fibers are now in force globally, with China having nearly doubled its trade coverage over five years, followed by South Korea, Malaysia, the US and Japan. The overlap of national and multilateral governance regimes is concentrating production, infrastructure and modelling capabilities in the hands of a small number of economies. Regulatory fragmentation is becoming a binding constraint on AI services trade, with ICT trade restrictiveness already explaining 5% of the cross-country variation in AI services flows. The EU and US are advancing incompatible regulatory frameworks, creating compliance friction that reinforces existing structural imbalances.

# An AI boom built in Asia

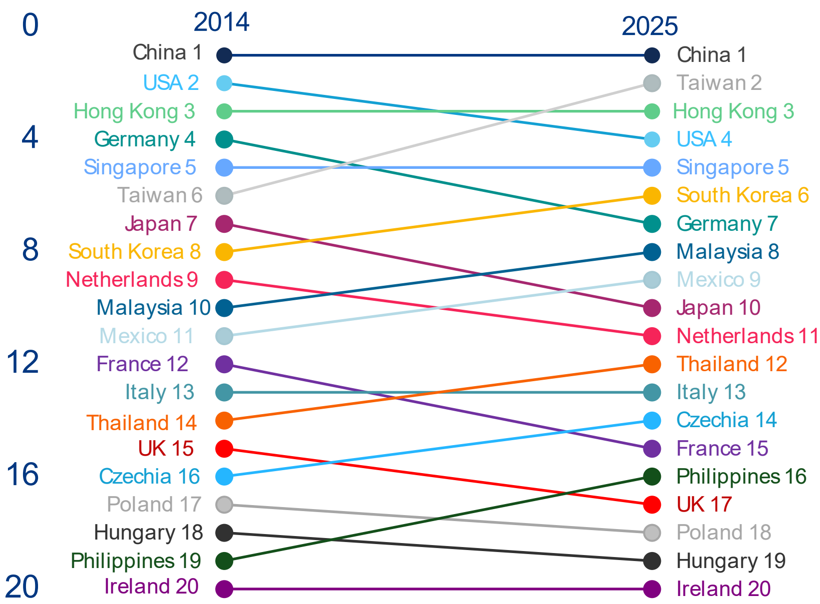
**AI fundamentally depends on global trade in goods and services, with a complex, internationally fragmented value chain spanning raw materials, semiconductors, computing hardware, cloud infrastructure, data and final applications (AI-enabling goods).** Trade enables access to critical inputs, such as advanced chips, specialized equipment and energy resources, while facilitating the cross-border exchange of intermediate goods, services and knowledge needed to build and scale AI systems. However, the strong geographic concentration of key stages of the AI supply chain coupled with heightening frictions tightening the exchange channel (tariffs, shutdown of the Strait of Hormuz) raises concerns about unequal access and the risk of a widening technological divide.

**Over the past decade, global trade in AI-related goods has doubled, far outpacing the growth of overall goods trade and of non-AI-related goods in particular.** Global AI-related goods trade has expanded rapidly over the past decade, doubling from USD1.9trn in 2014 to USD3.8trn in 2025. Meanwhile, overall global trade in goods increased more moderately, rising by +40% to USD26trn in 2025. This growth was increasingly supported by AI-related sectors, with trade in non-AI goods rising by only a third. This expansion has seen three distinct acceleration phases. The first acceleration occurred in 2017–2018 (+15% y/y and +11% y/y respectively), driven by fundamental developments in need for AI-related goods and the introduction of the transformative architecture, which underpins most large language models. A second surge followed in 2021 (+30% y/y), driven by the release of early large-scale models, including GPT-3 (2020) and DALL·E (2021) and paving the way for the emergence of generative AI in 2022, which needed more confounding technical infrastructure for model innovation and computation. However, momentum weakened in 2022 amid a challenging macroeconomic environment. Higher interest rates and elevated inflation combined with the introduction of US export controls on advanced AI chips and semiconductors, led to a marked deceleration in AI-related goods trade growth (+4% y/y in 2022), followed by a contraction in 2023 (-7% y/y), with export values falling below their 2021 peak. More recently, advances in multimodal and agentic AI systems have reignited investment momentum. As AI became increasingly available and embedded in the economic system, AI-related goods exports reached new heights in 2025

(+22% y/y). Early 2026 data, covering January and February, show that the boom continues, especially in countries such as Taiwan (+62% y/y) and South Korea (+105% y/y).

**Asia has firmly established itself as the center of global trade in AI-enabling goods.** The region accounts for 65% of global exports and dominates the entire value chain. The region is at the heart of an increasingly concentrated landscape, with seven of the top ten exporters representing 80% of global trade. China leads with a 18% share, followed by Taiwan (12%) and Hong Kong (11%). Singapore (7%), South Korea (6%), Malaysia (4%) and Japan (3%) also reinforce the region's strength. The US (8%), Germany (5%) and Mexico (4%) are the only non-Asian economies in this group, highlighting the strong presence of AI trade in Asia. This dominance has deepened over time as competitive dynamics have shifted in the region's favor, causing advanced economies to lose ground. Despite a slight decline in its share (-3pps), China remains the leading exporter, while Taiwan's rise from 6th place in 2014 to 2nd place in 2025 (+7pps) marks the most significant shift. This was supported by a +53% year-on-year export surge in 2025 (Figure 1). Hong Kong has also gained prominence, further reinforcing Asia's lead. At the same time, emerging players are gaining traction: Mexico recorded the fastest growth in 2025 (+62% year on year), while Malaysia, Thailand (+49% year on year) and the Philippines are expanding as alternative hubs. Together, they accounted for 11% of global trade (+3pps) and 18% of additional trade in 2025. By contrast, most advanced economies, including those in Europe, have lost ground. The US, for example, slipped from second to fourth place, reflecting a relative decline in manufacturing-based AI trade despite continued strength in high-value innovation.

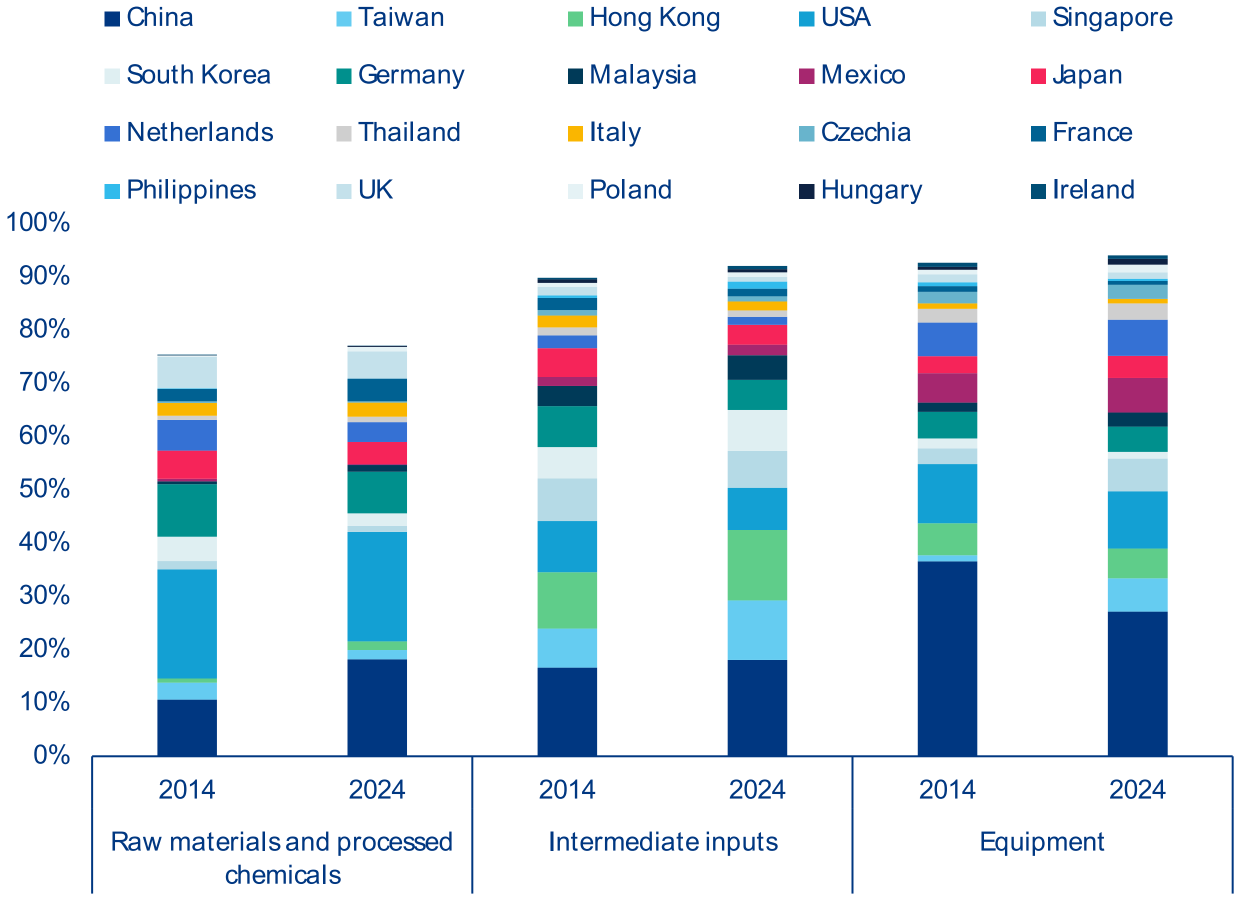
**Figure 1:** Top 20 AI-enabling goods exporters ranking (2014 vs 2025)




Sources: UN Comtrade, National Customs, Allianz Research

**Although AI-enabling goods trade is already highly concentrated globally, the value chain reveals even greater specialization.** Leading exporters cluster at distinct stages, particularly in raw materials, processed chemicals and equipment. AI-enabling goods can be categorized into three groups: raw materials and processed chemicals, intermediate inputs and equipment. Raw materials and processed chemicals constitute critical substances that are fundamental to the fabrication of AI equipment. At the aggregate level, this category represents the most diversified segment of the AI value chain. The top 20 exporters in this category account for three-quarters of global exports. This indicates significant concentration of the primary inputs for AI development, and this concentration only strengthens further upstream in the value chain, with the top 20 exporters representing over 90% of exports in the intermediate inputs (processed or partially manufactured goods) and equipment categories (specialized tools and machine components essential for AI development). China and the US have a particularly pronounced footprint at the extremes of the value chain. Together, they account for 39% of global exports of both raw materials and processed chemicals, as well as equipment (Figure 2). For intermediate inputs, the level of concentration remains high in absolute terms, but it is comparatively more evenly distributed, with China, Hong Kong and Taiwan collectively representing 43% of global exports, and each country holding a similar share. Notably, shifts in countries' global export shares of intermediate inputs have been linked to changes in the overall ranking. Taiwan and South Korea have gained +4pps and +2pps, respectively, reaching 11% and 8% of the global export share. Similarly, Hong Kong's share has increased by +3pps to 13% over the past decade. However, this has not allowed it to become the second-largest exporter, as Taiwan has made a notable entry at the high end of the value chain. It has gained +5pps to reach a 6% export share in equipment; it is the only country, alongside Singapore, to have increased its export share in this category. This has come at the expense of China, which has lost -9pps and stands at a 27% export share.

**Figure 2:** Decomposition of the AI value chain (% share of global trade for each category 2014 vs 2024)




Sources: UN Comtrade, National Customs, Allianz Research.

**An ultra-segmented AI supply chain, with Europe holding a modest role.** The global AI ecosystem is sharply segmented geographically. Upstream activities – chip design, semiconductor equipment, data-center infrastructure and telecoms – are concentrated in Western economies, with the US dominating GPU/CPU architecture and Europe and Japan leading in lithography and wafer-production machinery. Downstream, Asia takes over almost entirely: China holds a critical position in rare earth processing, battery manufacturing and electronic components; South Korea and Taiwan anchor the foundry landscape, the former specializing in memory chips and the latter as the undisputed leader in advanced logic manufacturing. Malaysia and Thailand have developed strong packaging and assembly capabilities, with Thailand increasingly focused on data-center hardware. Critical raw materials are similarly dispersed – nickel from Indonesia and Australia, cobalt from the DRC, lithium from Chile, helium from Qatar, silicon from China. Europe's position across this architecture is, overall, low to moderate, leaving the region with limited strategic leverage and meaningful exposure to supply-chain disruptions at any of its key upstream or downstream nodes.

**The trade of goods related to AI is dominated by intermediate inputs and equipment, reflecting a surging demand for high-performance AI infrastructure.** Although raw materials and processed chemicals account for only 2% of the trade in AI-enabling goods, exports are dominated by intermediate inputs (76% of total AI-related exports) and equipment (23%). This masks significant country-level specialization and exposure to supply risks. The UK stands out in the upstream sector, with a 7% share of raw materials (+5pps above the global average). At the intermediate stage, the strongest concentration is in Asia: South Korea and the Philippines rely heavily on this segment (95% and 91% respectively), as do Hong Kong and Taiwan, all of which are well above the global average of 75%. European exporters such as Italy (84%) and France (83%) exhibit comparable, albeit less pronounced, patterns. At the downstream stage, equipment exports are more prevalent in the Netherlands (57%), Mexico (49%), the Czech Republic (45%) and Thailand (43%). Meanwhile, China and the US have a more balanced approach, with each country having a 68% share of intermediate inputs and equipment shares of 31% and 28%, respectively.

## Box: The Middle East crisis could send semiconductor prices surging

**Over the past two years, two-thirds of the expansion in Asian semiconductor exports were due to price increases and one-third due to higher volumes.** Risks of energy shortages from the Middle East crisis could send prices even higher, given the already tight and extremely concentrated supply. Looking at the recent results of the biggest foundry worldwide located in Taiwan (70% market share in 2025) as a proxy for the advanced-node segment of the supply chain, wafer shipments grew +28% year-on-year in Q1 2026, a robust rate by historical standards. However, revenue in USD terms expanded by +40.6% over the same period, leaving a gap of approximately 12pps that cannot be attributed to volume alone. The explanation lies primarily in a structural shift in product mix rather than in nominal price increases on comparable goods. Taiwan's production capacity base has migrated progressively toward 3nm and 5nm process technologies, which carry substantially higher per-wafer prices than the legacy nodes they are displacing in the revenue composition; advanced technologies now account for 74% of total wafer revenue despite representing a modest share of physical units shipped. However, a look at China's trade statistics suggest that inflationary effects are rather broad-based as even for the less advanced chips used for industrial purpose – above 28nm – we observe a diverging trend between export volumes (downward) and value (upward) in Q1 (Figure 3, right). Chinese data show that strong volume on AI products is masking rising disruption risks in manufacturing production (over -10% contraction of global shipments of smartphone and computer estimated this year, Figure 3, left), as well as for some electric and mechanical devices (i.e., transistors, printed circuits, HVAC, batteries) that are critical for building up the data-center fleet.

**Figure 3:** China exports value, total & high-tech goods (left) and China exports value & volume, semiconductor in USD, 12-month trailing yoy% (right)


Sources: China National Customs, Allianz Research

**The sustained increase in semiconductor pricing is best understood as a demand-led phenomenon, with supply-side constraints serving as an amplifying mechanism.** The commercial deployment of large language models from late 2022 onwards ramped up the compute requirements of hyperscale cloud operators. Since then, the transition from AI model training to large-scale inference has extended and broadened the requirement for advanced silicon across a wider range of end-users. Naturally, this is squeezing supply: Leading-edge foundry capacity requires three to five years and capital commitments in the tens of billions of dollars to come online. The imbalance between demand urgency and supply inelasticity has given foundries meaningful pricing power, with Taiwanese semiconductor producers' gross margins expanding from 58.8% to 66.2% within a single year. Supply-side cost pressures related to critical materials, notably gallium, germanium, tungsten and antimony, all subject to successive rounds of Chinese export controls since 2023, are another increasingly consequential factor, with affected material costs rising in the range of 30-50%, though they are more of a medium-term geopolitical risk for now.

**The fact that the majority of AI-enabling goods exports are driven by a small group of countries also makes their trade model highly vulnerable to potential shocks.** Taiwan and Hong Kong are the most exposed to an AI bubble burst, with 74% and 59% of their exports being related to AI goods respectively (Figure 4). They are followed by Singapore and the Philippines (both 47%), Malaysia (43%) and South Korea (32%). In contrast, trade in AI-enabling goods accounts for 15% of US exports, indicating that American exports are relatively well diversified. Similarly, Mexico and Thailand (both 23%), China (19%), and Japan (18%) have a moderate level of export concentration in AI-enabling goods.

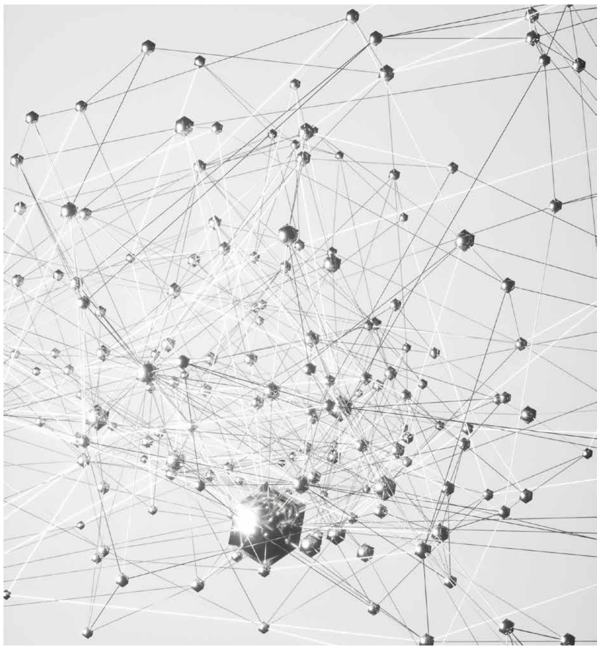
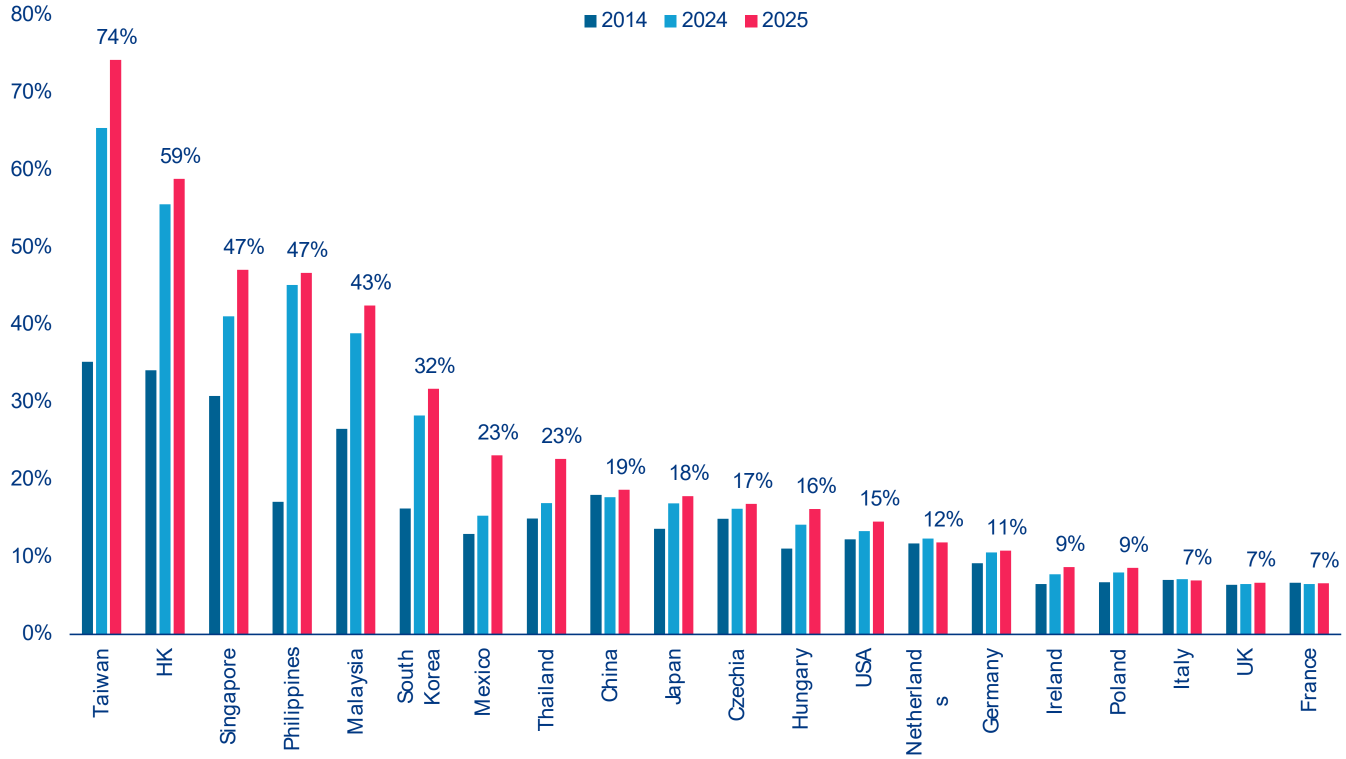


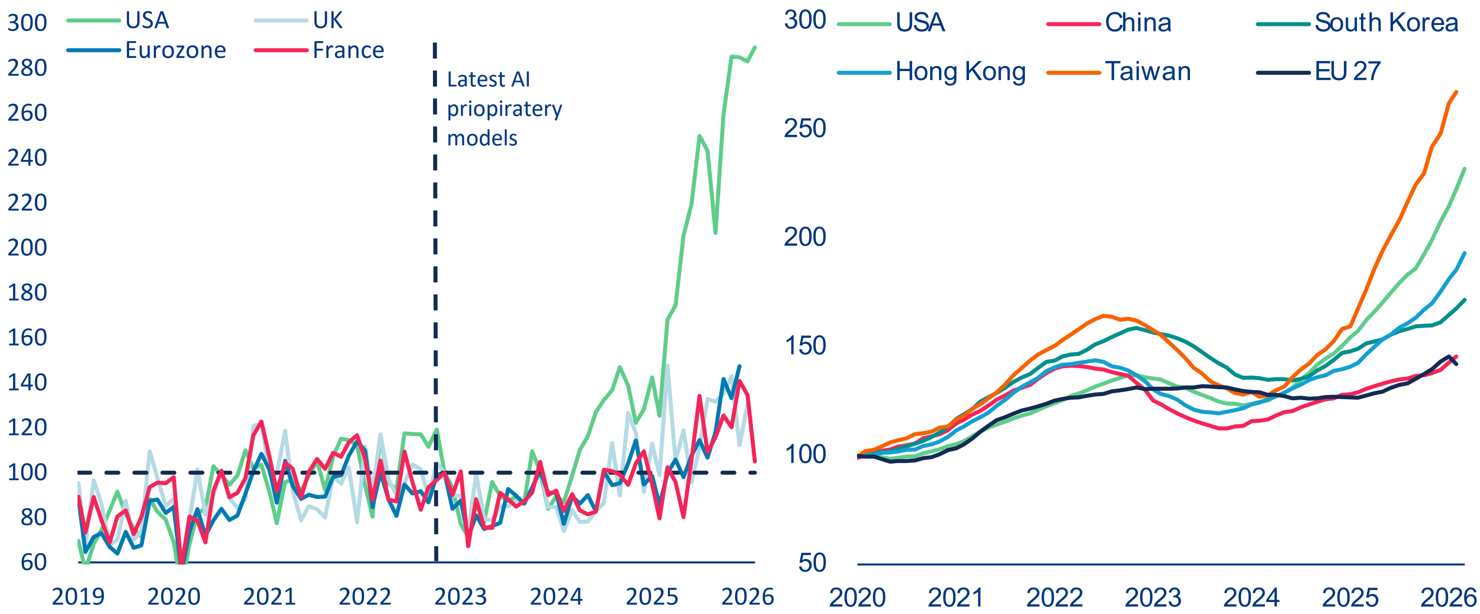
**Figure 4:** Share of AI-related goods (% of countries' total exports)



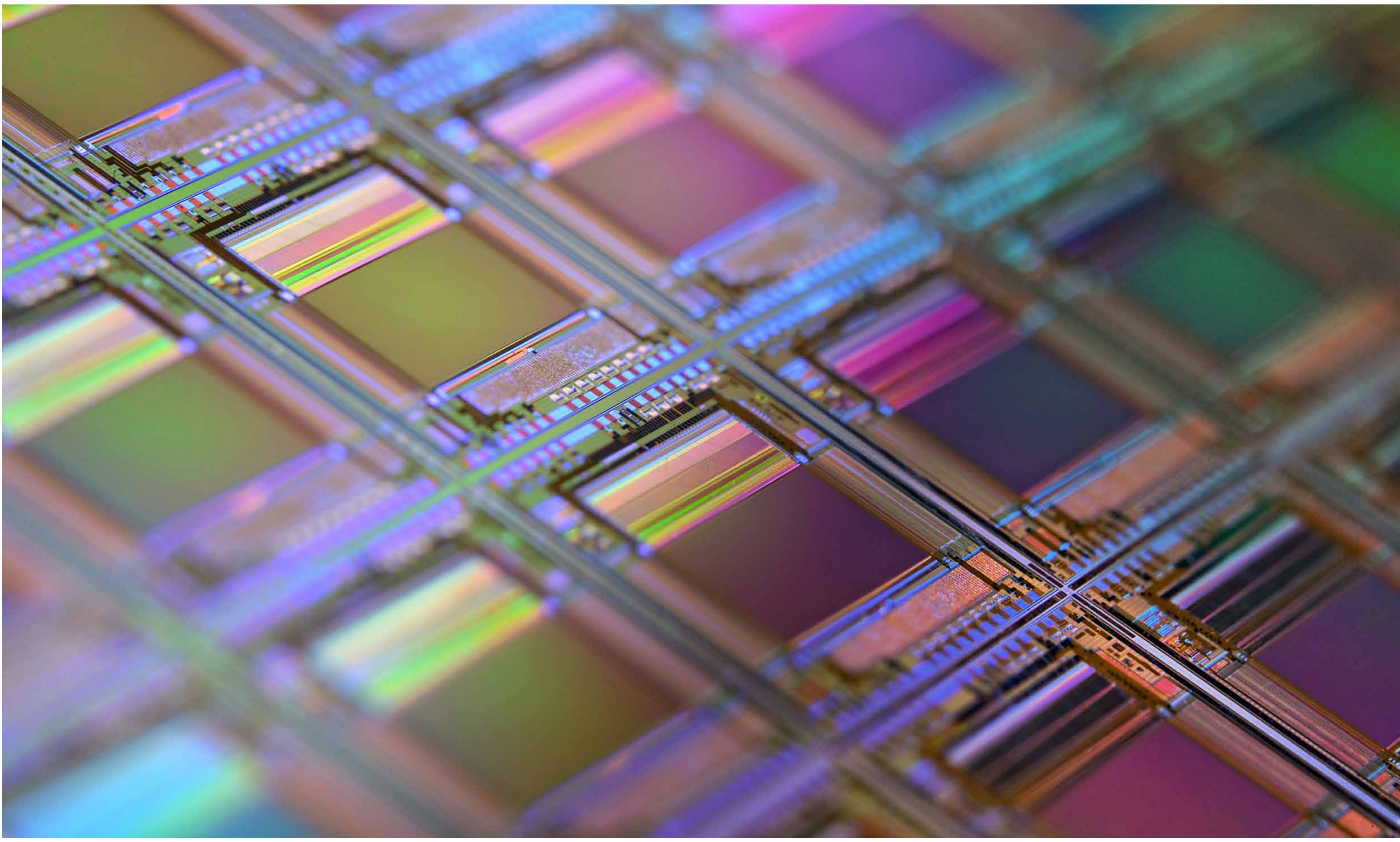

Sources: UN Comtrade, National Customs, Allianz Research

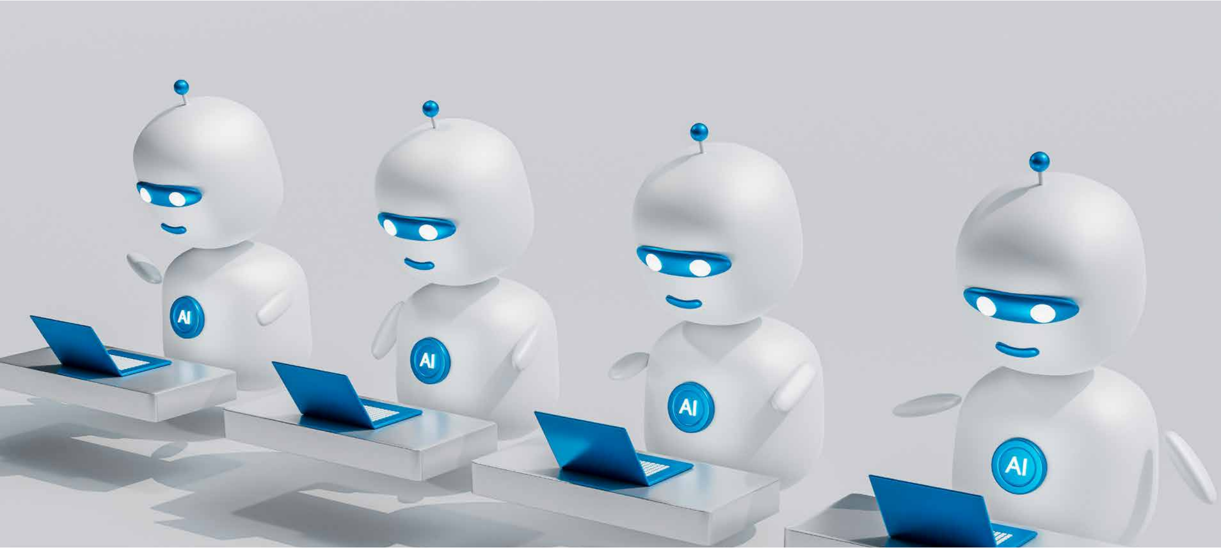
**However, AI-related import demand is highly concentrated in US and partially Asia.** Since 2023, Taiwan and the US have recorded the largest increases in AI-related imports, reflecting their pivotal positions in the AI supply chain: Taiwan through the import of capital equipment for semiconductor manufacturing and the US through demand linked to its dominance in AI services and data-center infrastructure. Since the mass-market deployment of large language models since 2023, the US has tripled its imports of advanced AI-related products (Figure 5, left), reflecting massive domestic investment in AI and continued reliance on foreign semiconductor supply. This surge has been reinforced by the rapid expansion of US data-center infrastructure, with 5,427 facilities accounting for 45% of global capacity in 2025 and 10 times the count of any other country. By comparison, Europe's AI-related imports have grown by only around 40% since 2023 (Figure 5, right), reflecting weaker investment in AI infrastructure, limited data-center capacity (Germany 2nd with 529, UK 3rd with 523, France 6th with 322 data-centers) and restricted access to advanced chips compared to US hyperscalers. Growth slowed further in early 2026. In Asia, beyond Taiwan, the strongest import growth has been seen in South Korea, reflecting its dual role in semiconductor production and consumers of advanced AI components, and Hong Kong, highlighting its important roles in semiconductor production and technology flows into China due to US export controls. Overall, the AI semiconductor market remains highly concentrated geographically: the US being the dominant end market for the most advanced generation of AI semiconductors, the EU and China as the other two large but structurally distinct demand pools. This creates significant exposure to shifts in demand, trade policy and regulation across a small number of economies: for AI hardware manufacturers any deterioration in US demand, tightening of trade and tariff policy, or regulatory disruption in Europe or China would have outsized consequences for the suppliers currently running at capacity to serve them.

**Figure 5:** US imports growth from selected AI products have skyrocketed, index (left) and Imports of overall AI-related products, in USDbn (right)




Sources: Trade Map, Allianz Research



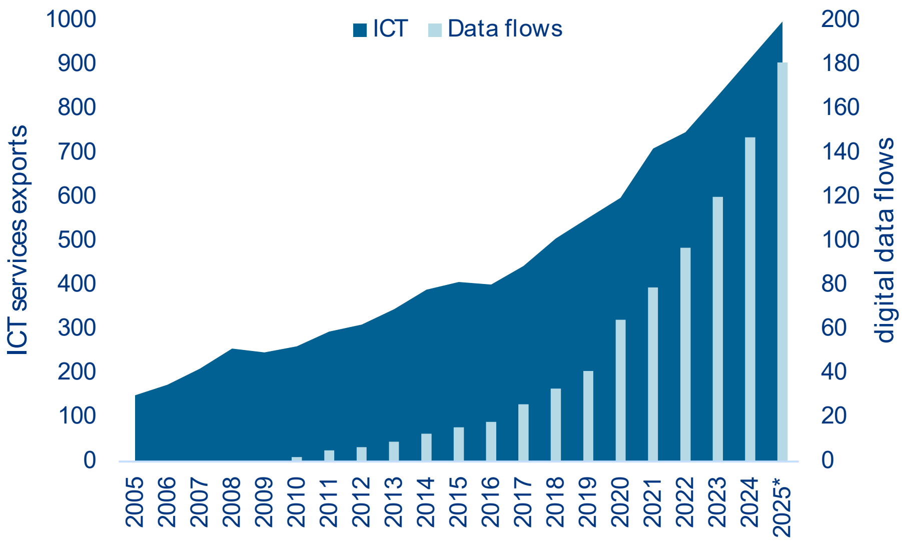


# The growth in trade in services and data flows is accelerating with the diffusion of AI

**Data flows and AI-related services trade are the backbone of transforming economic models into the future.** It differs structurally from the trade of goods in several key respects. Firstly, it is deeply integrated into broader digital and intra-firm flows, with much of its value being captured through cloud services, software licensing, data infrastructure and model access, which are frequently bundled together rather than being traded as individual services. This helps to explain the outsized role of hubs such as Ireland, which function less as traditional exporters and more as invoicing and coordination centers for global technology firms, particularly US multinationals. Consequently, bilateral imbalances, such as those between the EU excluding Ireland (''EU26'') and Ireland, understate the true extent of dependence on US-based AI ecosystems. Indeed, the EU26 entity runs a large ICT services trade deficit of USD45bn, mostly explained by the deficit with Ireland, at close to USD55bn. India and the UK are far behind in terms of ICT exports to the EU and the US comes fourth, exporting less USD5bn directly to the EU26. Secondly, AI services act as the operating system for the trade of AI goods: advanced chips, servers and hardware only generate value when combined with cross-border flows of computing power, software and continuous model updates. This makes services a prerequisite rather than an adjunct to the trade of goods. Thirdly, the high concentration of trade reflects platform-based business models rather than pure comparative advantage, with a small number of economies capturing disproportionate value through intellectual property, data and digital infrastructure. Finally, the trade of AI services is increasingly shaped by regulatory and data governance regimes, which influence where services are hosted, billed and exported from. This reinforces fragmentation across jurisdictions. Together, these features imply that not only is AI-related services trade concentrated, it is also foundational, determining how value is distributed across the global AI economy.

**An essential for the development and distribution of AI-systems are cross-border data flows.** As AI systems rely on large, diverse and continuously updated datasets that only generate value when combined with software, hardware and expertise, the flow of these data internationally is essential for AI development and adoption. As data is non-rivalrous, it can be reused by multiple users and applications, creating strong economies of scale and scope. Free data flows enable firms to combine information across markets, build more accurate and generalizable models and accelerate innovation. By contrast, restrictions limit scale, raise costs, reduce interoperability and slow diffusion. The growing importance of this issue is reflected in broader trends: data-driven services such as computing, telecommunications and finance now account for almost half of the global trade in services. ICT services have evolved rapidly over time making up 0.53% of nominal global GDP in 2015 and have increased by 56% to 0.85% in 2025, while cross-border data volumes have grown more than tenfold from 15.5 zettabytes in 2015 to 181 zettabytes in 2025 (Figure 6). Together, these dynamics demonstrate that seamless cross-border data flows are a key driver of AI performance and innovation, as well as being a fundamental pillar of the global digital economy and the rise in AI-related services. And although data is generated and flows globally, it is often processed, stored and monetized in a small number of digital hubs with advanced infrastructure, which concentrates value geographically.

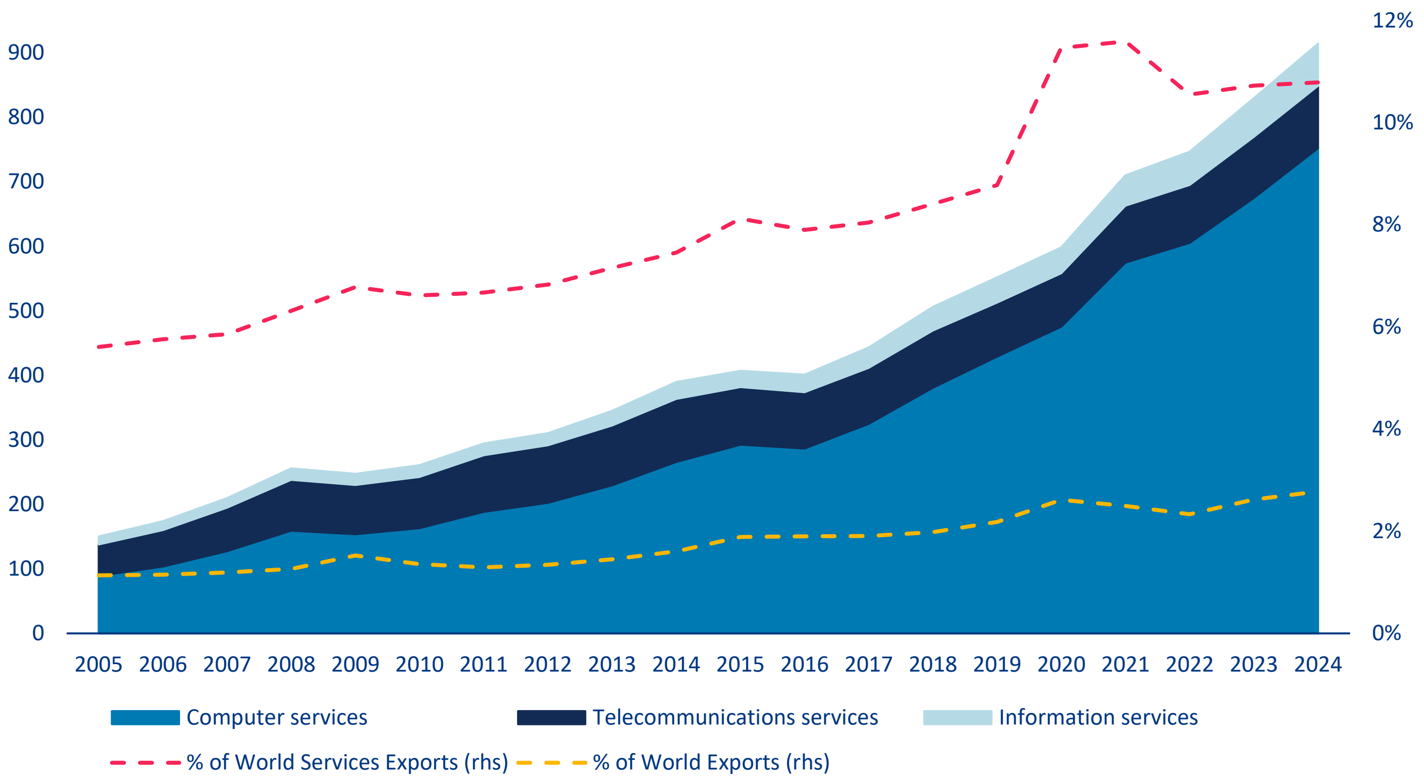
**Figure 6:** Cross-border global trade in ICT services (in USDbn) and digital data flows (in zettabytes)




Sources: OECD BATIS, Statista, Allianz Research. Notes: 2025* is an interpolated approximation for ICT services as no data is yet available.

**The value of international trade in ICT services is approaching USD1trn, growing more rapidly than overall services (11% of global services trade, up from 8% a decade ago).** This growth has been driven by rapid expansion since 2016–17, with the sector being highly concentrated in a few hubs. The sector, which is dominated by computer services such as software, consulting, infrastructure and support, reached approximately USD900bn in 2024 (around 3% of the total value of goods and services traded globally, and 11% of the value of services) (Figure 7). Ireland stands out as a key hub for US tech multinationals providing digital services to European core markets. It exported ICT services worth USD173bn in 2024, accounting for an exceptionally high 60% of its total exports. This is largely due to intra-firm flows linked to datacenters and licensing revenues. Consequently, Ireland has become the world's largest ICT services exporter, surpassing the EU (excluding Ireland) with USD142bn and the US with USD108bn of ICT exports to the rest of the world, respectively. Within the EU (for which we count both extra and intra-EU trade), major ICT exporters include Germany, the Netherlands and France. Meanwhile, India remains a major player, while China lags behind in providing digital services to the world.

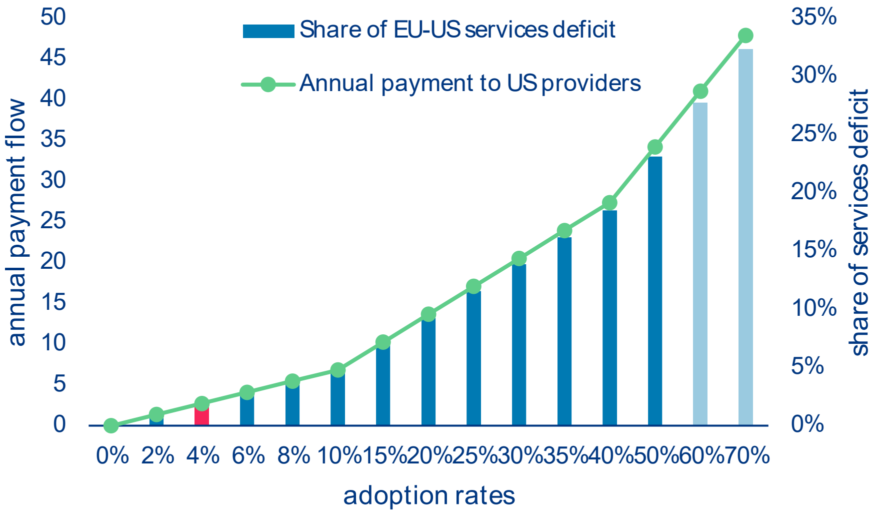
**Figure 7:** AI-related ICT services trade breakdown (in USD bn)




Sources: OECD, WTO, Allianz Research

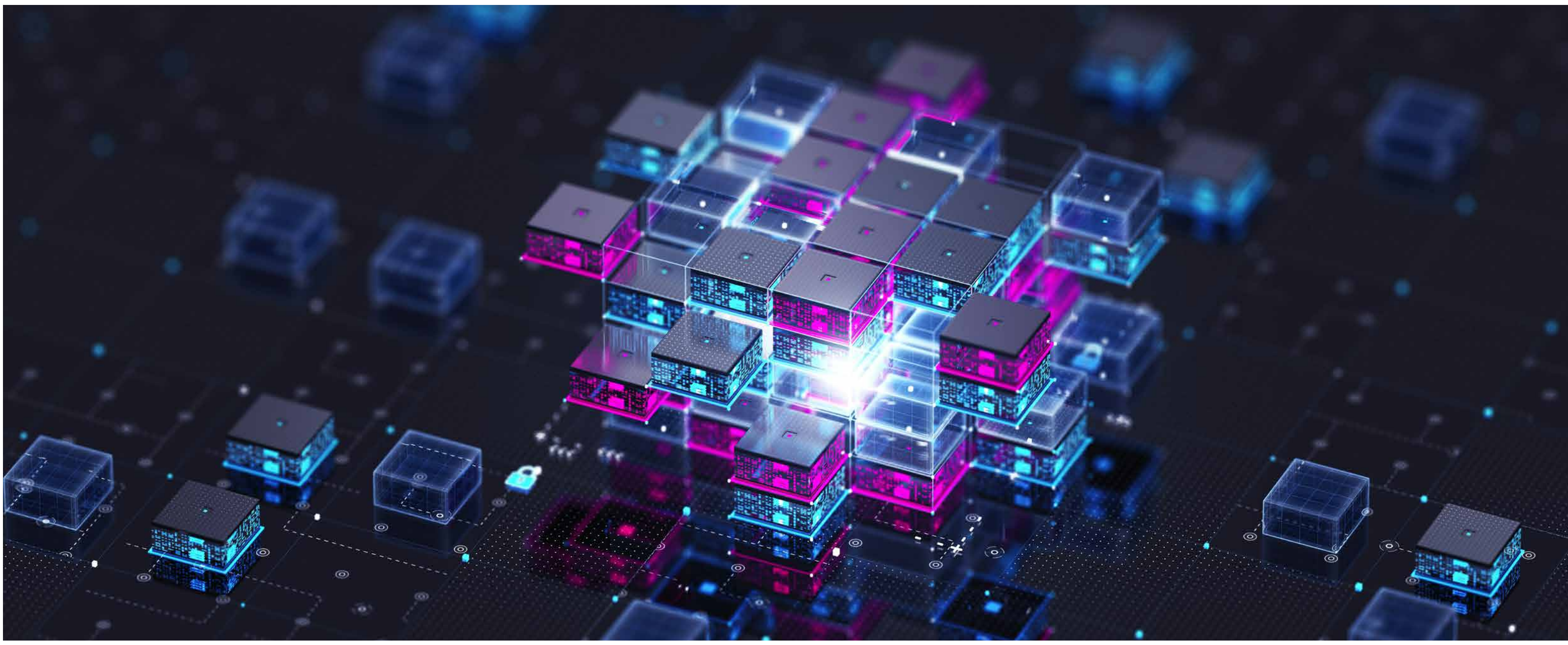
**AI diffusion will be the next big driver in EU-US services imbalances due to EU dependence on AI model provision.** Adopting AI in the Eurozone could significantly increase the transatlantic services imbalance through recurring subscription payments to US AI providers. Assuming an adult population of around 285mn in the Eurozone and a Claude Pro or ChatGPT Plus-style price of EUR20 per month (EUR240 per year, the current standard pricing for individual users), even modest adoption would result in significant cross-border services imports. With an estimated current penetration rate of around 4%, this equates to approximately EUR2.7bn per year in payments to US providers; already a significant portion of digital services imports, albeit still modest compared to the EUR148bn EU–US services deficit in 2024. Increasing adoption to a 10% penetration rate would raise outflows to approximately EUR6.8bn, and to around EUR17.1bn at a 25% penetration rate (Figure 8). At this point, AI subscriptions alone would be comparable to a tenth of the entire current services deficit. Under a high-adoption scenario in which 50% of adults subscribe, annual payments would total approximately EUR34bn, implying an additional impact of over one-fifth of the current EU–US services deficit, representing a significant increase in US services exports driven purely by AI diffusion.

**Figure 8:** Back-of-the-envelope calculation of annual payment flow from the EU to the US (in EURbn) and share of EU-US services trade deficit (in %) along AI penetration rates in the adult population




Source: Allianz Research. Notes : Assumes an adult European population of 285mn along hypothetical AI adoption rates, a subscription fee of EUR20/month (current standard pricing of ChatGPT Plus or Claude Pro for individual users) and a EU-US services trade deficit of EUR148bn.

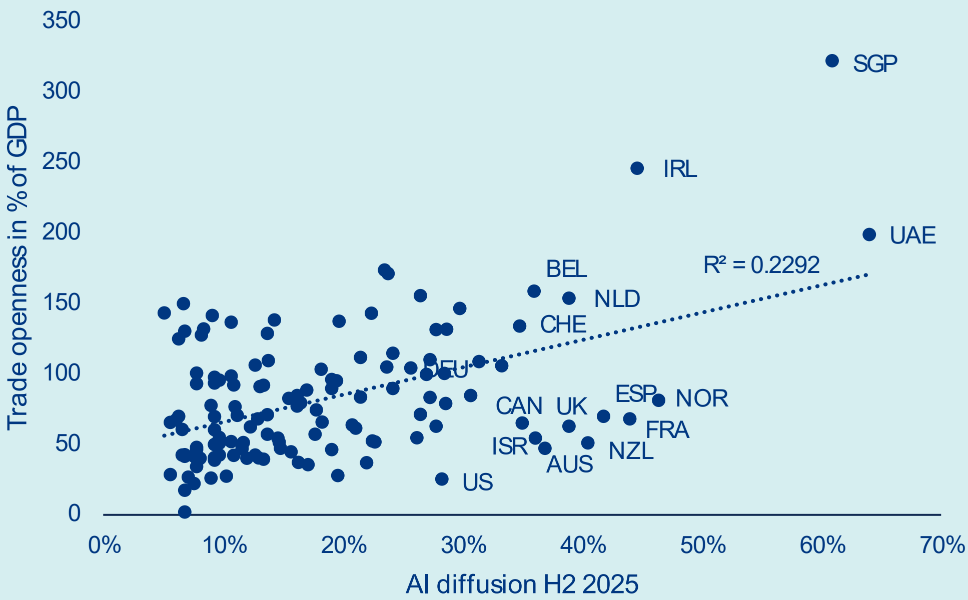
**More broadly, AI acts as a scalable, recurring channel for importing services to the US economy via software subscriptions, cloud computing, APIs, IP licensing and digital advertising.** These flows are intangible, highly scalable and concentrated in US hyperscalers. This matters because the EU's apparent digital surplus is already distorted by profit shifting and US tech revenues booked via Ireland. Stripping this out suggests a structural digital services deficit exceeding EUR100bn. AI accelerates this dynamic further as, unlike infrastructure-heavy digital services, subscriptions represent near-pure cross-border revenue transfers with limited domestic value creation. Furthermore, the impact is likely underestimated in the simple subscription model: enterprise AI usage (APIs, Copilot integrations and SaaS embedding) already exceeds consumer subscriptions, and productivity effects could also reduce Europe's own services exports. Therefore, unless offset by strong domestic AI platforms, European digital champions or localization of AI infrastructure and billing, AI diffusion risks becoming a persistent structural driver of widening EU–US services imbalances.



# Box: Europe struggles with strategic dependencies

**Despite its openness to trade, Europe faces uneven AI adoption, restricting its ability to compete with leading rivals.** Trade openness explains 23% of the variation in AI diffusion across countries (Figure 9), with highly open economies such as Singapore, the United Arab Emirates and Ireland also recording the highest diffusion rates. But Europe contradicts this trend. Though its economies are highly open and integrated into global value chains, AI adoption remains uneven due to strict data privacy and regulation, limiting the continent's ability to benefit from AI-driven productivity gains. Overall, while AI can significantly boost productivity and growth, its benefits are unlikely to be evenly distributed, potentially increasing inequality between capital and labour, high- and low-skill workers, and larger versus smaller firms.

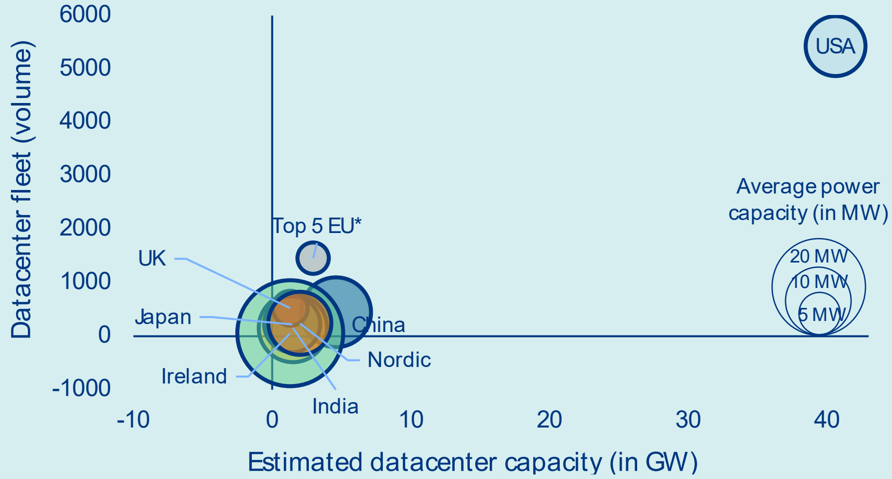
**Figure 9:** Brade openness (total trade as % of GDP) versus AI diffusion in H2 2025 (in %)




Sources: World Bank, Microsoft, Allianz Research

**When it comes to data centers, the US is racing far ahead.** With an estimated 40+ GW of installed capacity today (Figure 10), projected to surpass 100 GW by 2030, the US already has the world's largest computing infrastructure. This is underpinned by over USD600bn in capital expenditure committed by US hyperscalers alone this year. By contrast, Europe remains structurally underequipped. Its data-center capacity is heavily concentrated in five core markets (FLAP-D: Frankfurt, London, Amsterdam, Paris and Dublin), accounting for around 60% of the total European capacity. Meanwhile, emerging hubs such as the Nordic countries are only just beginning to attract significant interest, thanks to competitive energy costs and an increasing proportion of renewable energy sources. The challenge for Europe is not merely one of scale, but also of capital and political architecture. Unlike the US, where private hyperscalers flood the market with investment, or China, where the state substitutes for private capital, Europe suffers from a dual deficit: insufficient private investment and an absence of coordinated public policy. The fragmentation of Europe's regulatory landscape prevents the emergence of a clear, common framework dedicated to AI infrastructure, leaving European ambitions dangerously reliant on external resources.

**Figure 10:** Data-center fleet & estimated capacity (in GW) per region

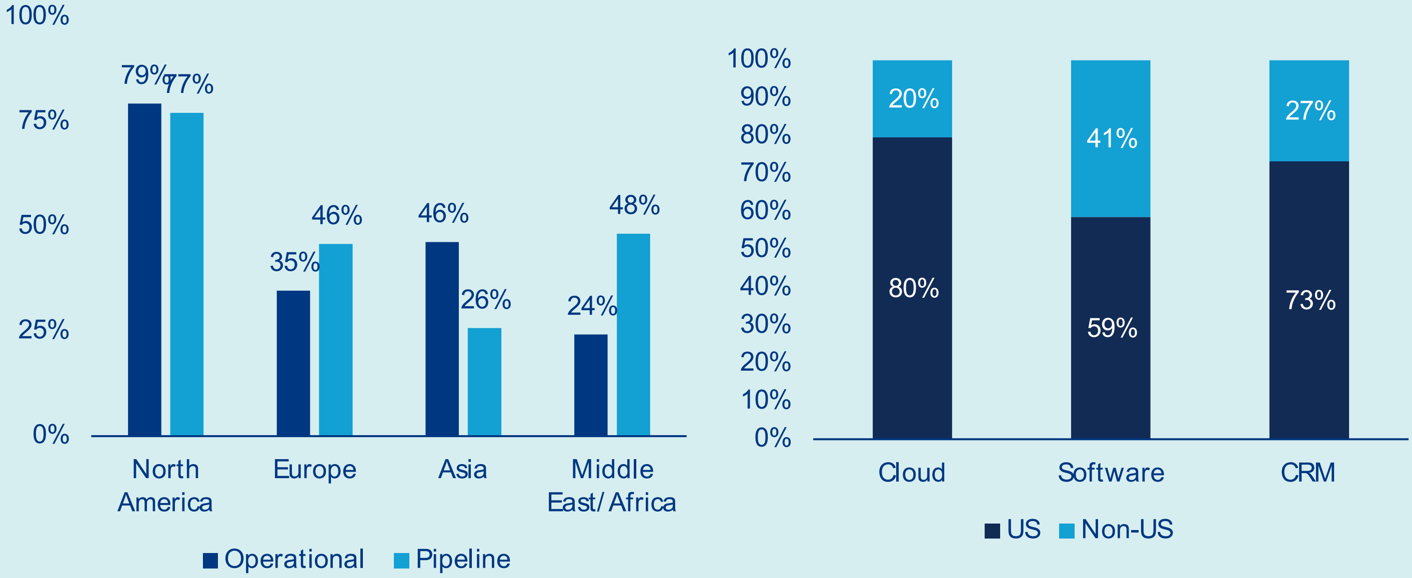



Sources: Cloudscene, Cushman & Wakefield (Datacenter H2 2025 update), Allianz Research. Notes: *Country group including Germany, France, Italy, Spain & Netherlands.

**To avoid falling into a technology dependency trap, regulatory and capital constraints need to be cleared.** Building data-center infrastructure in Europe is a formidable undertaking, constrained by a unique combination of structural, regulatory and financial hurdles that are unparalleled in the US or China. Limited land availability in urban areas, lengthy and complex permitting processes and increasingly stringent environmental regulations – particularly regarding water consumption and carbon footprint – can extend development timelines significantly, often stretching from planning to commissioning to over four years. Perhaps the most acute bottleneck is power connectivity: grid connection queues in key European markets have grown dramatically, with some operators reporting wait times of five years or more for adequate power access. This is a direct consequence of ageing grid infrastructure and competing demand from industrial and residential users. Financing conditions add another layer of complexity, as the capital-intensive and long-term nature of data-center projects does not align with the risk appetite of many European institutional investors. This is due to the absence of the deep and liquid private credit markets that underpin US hyperscaler investment.

**The consequences of this investment gap are already evident in the ownership structure of infrastructure.** As of 2025, US hyperscalers accounted for roughly 40% of total data-center operational capacity across Europe, but almost half of the pipeline (Figure 11, left) as large US tech firms are taking profits from weak private capital market in Europe and low competition from local players to consolidate their presence within a key market for them. The digital services layer tells an even starker story: US firms currently capture 80% of European cloud revenue, 59% of enterprise software revenue and 73% of CRM expenditure, leaving non-US players, including European vendors, to compete for the remaining margins Figure 11, right). This dominance is no accident. US hyperscalers' substantial financial resources enable them to absorb permitting delays, pre-fund grid upgrades and negotiate directly with national governments on energy and land access, structural advantages that no European operator can currently match on an equivalent scale. Their ample cash reserves effectively enable them to fill the void left by the lack of traction in private and public investment on the continent, as they steadily expand their physical and commercial footprint with each passing planning cycle.

**Figure 11:** Weight of US hyperscalers in datacenter capacity (current & future) across the globe (left) and market share of IT services and software applications revenue in Europe per firm nationality (right)




Sources: CBRE, Synergy Research, European Software & Cyber dependencies report (European Commission, Dec. 2025), Allianz Research. Notes: Data as of Q4 2025.

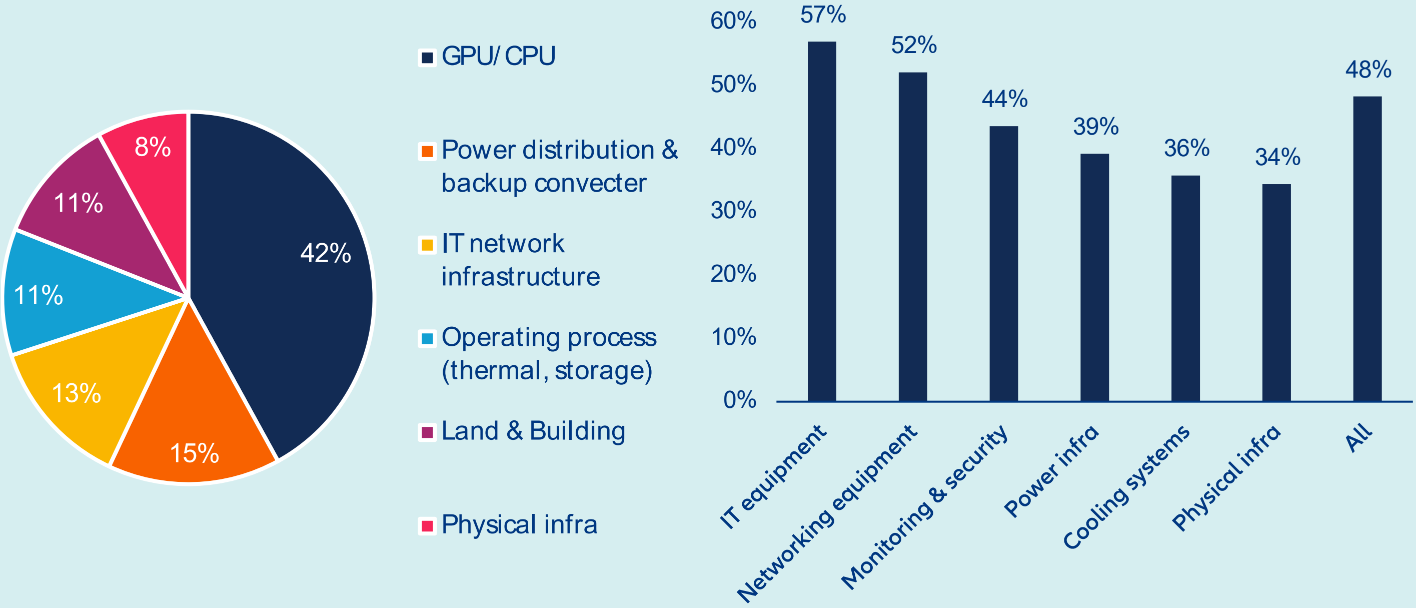
**Initiatives such as Mistral AI and sovereign computing capacity projects in France and Sweden are promising, albeit modest, counterweights.** Without decisive, coordinated action from Europe, including the mobilization of sovereign funds and development banks, as well as the establishment of a harmonized permitting framework, the continent risks becoming dependent on foreign infrastructure providers, losing not just market share, but also strategic autonomy over the digital backbone of its economy.

**Strong industrial-linked specialization and regulatory constraints undermine European ambitions to build up a domestic semiconductor ecosystem and take part to AI race.** Historically, European industrial strategy oriented domestic semiconductor production toward application-specific and industrial-grade chips, well suited to its automotive and manufacturing base but leaving the continent without meaningful capacity in the advanced nodes where AI demand is now concentrated. In the meantime, the relatively small weight of the ICT industry in Europe (less of 1% of GVA) but also the absence of a large domestic hyperscaler with ample cash reserves like in the US meant there was never sufficient demand to justify leading-edge foundry investment at scale, a gap that policy has failed to compensate for. Regulatory complexity and slow subsidy disbursement have further deterred the large infrastructure commitments that catching up would require. The cancellation last year of Intel's plan to build a foundry in Magdeburg is a good example Underlying both is a structural weakness in private capital markets: With only 6% of global AI funding flowing to European firms, the continent lacks.

**Finally, Europe's data-center ambitions depend on fragile, Asia-heavy supply chains.** Building AI-dedicated data-centers is, first and foremost, a hardware problem, and Europe currently owns very little of it. GPUs and CPUs alone account for around 42% of the total capital cost of a 1GW AI data center (Figure 12, left), and this expenditure largely flows outside of Europe: non-EU import ratios exceed 57% for IT equipment and 52% for networking gear (Figure 12, right). Five Asian economies (Taiwan, China, South Korea, Malaysia and Vietnam) collectively supply around two-thirds of this expenditure. Taiwan's near-monopoly on sub-3nm logic chips and South Korea's dominance in memory represent structural dependencies, not cyclical ones, and cannot be meaningfully substituted in the short term. The next tier of components – power distribution, IT network infrastructure and cooling systems, each representing 10–15%

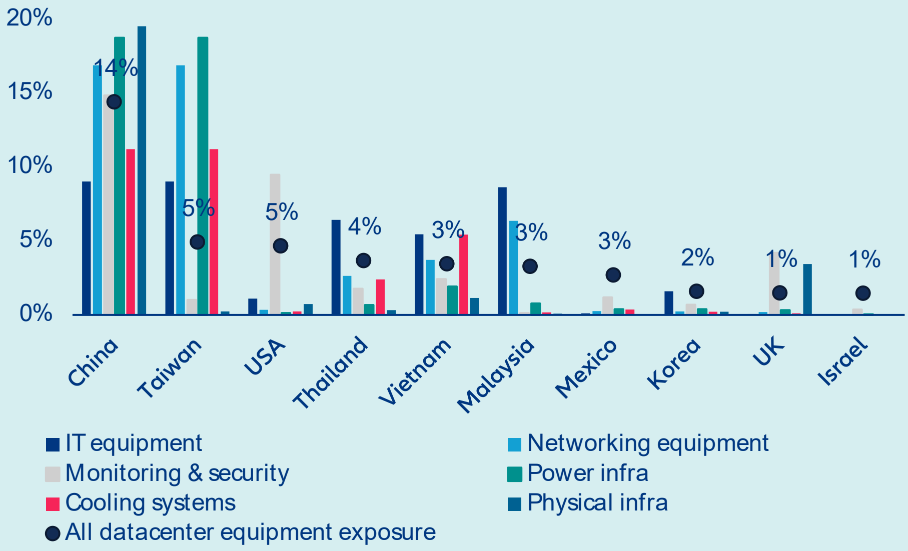
of datacenter value – shows only modestly better numbers: monitoring and security sits at 44% non-EU exposure, power infrastructure at 39%, and cooling systems at 36%. Even physical infrastructure, the least technology-intensive segment, runs at 34% external reliance. At the country level, China stands out with a total exposure share of 14% across all datacenter equipment categories (Figure 13), nearly three times the weight of both the US and Taiwan, reflecting its dominance in advanced electronics and physical and cooling inputs. Vietnam primarily supplies computers and electronic transmission devices, while Turkey is a notable partner for cooling systems. Despite its influence on the software and services stack through its hyperscalers, the US has a more modest hardware footprint concentrated in IT and networking. Overall, Europe has little domestic manufacturing depth in advanced electronics and limited battery and power storage capacity. Its supply chain is also acutely exposed to two live geopolitical risks: renewed trade tensions between Washington and Beijing (currently on pause but far from resolved) and any energy disruption in Asia stemming from Middle East instability. For a continent with sovereign AI ambitions, this is a precarious foundation.

**Figure 12:** Estimated cost breakdown of an AI-dedicated 1GW capacity data center* (left) and share of non-EU imports ratio for critical materials used in AI data-center infrastructure construction, 2025 (right)




Sources: Bernstein, Eurostat, Allianz Research.** GPU share based on Nvidia GB200 cost.

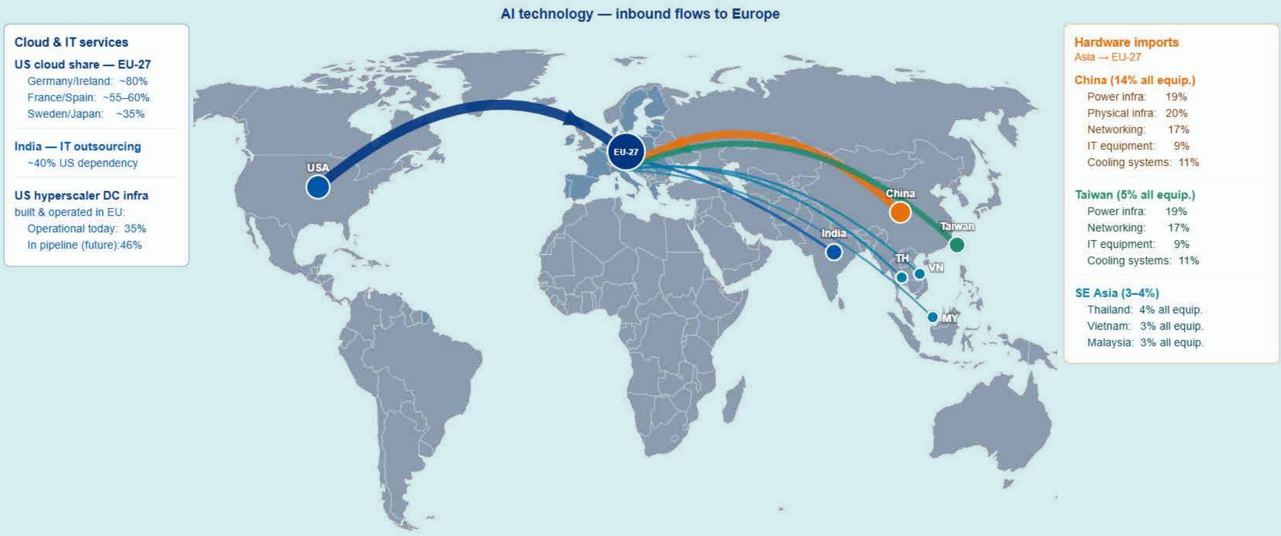
**Figure 13:** Top 10 non-EU suppliers of critical materials for AI data-center infrastructure construction




Sources: Eurostat, Allianz Research

**This structural dependency has direct macroeconomic consequences.** Europe's limited control over AI infrastructure and its reliance on imported hardware and foreign hyperscalers imply that a growing share of AI-related value added will be captured abroad, dampening domestic productivity multipliers and GDP gains from AI adoption. At the same time, higher data-center build costs, prolonged permitting cycles and constrained grid access raise the effective cost of AI deployment in Europe, slowing diffusion into the broader economy. This combination risks a dual drag: weaker productivity growth relative to the US and higher unit costs for digital and industrial transformation. Over time, it could also widen the transatlantic gap in investment intensity, reinforce Europe's structural current account outflows in digital services and increase vulnerability to external price-setting in critical digital inputs.

**Figure 14:** Top 10 non-EU suppliers of critical materials for AI data-center infrastructure construction

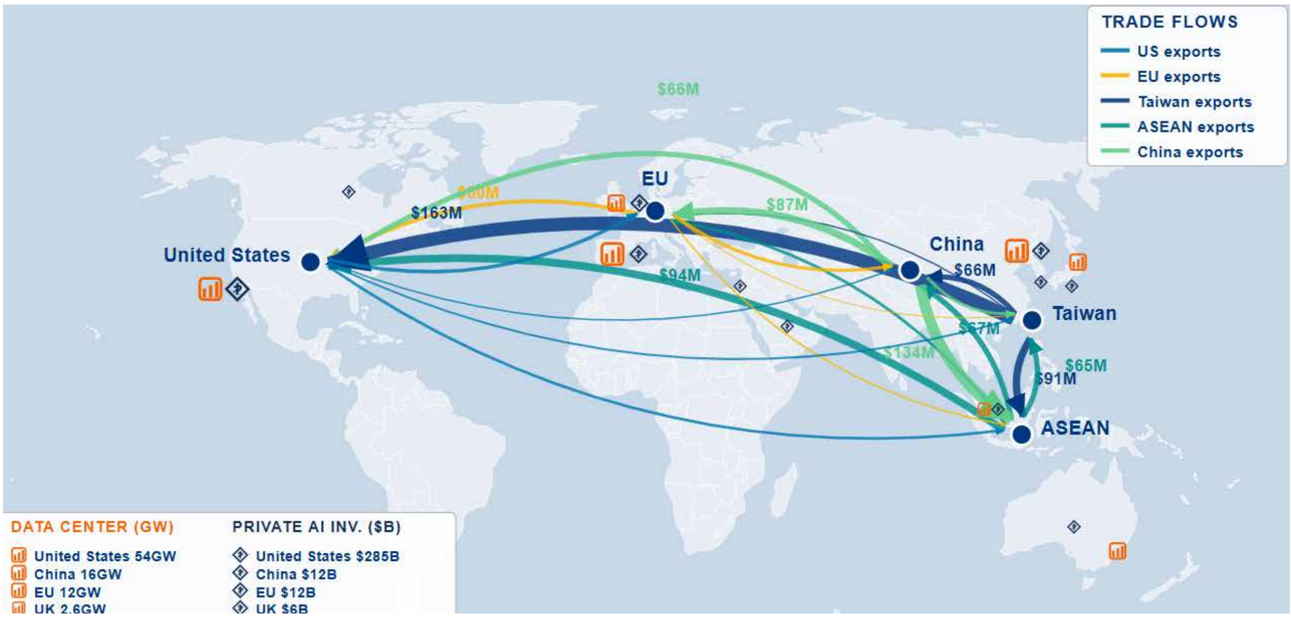



Sources: Allianz Research

# The world is flat: AI reconfigures the geography of trade and dependencies

**The AI supply chain is becoming more geographically distributed in assembly and packaging, but more concentrated and fragile at the technological frontier, with diversification creating the appearance of resilience while reinforcing dependency on a small number of critical nodes.** The most critical parts of the AI supply chain (advanced chips, high-bandwidth memory, and the equipment to produce them) remain highly concentrated in a small set of countries, especially Taiwan, South Korea, and the Netherlands, and this concentration has intensified since 2022 due to export controls and the AI buildout. This creates systemic single points of potential failure in the global AI system. Partial diversification took place only in the middle layers of the supply chain. Server assembly, packaging and networking are shifting geographically under the influence of US policy and export controls, with gains for Vietnam, Mexico, Japan, Taiwan and parts of Eastern Europe. But this friendshoring is still limited in scale and far from replacing established hubs.

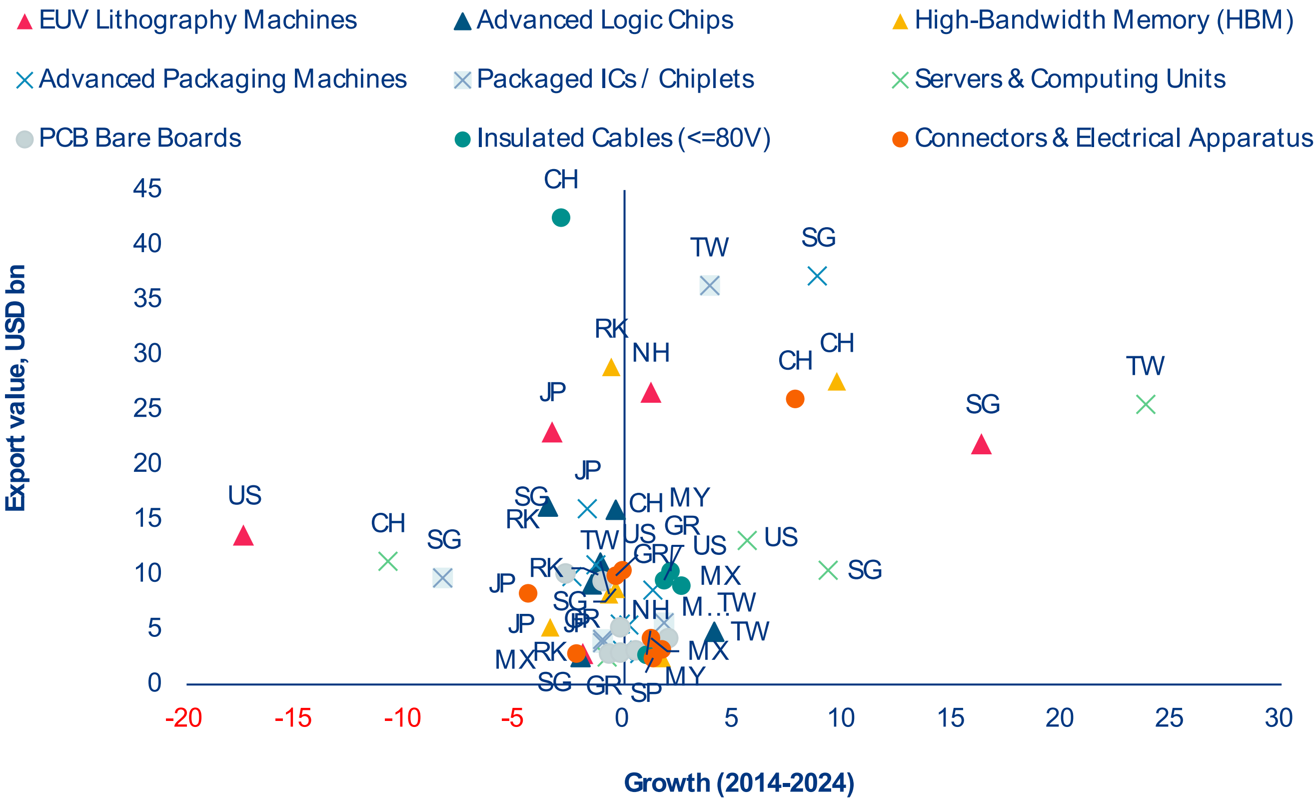
**Figure 15:** AI supply chain concentrated in several key geographies




Sources: Stanford HAI 2026, Synergy Research, Trade Map, Allianz Research

**Despite friendshoring efforts, diversification away from China remains limited.** A decade of diversification investment has produced incremental shifts in specific sub-segments but no geography has emerged as a credible large-scale alternative. Vietnam and Mexico have attracted investment and attention as potential alternative bases for electronics assembly, and there are signs of incremental share gains in specific sub-segments, but the scale of Chinese capacity in these categories means that meaningful diversification remains a multi-year undertaking for most supply chains. China continues to dominate low-value, high-volume components that are essential but underappreciated (Figure 16). Diversification away from China in these segments is slow, meaning supply chains remain exposed to geopolitical shocks despite rhetoric of decoupling. The risk here is overexposure to Chinese production in case of a new geopolitical shock or policy change in Bejing.

**Figure 16:** Concentration towards Asia in the higher value added parts of the AI supply chain, while China further integrates low-value, high-volume



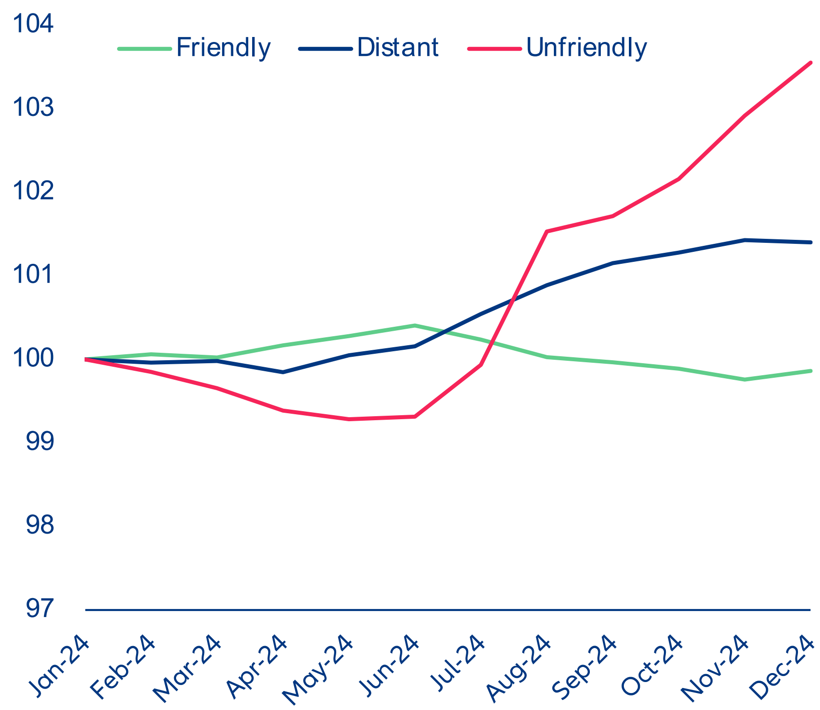

Sources: UNComtrade, Allianz Research. Note: Shapes of the markers correspond to different sections of AI-supply chain: triangle = higher value, cross = middle value, circle = lower value.

**Hidden upstream vulnerabilities emerge with the current conflict in the Middle East**. Shortages of key inputs for semiconductor production, notably helium, would impact the most leading-edge fabs, which could translate to losses above USD10bn a month. Qatar accounts for a third of global production of helium, an upstream input of the AI supply chain that is also almost entirely shipped through Gulf terminals. While the inflationary cost impact is relatively small on semiconductor production compared to the rest of the supply chain, the production shortage would be much more impactful, with an asymmetric impact. Semiconductor producers in South Korea are most at risk, given their exposure of imports from Qatar, and early estimates suggest 4 to 6 months of reserves. However, everything hinges on gas storage in non-producing countries. The US holds the largest known reserves and has historically been the dominant producer, but its federal helium reserve has been progressively wound down under legislation mandating its disposal. Japan, one of the world's largest helium consumers given its electronics and semiconductor manufacturing base, holds no domestic reserves and is almost entirely import-dependent. New countries could increasingly offer their helium reserves to the market, including Russia, Australia and Algeria, as well as Tanzania, which holds what may be one of the world's largest undeveloped helium deposits in the Rift Valley, though it remains years away from commercial production.

**US trade policy has reinforced rather than fundamentally reshaped semiconductor supply chains.** China now accounts for just 0.7% of US semiconductor imports in tariffed categories, subject to a 25% tariff, while production (or rerouting) shifted toward Taiwan, Vietnam, Malaysia, Thailand and Mexico. Despite headline tariff threats of up to 46% on Vietnam and 36% on Thailand, semiconductors remain largely exempt from new trade measures, with effective tariff rates near zero for most major suppliers and around 1% for Malaysia. As a result, the post-2018 semiconductor supply chain continues to operate under near free-trade conditions. The more ambitious reshoring effort has been the CHIPS Act which redirected USD52bn toward domestic manufacturing and research. Its investment allocation has been substantial, but the production results remain limited and represent only a small fraction of Taiwan's output.

**Geopolitics is changing where semiconductors are manufactured rather than with whom they are traded.** The commercial logic of the supply chain remains largely intact: Taiwan, South Korea and the broader Southeast Asian production ecosystem continue to sell to every major economy without any need of political alignment. This creates a form of structural interdependence that cuts across geopolitical blocs in ways that policy narratives tend to obscure: China remains deeply integrated into the commercial semiconductor market for everything below the frontier of advanced logic and memory, and continues to source the vast majority of its chip requirements from the same Taiwanese and Korean suppliers that serve US hyperscalers. The meaningful exceptions are narrow but strategically significant, US export controls have effectively excluded China from sub-7 nanometre logic, high-bandwidth memory for AI applications, and the lithography equipment required to produce either domestically, creating a two-tier market in which the frontier is controlled and the mainstream is not. This commercial logic is increasingly visible in the trade data. For most goods, trade has fragmented along political lines, rising between geopolitically „friendly“ countries and declining between „unfriendly“ ones, and until recently AI-related goods followed the same pattern. Since 2024, however, this trend has inflected, with trade in AI-related goods between „unfriendly“ countries stabilising and subsequently rising, even as broad goods trade between the same countries has continued to fragment (Figure 17). This divergence suggests that, outside the narrow frontier segment subject to export controls, commercial interdependence in the AI supply chain is proving more resilient than political alignment.

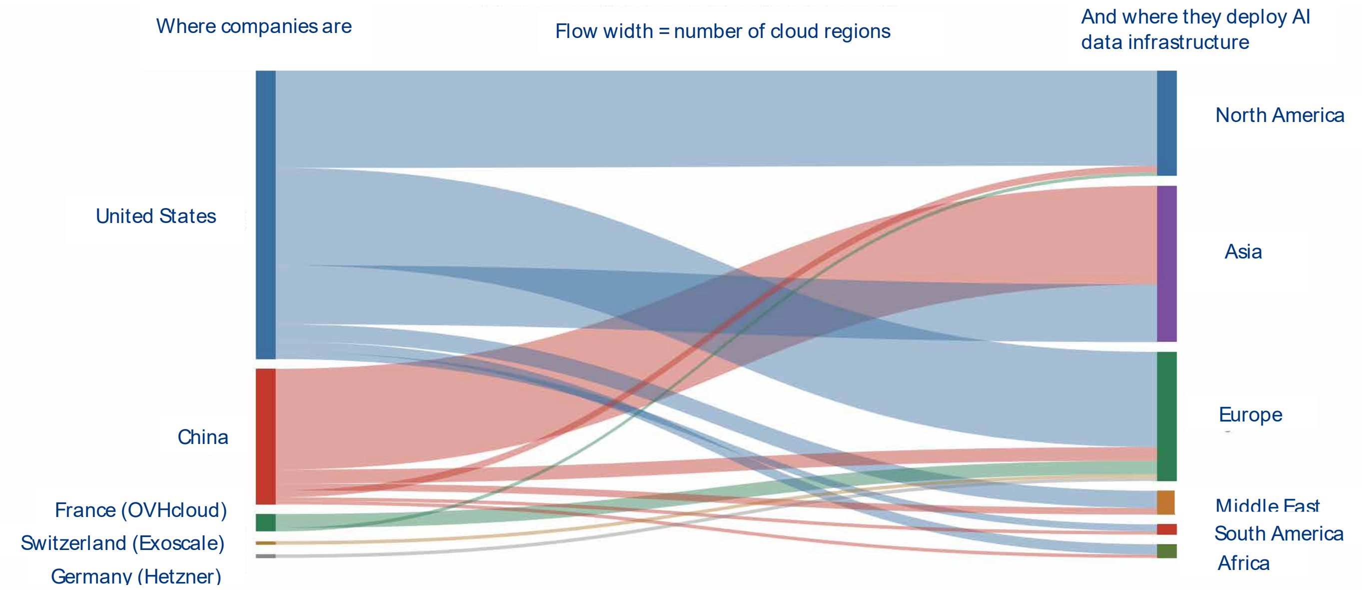
**Figure 17:** Growth of AI-related goods between unfriendly countries experiences the largest increase during 2024



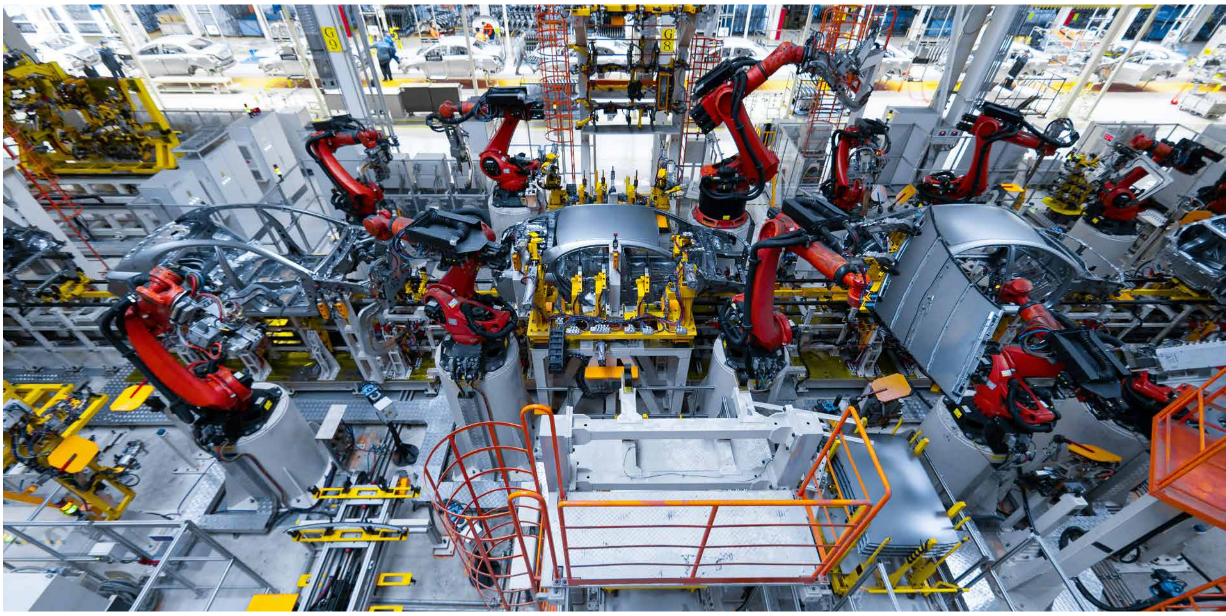

Sources: UN Comtrade, Allianz Research

**AI-related production and infrastructure provision will remain concentrated.** Geopolitics is reshaping semiconductor investment globally, but despite massive subsidy programs across the US, Europe, China, Japan and India, the likely outcome is not full supply security but a more geographically distributed industry that remains highly dependent on a small number of leading-edge producers for the most critical AI and defense technologies. The geographic concentration of large-scale AI-capable data centers is becoming a new source of geopolitical power, as control over compute infrastructure increasingly determines which economies can develop, export and benefit from AI services. The US dominates the global AI infrastructure landscape in the Western hemisphere, while China dominates Asia (Figure 18). Large parts of Africa and Latin America remain largely excluded from the compute capacity and so far depend half-half on the US and China with no dominance decided yet. This growing concentration risks reinforcing technological dependence, widening digital divides and shifting economic and strategic influence toward the countries that control cloud infrastructure, data flows and AI platforms.

**Figure 18:** Distribution of headquarters of AI companies and AI data infrastructure deployment




Sources: WTO (2025), Lehdonvirta, Wu and Hawkins (2024), Allianz Research

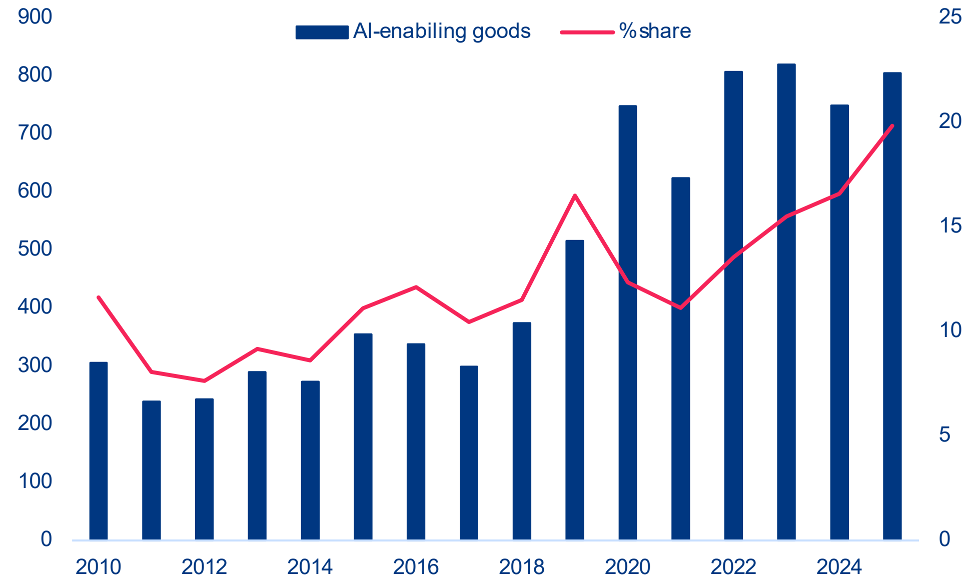


# The race for AI dominance is increasingly also being fought through industrial policy

**Tariffs on AI-enabling products remain low.** Industrial policy and trade rules are becoming central to competition in the field of AI, influencing access to critical minerals, semiconductors, cloud infrastructure, IT services and cross-border data flows, all of which are essential for the development and deployment of AI. Open and predictable trade regimes support innovation, productivity and the integration of AI into global value chains. However, these tools are increasingly being weaponized in an increasingly geopolitical world. Despite rising tensions, tariffs on AI-enabling products remain relatively low. Worldwide, most-favored-nation (MFN) applied tariffs on AI-related goods fell from 5.6% in 2015 to 2.8% in 2025, compared with 7.8% on manufacturing goods overall. The US maintains an average tariff of 1.4%, the EU of 2.0%, the UK of 1.3%, and EFTA countries effectively zero, while China applies the highest average tariff in the group at 3.5%. Overall, AI-related goods continue to be traded under relatively open conditions, reflecting the strong interdependence of global AI supply chains and the strategic importance of maintaining access to computing infrastructure, digital technologies and knowledge flows.

**Non-tariff measures (NTMs) on AI-enabling goods account for a fifth of all harmful NTMs in force.** Over the last decade, NTMs rose from 355 in 2015 to 805 in 2025 (Figure 19) with their share of the total constantly increasing since the deployment of AI to the general public in 2022. These measures, including export restrictions and financial assistance, increasingly reflect efforts to protect technological leadership and secure strategic industries. China has been the top player in implementing trade restrictions with a peak 2020, while the US has more recently picked up since 2022 overtaking China only in 2025. The US and China implemented the most restrictions in 2025 with 208 new restrictions from the US and 159 from China. At the same time, the US is the most affected nation as well, followed by Germany, Malaysia, Thailand, China, the Netherlands and Japan. While the most common new measure implemented in 2025 were import tariffs, this has shifted clearly from financial assistance in foreign markets which was top of the list since 2011 and peaked in 2020 with 431 new measures, followed by tariffs and trade finance and the occasional export ban on AI-enabling goods. While such restrictions may slow technology diffusion in the short term, they also carry significant costs, are difficult to enforce and can accelerate innovation and diversification elsewhere, limiting their long-term effectiveness unless backed by strong domestic industrial capabilities and coordinated policy frameworks.

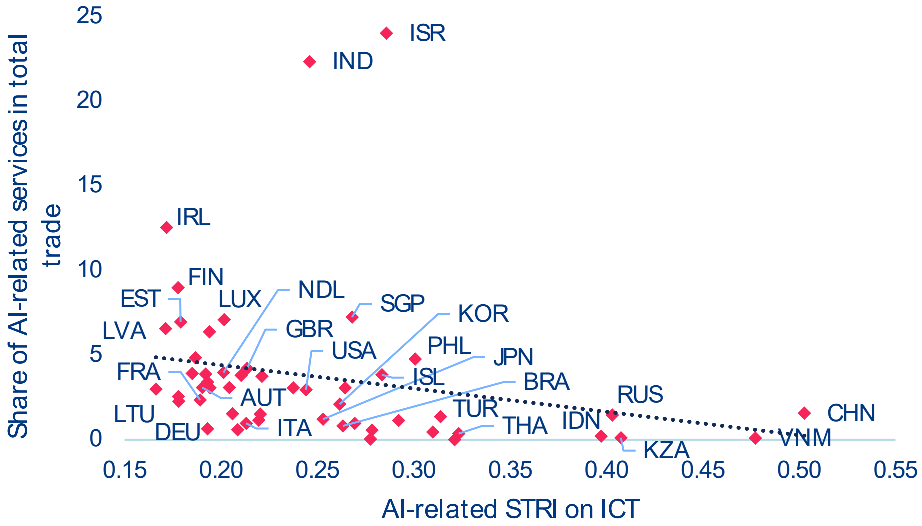
**Figure 19:** NTMs on AI-enabling goods, number of cumulative measures implemented (left) and percentage share (right)




Sources: GTA, Allianz Research

**Trade in AI-enabling services is limited by restrictive regulations.** The trade of AI-related services is increasingly being shaped by restrictions on digital services and cross-border data flows. These restrictions are becoming a key determinant of global AI market share. Although AI is making services such as finance, software and telecommunications more tradable, these sectors are still limited by foreign equity restrictions, licensing regulations, localization requirements and barriers to cross-border delivery. Economies with more restrictive ICT and digital trade regimes capture smaller shares of AI-enabled services exports. In fact, ICT trade restrictiveness explains around 5% of a country's share of AI-related services trade (Figure 20). There is significant regulatory fragmentation: the OECD Digital Services Trade Restrictiveness Index (STRI) records 78 economies requiring a local presence out of 129 countries covered and 46 imposing data localization rules in 2025. The number of countries requiring data to be processed or stored locally nearly doubled in the last decade. These barriers disproportionately affect AI-intensive sectors such as telecommunications, banking, insurance and software, where cross-border supply and commercial presence remain tightly regulated. Furthermore, fragmented data governance amplifies divergence by raising compliance costs, limiting access to cloud infrastructure and global datasets, and weakening participation in AI innovation networks, particularly for smaller economies. While some regulation is necessary for trust and security, openness to digital trade and data flows is now a key factor in achieving AI competitiveness. Economies with restrictive regimes are increasingly failing to capture AI-driven growth.

**Figure 20:** AI-related services share in total trade (in %) to AI-related services trade restrictiveness index

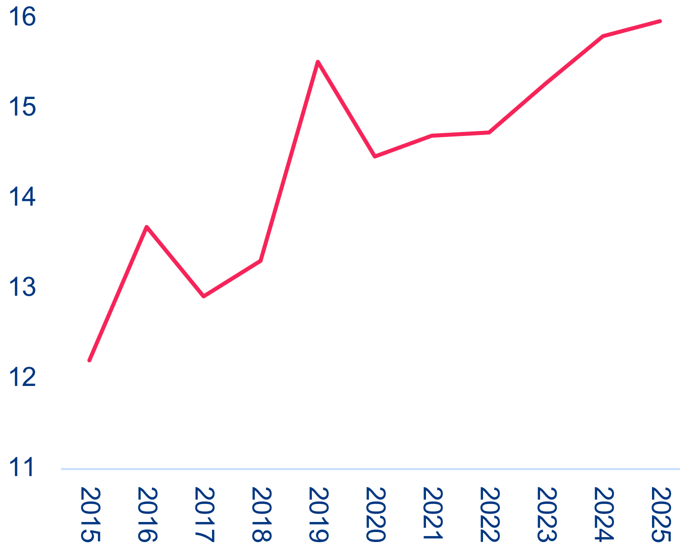


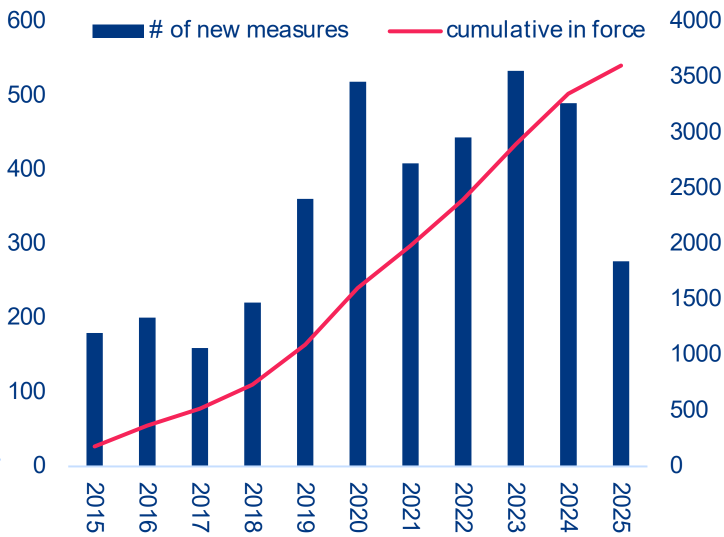

Sources: OCED TIVA, OECD STRI, Allianz Research. Notes: Sector shares are as reported in OECD TiVA VA exports for 2022. AI-related sectors are ICT services. Services restrictiveness based on 2025 data only on ICT services sectors.

**The growing financial support for AI is exacerbating global technological disparities.** Higher-income economies are implementing large-scale AI strategies with the backing of fiscal resources, industrial policy, and targeted support for domestic champions. In contrast, lower-income economies are hampered by fiscal and infrastructure constraints, which limit their ability to take comparable action. This is widening the structural gaps in innovation and trade capacity. The scale of AI-related subsidies is also expanding: the proportion of AI-enabling goods covered by subsidies increased from 12% in 2015 to almost 16% in 2025 (Figure 21, left), and the total number of AI-related subsidy measures now exceeds 3,600 worldwide (Figure 21, right). Although the introduction of new measures slowed in 2025, the overall stock continues to rise. China is in the lead by a large margin, with its subsidy coverage of AI-related goods trade increasing from 14.5% in 2015 to 55% in 2020 and continuing to expand nearly twofold since then. It is followed by South Korea, Malaysia, the US and Japan as the top five AI-related subsidy measure players to expand or protect their position in the AI value chain. These measures primarily target critical materials, advanced semiconductors, AI accelerators (GPUs), semiconductor manufacturing equipment, optical fibres and data-centre components. They are funded through grants, tax credits, loans, state-backed equity injections, electricity subsidies, preferential financing and public support for R&D and infrastructure.

**Initiatives to develop and regulate AI are expanding rapidly, but they are also creating a fragmented policy landscape.** Regulation now extends beyond algorithms to encompass the entire physical and digital infrastructure of AI systems, including data centers, semiconductor supply chains, energy-intensive computing facilities, and cross-border data flows that facilitate AI-driven trade. Governments are no longer merely shaping trade in AI-enabling inputs; they are also actively building industrial ecosystems to secure positions in AI value chains. 23% of countries have set up national AI governance bodies (32 in Europe, 10 in Asia, four in Africa, five in North America, four in Central and South America, and two in Oceania), and 28% have adopted AI regulations, guidelines or standards. These frameworks are increasingly integrating regulation, intellectual property, competition policy, skills development, labor markets and infrastructure investment in order to foster innovation while managing concentration risks. Regional approaches differ significantly. North America has traditionally relied on lighter regulatory frameworks, allowing faster innovation with fewer constraints. China's model is more state-driven, with AI governance closely tied to industrial policy and oversight. Europe occupies a middle ground: while regulation strengthens trust, transparency and adoption, key factors in scaling AI, it also introduces

**Figure 21:** Share of cumulative subsidies on AI-enabling goods in total subsidies, cumulative in force as % of total (left) and number of new subsidy measures in force (right)




Sources: GTA, Allianz Research

compliance and security requirements that can slow innovation cycles, even though many processes are becoming increasingly automatable. Overall, however, regulation is not currently a binding constraint in Europe. AI governance is emerging as a global trade regime in its own right, which could further reinforce the existing dominance of a select few. At the multilateral level (Table 1), frameworks such as the OECD AI Principles and the G7 Hiroshima Process promote interoperable standards and trusted data flows in order to reduce friction in the trade of AI services. At the same time, national strategies are becoming increasingly geopolitical. The US ‚Pax Silica' approach links AI directly to trade security through export controls, investment screening, and domestic incentives for the production of chips and computing power.

Meanwhile, China's Digital Silk Road initiative integrates AI infrastructure exports, such as datacenters and cloud platforms, into broader trade and connectivity strategies. Middle Eastern economies are leveraging their energy abundance and sovereign wealth to establish themselves as global compute hubs with AI-friendly regulatory environments. Other initiatives, including the UNESCO AI Ethics Framework and the UN Global Digital Compact, seek to establish shared norms for data governance, model oversight and infrastructure efficiency. Together, these overlapping regimes are increasingly determining where AI is produced, how it is governed and how AI-enabled goods and services flow globally, effectively turning AI governance into an emerging trade regime in its own right.

**Table 1:** Major AI-trade-related international policy initiatives

| Initiative | Lead Actors | Core Focus Areas | Trade Relevant Impacts |
|---|---|---|---|
| **OECD AI Principles (2023)** | OECD countries | Responsible AI, transparency, data governance | Supports interoperable standards for cross border data flows and cloud services; reduces regulatory fragmentation. |
| **G20 AI Principles (2019)** | G20 economies | Human centric AI, innovation, digital economy | Encourages harmonized data governance approaches that facilitate AI enabled services trade. |
| **UNESCO AI Ethics (2021)** | UNESCO members | Ethical standards, inclusiveness, data rights | Shapes global norms for data handling, influencing access to datasets and model training resources. |
| **G7 Hiroshima Process & AI Code of Conduct (2023)** | G7 | Frontier model safety, AI supply chain security | Promotes common benchmarks that support trusted AI supply chains; potential trade conditions for high risk AI systems. |
| **Bletchley Declaration (2023)** | 28 countries incl. US, UK, EU, China | Safety of frontier models | Introduces shared risk management expectations, influencing compliance requirements for cross border AI deployment. |
| **UN General Assembly AI Resolution (2024)** | UN member states | Safe, inclusive global AI governance | Calls for capacity building in digital infrastructure; facilitates AI adoption in developing countries. |
| **Council of Europe Framework Convention on AI (2023)** | Council of Europe | Human rights, democratic governance of AI | Sets obligations that may become preconditions for AI trade with EU markets. |
| **International Network of AI Safety Institutes (2024)** | Multinational | Safety research, model evaluations | Could lead to standardized audits that function as "technical passports" for AI model trade. |
| **UN AI Advisory Body Final Report (2024)** | UN | Global governance mechanisms | Recommends infrastructure investment frameworks and compute governance strategies. |
| **UN Global Digital Compact (2024)** | UN | Digital cooperation, data flows, public infrastructure | Promotes open digital markets and trusted cross border data exchange. |
| **International Scientific Report on Advanced AI Safety (2024)** | Multilateral science groups | Scientific evaluation of AI risks | Provides technical benchmarks and safety thresholds that may condition AI exportability. |
| **US Pax Silicia Strategy** | United States | Chip controls, AI supply chain security, cloud regulation | Reshapes global trade in semiconductors, cloud capacity, and advanced models; creates new dependencies. |
| **China's Digital Silk Road** | China | Digital infrastructure, cloud, AI services exports | Embeds AI capabilities into BRI corridors, strengthening China centric digital trade routes. |
| **Middle East AI Hub Strategy** | UAE, Saudi Arabia, Qatar | Compute clusters, cloud zones, energy powered AI | Turns the region into a global exporter of compute and AI services; attracts multinational AI firms via regulatory free zones. |
| **OECD Cross Border Data Flows** | OECD | Data governance, interoperability, trust frameworks | Provides stock taking of global data flow policies; supports common G7/G20 approaches for enabling AI driven digital trade. |
| **OECD Going Digital: Trade & Data Flows** | OECD | Policy landscape for international data flows | Identifies how data regulations affect international trade and AI-enabled services; informs trade negotiators. |
| **OECD aligned research on AI & digital trade rules** | OECD affiliates, legal scholars | Next gen rules for AI, digital trade, cross border data | Explores regulatory divergence, national security restrictions, and future international digital trade agreements. |

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Allianz Research | 19 May 2026

# Transatlantic spread – from monetary signal to fiscal mirror

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## In Summary

- **Is one of the world's most important pricing signals entering a new era?** Since 2015, the 10-year spread between US Treasury yields and Eurozone German Bunds has ranged between 100bps and 270bps. Since early 2025, pushed first by Germany's fiscal pivot and then by converging rate expectations (hawkish ECB vs Fed easing bias), the 10y US–Bund spread has narrowed by roughly 90bps. If this continues, the transatlantic spread could finally break through the lower trading range that has held for more than a decade.
- **The latest bond sell-off suggests that the convergence trade has run its course: We expect a gradual reversal rather than a new tighter-for-longer regime.** Over the last few weeks, the spread has stopped narrowing and rewidened to around 140bps. In the short run (next 6–12 months), the move should remain modest (+20bps) and still driven by rate expectations as markets reprice a less hawkish ECB against a Fed that stays on hold longer than consensus expects or even proceeds to hike.
- **In the medium term (3–5 years), the transatlantic drivers will shift decisively from rate expectations to risk premia**. We see the US neutral rate rising by 10–20bps on stronger AI-driven productivity gains while the Eurozone neutral rate drifts 10bps lower, consistent with a historical 20–50bps widening of the real-rate spread. On top of that, the US Treasury term premium looks underpriced relative to Bunds: the US convenience yield has flipped from -30bps to +20bps as Treasury supply erodes the safety premium, while the Bund retains a -40bps convenience yield anchored in collateral scarcity. The ECB's faster QT has already cheapened Bund duration by 35bps versus 10bps for Treasuries and we expect that gap to close. Adding a structural +10bps from the OIS/swap-spread channel, we see the spread widening by ~75bps to around 200bps.
- **Ultimately, the transatlantic spread is evolving from a pure monetary-cycle gauge to a barometer of fiscal credibility and sovereign liquidity.** For investors, this means the transatlantic spread should be monitored not only through the lens of rate differentials, but equally as a mirror of fiscal sustainability and sovereign market liquidity. For duration positioning, this means active positioning in Bunds versus US Treasuries not only in terms of rate expectations but also where the term premium repricing will be most pronounced (longer maturities). In addition, the marginal buyer of Treasuries will increasingly demand a higher term premium, supporting a steeper US curve and a structurally weaker dollar against the euro. For corporate issuers, the EUR–USD funding-cost differential is expected to widen, favoring euro issuance (reverse Yankee issuance) that may support EUR investment-grade bonds over their US counterparts.

## The transatlantic spread reaches the lower bound of its long-term trading range

**Is transatlantic spread entering a new lower-for-longer era?** The transatlantic spread, which refers to the differential between US Treasury yields and Eurozone benchmark yields (German Bunds), is among the most important pricing signals in global financial markets. It influences two of the largest fixed-income markets and moves the EURUSD, the most important currency pair in global trade and financial transactions.

Any sustained shift in the transatlantic spread carries transmission effects across global capital flows, equity valuations, corporate funding and hedging costs. Since the 1980s, transatlantic spread has experienced three distinct phases. From 1980 to 1995, covering the Volcker Era and the post oil-shock disinflation, it was characterized by a high and volatile spread, fluctuating between +500bps and -150bps for the ten-year maturity. The period from 1995 to 2015 marked a phase of moderation and convergence, during which the spread appeared capped at an upper bound of 100bps and rarely fell below -100bps. With the onset of ECB quantitative easing in 2015, which pushed long-term yields in the Eurozone persistently lower, a third phase began, with the ten-year spread ranging between 100bps and 270bps (Figure 1). Since 2025, the transatlantic spread has narrowed by ~90bps, approaching the lower bound of the third phase trading range. This raises the question of whether we are at the start of a new lower-for-longer regime or should expect a near-term reversal of the narrowing trend (Figure 2).

Figure 1: Long-term 10y transatlantic spread
*Spread in %*

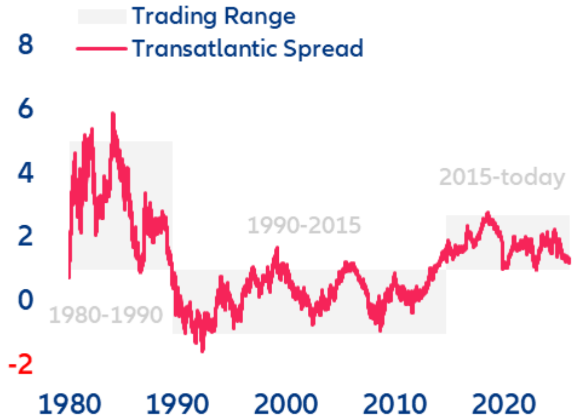
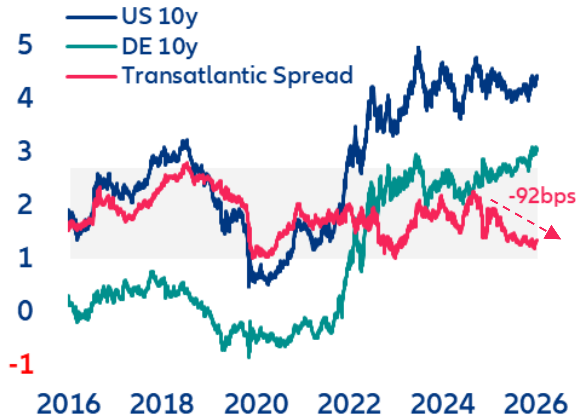



Figure 2: Transatlantic spread at lower bound
*Yield and spread in %*




*Sources: LSEG Workspace, Allianz Research*

## US and Eurozone yields: from term-premium suppression to repricing rate expectations

**At its core, the transatlantic spread signals divergence in the monetary policy stance as well as in the growth and inflation outlooks.** But it also embodies risk premia for the uncertainty around future inflation and neutral rates in addition to premia for fiscal risk and market liquidity. Term-structure models help to refine this view by decomposing observed market yields and rates into three components:

- **Rate expectations** capture growth and inflation expectations as expressed by expected nominal short-term rates.
- **Term premium** reflects the uncertainty around the future path of short-term rates and inflation expectations. It also captures duration supply/demand pressure with fiscal policy (debt issuance) on the supply side and changes in central bank bond purchases and holdings – Quantitative Easing (QE) or Quantitative Tightening (QT) – as a major factor on the demand side.
- **OIS spread**, defined here as the difference between government bond yields and the swap rate, represents the premia for market liquidity. Its level and sign signal whether government bonds are scarce or abundant, and whether they are cheap or expensive to finance and hedge (market liquidity). The OIS spread is an indicator of imbalances in government bond supply and demand. The higher the OIS spread, the more constrained governments are in issuing additional debt without paying a liquidity premium to the market.

**The weight of these components varies over time.** If we look at the composition of US and German 10y yield over the last 10 years, we can clearly see these shifts. Pre-Covid, both yields were subject to prolonged suppression of term premia (negative), reflecting QE duration demand, while rate expectations reflected different monetary policy stances: tightening in the US versus stable in the Eurozone. Post-Covid, both yields surged, driven by rapidly rising rate expectations. During this time, the term premium shifted from a dampening to an amplifying factor once central banks reduced duration demand through QT while governments increased supply by fiscal expansion. Since 2025, US rates have held steady as a rising term premium was offset by falling rate expectations, reflecting the Fed's easing bias amid a late-cycle slowdown. German Bund yields, however, rose steadily. First this was driven by a higher term premium following Germany's fiscal pivot (*Sondervermögen*). In a second phase, from mid-2025 onwards, the yield increase was driven by a shift in rate expectations as markets priced upside risk to euro short-term rates on the back of an emerging economic recovery. Over this period, market liquidity played a minor role for US Treasuries in absolute terms. Structurally, however, OIS spreads switched into positive territory in 2025, marking a shift from the US government being a receiver of a liquidity premium to being a payer. For German Bund yields on the other hand, OIS spreads were a major driver – at times even equal to the term premium. Until 2022, this reflected the relative scarcity of German government debt in the euro sovereign market. Investors paid a substantial premium to hold these ultra-safe assets, contributing to structurally low Bund yields. With the fiscal pivot, this privilege has at least diminished, reflecting the increased supply of German government bonds (Figure 3 and 4).

Figure 3: Decomposition of US 10y Treasury yield
*Yield and components in %*

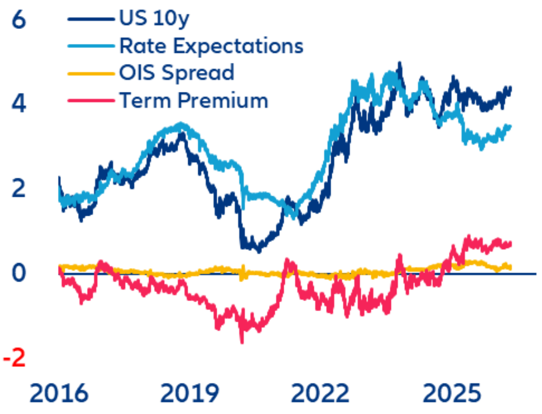
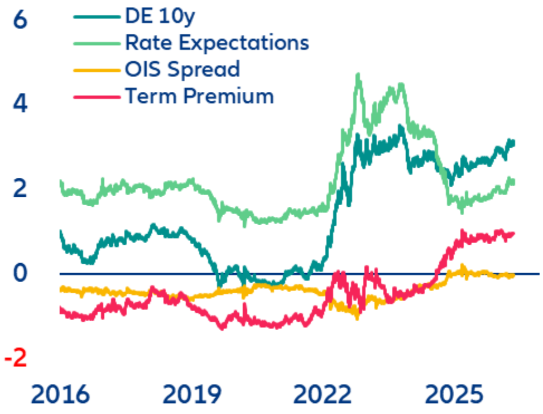



Figure 4: Decomposition of 10y German Bund yield
*Yield and components in %*

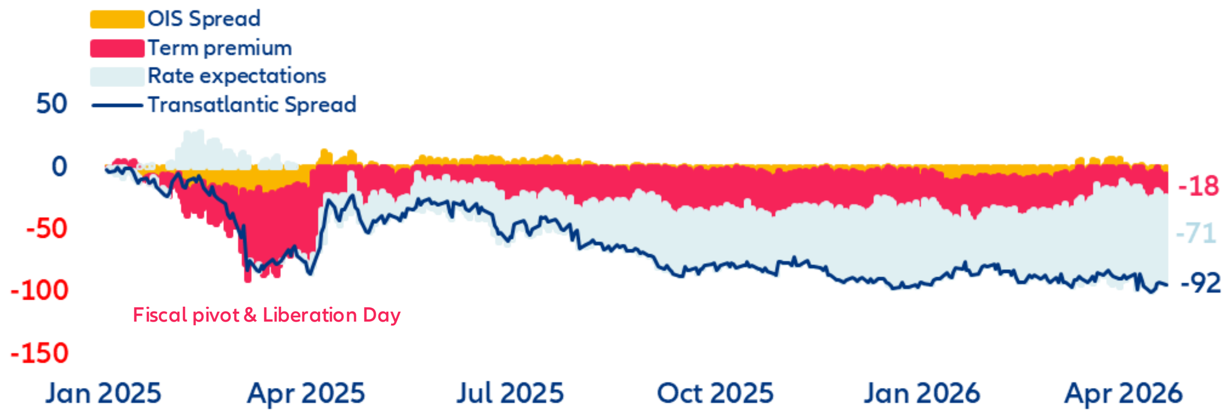



*Notes: Term structure decomposition based on Abrahams et al. (2016)*
*Sources: LSEG Workspace, Allianz Research*

## Latest transatlantic spread narrowing driven by rate expectations not risk premia

**The latest narrowing of the transatlantic spread (-92bps since 2025) was initially driven by a term-premium repricing after the German fiscal pivot, reinforced by the "Liberation Day" trade shock.** After that, however, it increasingly became the product of converging rate expectations (reflecting the easing bias of the Fed versus an ECB on standby) and divergent growth dynamics (slowdown in US vs early recovery in Eurozone). The energy-price shock triggered by the Iran war reinforced this trend but inverted the signs of the rate expectations. Markets now expect the Fed to stay on pause while pricing at least two ECB rate hikes by end-2026 (Figure 5). If this trend continues, the transatlantic spread could finally break through the lower trading range of the last 10 years.

Figure 5: Contributions to the 10y transatlantic spread since 2025
*Yield and components in bps*




*Notes: Term structure decomposition based on Abrahams et al. (2016)*
*Sources: LSEG Workspace, Allianz Research*

**However, we believe the current convergence trend is nearing its end and is about to reverse.** In the near term, markets are pricing in 25bps of excess ECB rate hikes through year-end, in our view. This should push the transatlantic spread wider again in H2 2026 as long-term inflation expectations remain largely aligned, offering little offset. In the medium term, we see real rates as a major driver for a wider transatlantic spread. The expected real rate differential currently sits at its long-term average but we estimate the US neutral rate will rise by 10-20bps over the next five years, driven by stronger productivity growth. In contrast, we see the Eurozone neutral rate declining by 10bps. Historically, such divergence in neutral rates has been followed by a widening of the expected real rate spread between US and Eurozone of at least 20–50bps

Figure 6: Difference in neutral rates and expected rates
*Real yield in %*

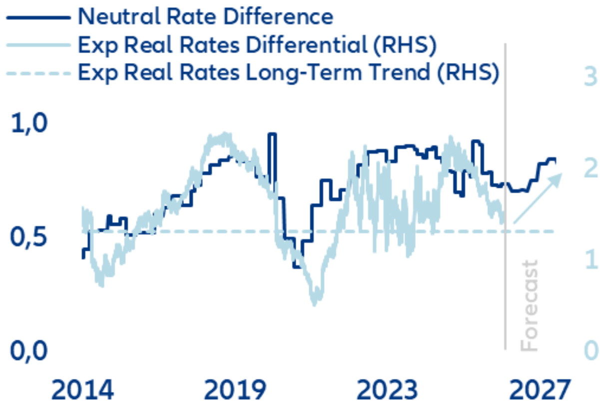
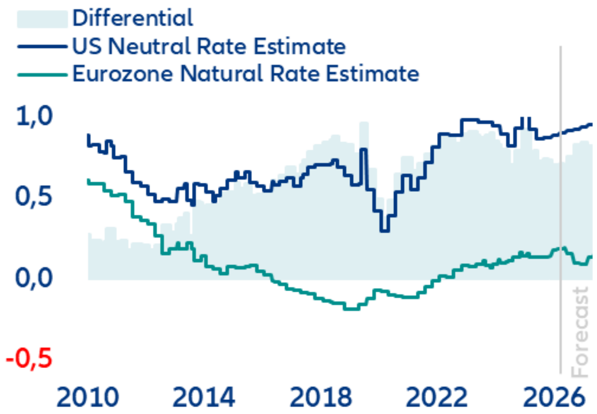



Figure 7: US vs Eurozone neutral rates
*Real yield in %*




*Notes: US neutral rates estimate as median value of Lubik & Matthes (2015), Del Negro et al. (2020), Davis, Fuenzalida, Huetsch, Mills, and Taylor (2024), Ferreira & Shousha (2023), Holston, Laubach and Williams (2017) and Ferreira, Lott and Richards (2025)*
*Eurozone neutral rate estimate as median of Lubik & Matthes (2015), Brand et al. (2024), Davis, Fuenzalida, Huetsch, Mills, and Taylor (2024), Ferreira & Shousha (2023), Holston, Laubach and Williams (2017) and Ferreira, Lott and Richards (2025)*
*Sources: LSEG Workspace, Allianz Research*

## Diverging QT pace drives term premium

**On the term premium side, we also see no further impulse for convergence in US and Eurozone sovereign yields.** The term premiums for 10y Bund and 10y US Treasury tracked closely for an extended period. From mid-2024, however, the Bund term premium began rising significantly faster than its US counterpart. Today, the German term premium stands 60bps above the US level (Figure 8). We consider this gap excessive. Notably, half of the divergence is attributable to the ECB's faster pace of quantitative tightening (QT). We estimate that the run-off of ECB holdings has cheapened 10y Bund duration by 35bps, versus only 10bps in the US (Figure 9).

Figure 8: German term premium exceeds US by 60bps
*in %*

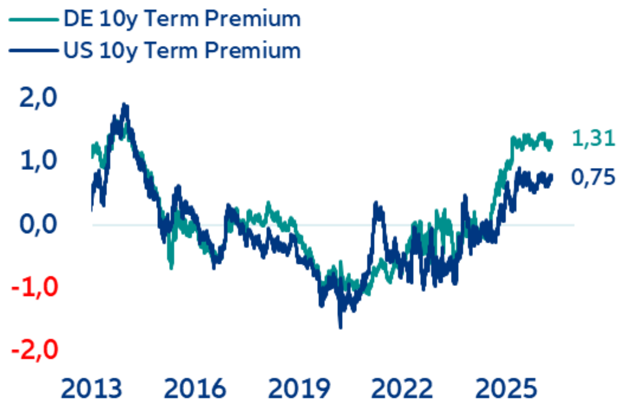
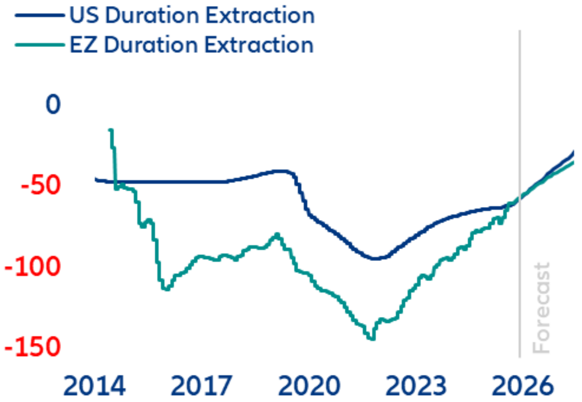



Figure 9: QE/QT effect on 10y term premium
*in bps*




*Notes: Term structure decomposition based on Abrahams et al. (2016)*
*Notes: for US: Li & Wei (2013) ; Eurozone: Eser at al. (2019)*
*Sources: LSEG Workspace, Allianz Research*

**We expect this QT effect to equalize over the next years.** In particular, recent Fed T-Bill purchases do not cause a renewed yield-dampening effect. Confined to very short-dated securities, these purchases do not drain duration from the market, therefore the upside pressure on US duration pricing from of Fed portfolio run-off remains intact. The elevated German term premium relative to the US also appears stretched given the US's higher risk profile. Strong expansion of Treasury supply (fiscal deficit) has eroded the US convenience yield – the premium investors pay for an asset's combined safety and liquidity properties. US Treasuries still carry a significant liquidity privilege based on the USD-based collateral (repo) ecosystem. But the safety component has deteriorated to a point where the US convenience yield flipped from a dampening effect of -30bps at end-2024 to +20bps most recently. The Bund convenience yield has also declined with rising issuance (fiscal pivot) but remains at -40bps, which is 60bps below US Treasuries (Figure 10). The scale of the US Treasury risk premium is also visible when comparing the term premium of the US 10y Treasury bond with the one implied in the USD 10y swap rate. The two tracked almost identically until "Liberation Day", after which markets have priced an additional +50bps premium for the sovereign (Figure 11). In our view, this is not yet sufficiently priced relative to the Bund. On this basis, we see clear upward pressure on the US Treasury term premium relative to the Bund.

Figure 10: Convenience yield US vs Germany, 10Y maturity
*in %*

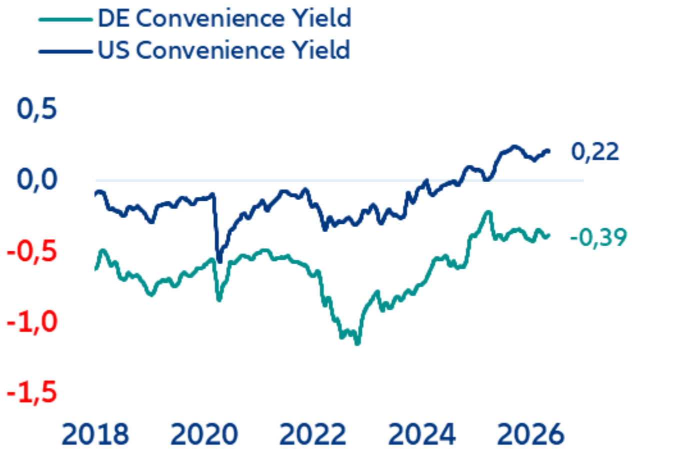



Figure 11: US 10Y Treasury risk premium vs swap
*in %*

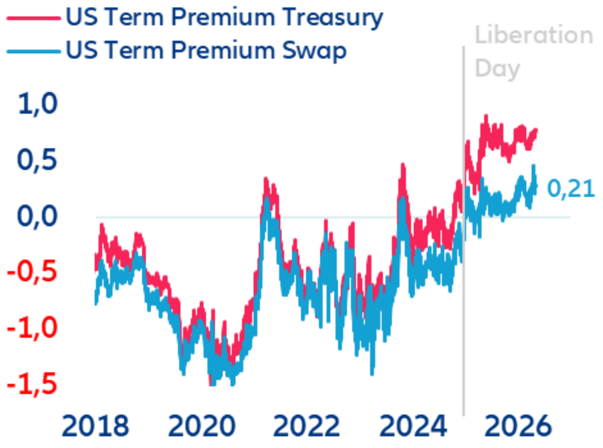



*Fig.10 Notes: to illustrate the effect on the total yield, the sign of the convenience yield is inverted. A negative value represents a yield dampening effect, a positive value represents an upside pressure on the total yield. Convenience yield is calculated as the average of synthetic proxies mentioned in Lustig et al. (2024) and the intersovereign convenience yield in Du et al. (2018) Fig 11. Notes: Term structure decomposition based on Abrahams et al. (2016)*

*Sources: LSEG Workspace, Allianz Research*

## The transatlantic liquidity divide will drive OIS spreads

**The transatlantic OIS spread differential remains structurally biased toward widening in Europe's favor, driven by fundamentally asymmetric liquidity and supply dynamics.** In the US, persistently high net Treasury issuance (USD2-3trn/year), structurally constrains dealer balance sheets and weaker foreign demand keeps US swap spreads biased wider. Near-term regulatory relief (Supplementary Leverage Ratio adjustment) could provide a temporary tightening impulse, but the underlying trend remains intact. In the Eurozone, the picture is the mirror image. Despite the step-up in Bund supply (EUR170-180bn additional issuance), superior bank absorption capacity, ample excess reserves and persistent high-quality collateral scarcity in repo markets support a structural tightening bias on euro swap spreads. The specialness[1] of German Bunds is still driven by collateral scarcity rather than credit. This is the repo market signal for of the higher Bund convenience yield. It reinforces this dynamic, as does solid foreign demand on the back of euro strength and a more favorable hedging profile for non-US investors. The net result is a widening US–EU liquidity differential, with euro swap spreads tightening relative to their US counterparts over the medium term. We see potential of around +10bps effect on the transatlantic spread.

[1] Specialness being the premium of a strongly desired collateral which allows cheaper borrowing in repo transactions.

**We expect the 10Y spread to widen by approximately +20bps in the short run. In the medium term (next 3 to 5 years) we see a widening of +75bps to around 200bps, reversing the bulk of its compression since 2025** (Table 1). However, we expect a clear shift in the drivers of that move. The transatlantic spread is transitioning from a predominantly rate-expectations-driven signal reflecting monetary policy divergence to one equally shaped by risk premia repricing.

Table 1: Transatlantic spread components

| Component | Sub-component | Since 2025 | Short-term Outlook | Mid-Term Outlook |
|---|---|---|---|---|
| **Rates Expectations** | Real Rates Expectations | -67bps | +15bps | +30bps |
| | Inflation Expectations | -5bps | +5bps | +10bps |
| **Term Premium** | QT Effect | -27bps | | |
| | Risk premium | +12bps | | +25bps |
| **OIS Spread** | OIS Spread | -5bps | | +10bps |
| **TOTAL** | | -92bps | +20bps | +75bps |

*Sources: LSEG Workspace, Allianz Research*

**We find the US Treasury term premium underpriced relative to the Bund, given the scale of US fiscal deficits, eroding convenience yield and tighter liquidity.** By contrast, the Eurozone benefits from a supportive convenience yield and more favorable liquidity conditions. This expected change in the composition of the transatlantic spread shows how it is evolving from a pure monetary cycle signal into a barometer of sovereign fiscal credibility. Treasury supply dynamics will increasingly set the floor for US long-end yields as the US is forced to offer elevated real rates to retain foreign capital inflows. Europe, meanwhile, stands to benefit from a relative repricing: even as Bund supply rises, the Eurozone's stronger structural liquidity position and lower fiscal risk premium provide a durable anchor. For investors, this means the transatlantic spread should be monitored not only through the lens of rate differentials, but equally as a gauge of fiscal sustainability and sovereign market liquidity. A cross-curve duration positioning between Bunds and US Treasuries should therefore not only reflect rates expectations but take into account which curve segment will see the most pronounced term premium repricing. A structurally higher term premium in the US should also add steepening pressure on the US curve and act as a weakening force on the USD versus EUR. On credit and issuance, we expect the EUR–USD funding cost differential to widen, favoring euro issuance for global borrowers (reverse Yankee supply). This could have a tightening effect on EUR investment-grade spreads relative to their USD counterparts. In short, the transatlantic spread is moving from a "*Fed versus ECB*" story to a "*Treasury versus Bund*" story, and portfolios built for the former need to be repositioned for the latter.

These assessments are, as always, subject to the disclaimer provided below.

## FORWARD-LOOKING STATEMENTS



## NO DUTY TO UPDATE